

Department of Software Engineering

University of Gujrat

LegisCounsel



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STATEMENT OF SUBMISSION

This is to certify that **Reham Batool** Roll No. **22011598-014, Nayab Gul** Roll No. **22011598-030** and **Ramiz Ilyas** Roll No. **22011598-051** have successfully completed the final project/thesis named as: **LegisCounsel**, at the Faculty of C & IT, University of Gujrat, to fulfill the partial requirement of the degree of **Bachelor of Science in Software Engineering**.

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Abstract

Legal research in Pakistan remains time-intensive and inaccessible to non-lawyers due to manual search methods and complex legal language. Existing digital solutions are expensive and designed exclusively for legal professionals, leaving the general public without adequate access to simplified legal understanding. LegisCounsel proposes an AI-powered intelligent legal argument searching system that uses Natural Language Processing and semantic search to interpret user queries contextually. The system delivers dual-interface outputs – technical, detailed results for lawyers and simplified, easy-to-understand explanations for clients. Feasibility analysis confirms technical, economic, and operational viability using React.js, Node.js, MongoDB, and transformer-based NLP models. Thirty-two functional requirements were specified across lawyer, client, and admin roles. Comprehensive testing achieved a 100% pass rate on forty-seven test scenarios, with search response times averaging 6.3 seconds. The system is initially scoped to criminal law in Pakistan, with scalability for additional legal domains. LegisCounsel demonstrates how AI can improve access to justice by making legal research both efficient for professionals and understandable for the public.

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Chapter 1: Project Feasibility Report.....9
1.1. Introduction.....9
1.2. Project/Product Feasibility Report.....9
1.2.1. Technical Feasibility.....9
1.2.2. Operational Feasibility.....9
1.2.3. Economic Feasibility.....10
1.2.4. Schedule Feasibility.....10
1.2.5. Specification Feasibility.....10
1.2.6. Information Feasibility.....10
1.2.7. Motivational Feasibility.....10
1.2.8. Legal & Ethical Feasibility.....11
1.3. Project/Product Scope.....11
1.4. Project/Product Costing.....11
1.4.1. Project Cost Estimation By Function Point Analysis.....11
1.4.2. Project Cost Estimation by using COCOMO'81 (Constructive Cost Model).....15
Basic COCOMO Estimation.....15
1.4.3. Activity Based Costing.....16
1.5. Task Dependency Table.....17
1.6. CPM - Critical Path Method.....18
1.7. Gantt chart.....24
1.8. Introduction to Team member and their skill set.....24
1.9. Task and Member Assignment Table.....25
1.10. Tools and Technology with reasoning.....28
1.11. Vision Document.....29
1.12. Risk List.....30
1.13. Product Features/ Product Decomposition.....30
Chapter 2: Software Requirement Specification (For Object Oriented Approach).....31
2.1 Introduction:.....31
2.1.1 System Specifications.....32
2.1.2. Identifying External Entities.....35
2.1.3. Context Level Data Flow Diagram:.....36
2.1.4. Capture "shall" Statements:.....36
2.1.5. Allocate Requirements:.....38
2.1.6. Prioritize Requirements:.....41
2.1.7. Requirements Trace-ability Matrix:.....44
2.1.8. High Level Use Case Diagram:.....48
2.1.9. Analysis Level Use Case Diagram:.....49

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2.1.10. Use Case Description....50
Chapter 3: Design Document (For Object Oriented Approach)....64
3.1. Introduction:....64
3.2. Domain Model....65
3.3. System Sequence Diagram....66
3.4. Sequence Diagram....67
3.5. Collaboration Diagram....80
3.6. Operation Contracts....83
3.7. Design Class Diagram....89
3.8. State Chart Diagram....90
3.8.1 For Admin....90
3.8.2 For Lawyer/Client....91
3.9. Data Model....92
Chapter 4: User Interface Design....93
4.1. Introduction....93
4.2. Site Maps....93
4.2.1 Landing Page....93
4.2.2 Login Page....93
4.2.3 Admin Panel....93
4.2.4 Client Dashboard....93
4.2.5 Lawyer Dashboard....94
4.2.6 Settings Page....94
a) Profile Page....94
b) Security Page....94
4.2.7 Case Management Page (Optional Module)....94
4.2.8 Communication Page (Optional Module)....94
4.3. Story boards....96
4.3.1 Login....96
4.3.2 Register....97
4.3.3 Admin Dashboard....97
4.3.4 Lawyer Legal Search(Technical View)....98
4.3.5 Client Legal Search(Simplified View)....99
4.3.6 Settings Management(Profile, Security)....100
4.3.7 Lawyer Case Management(Optional)....100
4.3.8 Client Case Tracking(Optional)....101
4.3.9 Communication(Optional)....102
4.4. Navigational maps:....103
4.5 Trace-ability Matrix....111
4.5.1 Login....111

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4.5.2 User Registration....111
4.5.3 Lawyer Dashboard (Technical View)....112
4.5.4 Client Dashboard (Simplified View)....112
4.5.5 Settings....113
4.5.6Admin Dashboard....113
4.5.7 Lawyer Case Management (Optional)....114
4.5.8 Client Case Tracking (Optional)....114
4.5.9 Communication (Optional)....115
Chapter 5: Software Testing....116
5.1 Introduction:....116
5.2. Test plan....116
5.2.1. Purpose....116
5.2.2. Outline....117
5.2.2.1. Test Plan Identifier....117
5.2.2.2. Introduction....117
5.2.2.3. Test Items....118
5.2.2.4. Features to be Tested....119
5.2.2.5. Features not to be Tested....123
5.2.2.6. Approach....124
5.2.2.7. Item Pass/Fail Criteria....125
5.2.2.8. Suspension Criteria and Resumption Requirements....126
5.2.2.9. Test Deliverables....127
5.2.2.10. Testing Tasks....127
5.2.2.11. Environmental Needs....129
5.2.2.12. Responsibilities....130
5.2.2.13. Staffing and Training Needs....132
5.2.2.14. Schedule....132
5.2.2.15. Risks and Contingencies....133
5.2.2.16. Approvals....134
5.3. Test Design Specification....134
5.3.1. Purpose....134
5.3.2. Outline....135
5.4. Test Case Specification....135
5.4.1. Purpose....135
5.4.2. Test Case Specification Identifier....135
5.4.3. Outline....136
5.5. Test Procedure Specification....147
5.5.1. Purpose....147
5.5.2. Outline....147
5.5.2.1. Test Procedure Specification Identifier....148

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5.5.2.2. Purpose....148
5.5.2.3. Special Requirements....148
5.5.2.4. Procedure Steps....149
5.5.2.4.1. Log....149
5.5.2.4.2. Set Up....149
5.5.2.4.3. Start....149
5.5.2.4.4. Proceed....150
5.5.2.4.5. Measure....150
5.5.2.4.6. Shut Down....151
5.5.2.4.7. Restart....151
5.5.2.4.8. Stop....151
5.5.2.4.9. Wrap Up....152
5.5.2.4.10. Contingencies....152
5.6. Test Item Transmittal Report....153
5.6.1. Purpose....153
5.6.2. Outline....153
5.6.2.1. Transmittal Report Identifier....153
5.6.2.2. Transmitted Items....153
5.6.2.3. Location....154
5.6.2.4. Status....154
5.6.2.5. Approvals....155
5.7. Test Log....155
5.7.1. Purpose....155
5.7.2. Outline....156
5.7.2.1. Test Log Identifier....156
5.7.2.2. Description....156
5.7.2.3. Activity and Event Entries....156
5.7.2.3.1. Execution Description....156
5.7.2.3.2. Procedure Results....157
5.7.2.3.3. Environmental Information....157
5.7.2.3.4. Anomalous Events....158
5.7.2.3.5. Incident Report Identifiers....158
5.8. Test Incident Report....158
5.8.1. Purpose....158
5.8.2. Outline....158
5.8.2.1. Test Incident Report Identifier....159
5.8.2.2. Summary....159
5.8.2.3. Incident Description....159
5.8.2.4. Impact....160
5.9. Test Summary Report....160

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5.9.1. Purpose....160
5.9.2. Outline....160
5.9.2.1. Test Summary Report Identifier....160
5.9.2.2. Summary....161
5.9.2.3. Variances....161
5.9.2.4. Comprehensiveness Assessment....162
5.9.2.5. Summary of Results....162
5.9.2.6. Evaluation....163
5.9.2.7. Summary of Activities....164
5.9.2.8. Approvals....165
Chapter 6: User Technical Manual....166
6.1. General Information....166
6.1.1 System Overview....166
6.2. System Summary....167
6.3. Logging On....167
6.3.1 System Menu....168
6.3.1.1 Login Function....168
6.3.1.2 Dashboard....169
6.3.2 Changing User's Credentials....171
6.3.3 Exit System....171
6.4. Using the System....171
6.4.1 Landing Page....172
6.4.2 Registration Page....173
6.4.3 Login Page....174
6.4.4 Client Dashboard (AI Legal Assistant)....175
6.4.5 Lawyer Dashboard (Legal Search System)....176
6.4.6 Admin Dashboard....177
6.4.7 Settings Page....178
6.5 Special Instructions for Error Correction....179
6.6 Caveats and Exceptions....180

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Chapter 1: Project Feasibility Report

1.1. Introduction

Legal research is one of the most time-consuming and complex tasks for lawyers, judges, and even clients who wish to understand their rights. Traditional research methods rely on manual searching through extensive collections of judgments, statutes, and case laws, which is not only slow but also difficult for non-lawyers to interpret due to technical legal language. Furthermore, existing systems are often designed exclusively for legal professionals and are expensive, leaving clients without proper access to simplified legal understanding.

LegisCounsel proposes an **AI-Powered Intelligent Legal Argument Searching System** that utilizes **Natural Language Processing (NLP)** and **Semantic Search** to understand the meaning of user queries and provide the most relevant legal arguments and judgments.

Unlike keyword-based search engines, LegisCounsel interprets context, summarizes complex cases, and presents results in dual formats: one for lawyers (technical, detailed) and one for clients (simplified, easy to understand).

Initially, the system focuses on selected laws but is scalable to include more laws and legal domains. Optional modules will allow lawyers and clients to connect, manage cases, and schedule appointments, making the system practical, accessible, and intelligent.

1.2. Project/Product Feasibility Report

When initiating a project, assessing its feasibility ensures it can be developed effectively with available resources. The feasibility of LegisCounsel was analyzed under the following types:

1.2.1. Technical Feasibility

LegisCounsel is technically feasible. The project employs widely available technologies and frameworks:

- **Frontend:** React.js
- **Backend:** Node.js +Express.js
- **Database:** MongoDB
- **AI/NLP:** Python with Transformer models (Ollama, GPT) for semantic search and summarization
- **Search Engine:** VectorSearch for fast and intelligent semantic search
- **Deployment:** Cloud platforms like AWS or GCP

The development team possesses expertise in full-stack development, machine learning, and NLP model integration, making the project well within the team's technical capabilities.

1.2.2. Operational Feasibility

The system aims to simplify legal research for both professionals and clients. The workflow and operations have been carefully designed for practical use:

- Lawyers can use it to quickly find technical arguments, statutes, and precedents.
- Clients can obtain simplified legal summaries and understand case contexts.
- The AI model and interface are user-friendly and intuitive, requiring minimal training.

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Since the solution automates an existing manual process, the operational feasibility is high and the problem-solving approach is justified.

1.2.3. Economic Feasibility

Economic feasibility considers the balance between cost and benefit.

Estimated Costs:

- **Development Costs:** Hardware, cloud hosting, and development tools.
- **Operational Costs:** Maintenance, server usage, and model retraining.

Estimated Benefits:

- Reduced research time for lawyers.
- Affordable legal guidance for clients.
- Increased accessibility to legal information.

Tangible Benefits: Time efficiency, accuracy, and reduced dependency on manual research.
Intangible Benefits: Improved access to justice, transparency, and enhanced user satisfaction.
Given these factors, the overall economic feasibility of the project is **strongly positive**.

1.2.4. Schedule Feasibility

The proposed timeline ensures the project can be completed within the academic year. A **Gantt chart** (see section 1.7) divides development into manageable phases such as Feasibility Study, Requirement Analysis, System Design, Frontend Development, Backend Development, Database Design, AI model Integration, Testing and Deployment.
Each milestone has defined deliverables, and the workload is distributed evenly among the team. With proper time management, deadlines are realistically achievable.

1.2.5. Specification Feasibility

The requirements of LegisCounsel are clearly defined and achievable:

- AI-based semantic search
- Dual result views (lawyer and client)
- Legal document database and summarization engine
- Role-based access (lawyer/client)

Scope boundaries are well established, with possible extensions (mobile app, more laws) planned for future iterations.

1.2.6. Information Feasibility

All required information such as judgments, statutes, and case laws can be collected from publicly available legal databases and open-access case repositories.
Data will be structured and stored efficiently for vector storage in **VectorSearch**, ensuring quick and meaningful retrieval. Therefore, the information feasibility is confirmed.

1.2.7. Motivational Feasibility

The development team is motivated and skilled, with clear role distribution and commitment to project goals.
Moreover, the project addresses a real-world societal issue with accessibility to justice which provides intrinsic motivation and professional growth opportunities for the team.

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1.2.8. Legal & Ethical Feasibility

LegisCounsel will strictly comply with ethical standards and data privacy laws. It will:

- Use only publicly available legal data.
- Not provide or replace professional legal advice.
- Display disclaimers to ensure users understand the informational purpose.

There are **no anticipated legal infringements or liabilities** arising from the project.

1.3. Project/Product Scope

Scope Definition:

LegisCounsel will serve as a web-based AI-driven platform providing intelligent legal argument search, semantic analysis, and dual-level output for lawyers and clients.

System Boundaries:

- **Inside Scope:** NLP-based search, summarization, user management, and optional modules (chat, scheduling).
- **Outside Scope:** Billing, live audio/video consultation, and legal advice delivery.

Primary Users:

- Lawyers and legal professionals
- Clients/general public/Law Students

The project's scope is designed to fit available resources and time, with scalability for future enhancement.

1.4. Project/Product Costing

1.4.1. Project Cost Estimation By Function Point Analysis

Function Point Analysis (FPA) is a standardized method used to evaluate the total cost and work required to build the LegisCounsel System. It evaluates major functional components including user inputs, outputs, files and external interfaces to assess the system's size and complexity.

1) **Calculate the Unadjusted Function Point (UFP) :**

   The total UFP for LegisCounsel was calculated by multiplying five measurable elements by their complexity weights. LegisCounsel falls in the category of average Complexity.

Table 1.4.1.1- Unadjusted Function Points

Functional Components	Count	Complexity	Weight	Total FP
User Inputs	9	average	4	36
User Outputs	5	average	5	25

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User Inquiries	5	low	3	15
Internal Files	5	average	10	50
External Interfaces	**4**	low	5	20
Total UFP				**146**

- **External Inputs:**
  - Login
  - Registration
  - Updating profile
  - Uploading legal cases (Admin)
  - Saving legal information (Client)
  - Bookmarking legal information (Lawyer)
  - Lawyer/Client searching query
  - Manage Users (Admin)
  - Logout

  Total External Inputs = 9

- **External Outputs:**
  - Technical legal results for lawyers
  - Simplified summaries for clients
  - Displaying legal judgments
  - Case/statute details
  - Activity logs for admin

  Total External Outputs = 5

- **External Inquiries:**
  - Search history view
  - Viewing previous searches
  - Viewing bookmarked info
  - View saved items
  - Viewing user profile

  Total External Inquiries = 5

- **Internal Logical Files:**
  - Users table (Lawyers, Clients, Admin)
  - Search logs
  - Saved/bookmarked arguments
  - System activity logs

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- Role/permission table

Total Internal Logical Files = 5

- **External Interface Files:**
  - External Legal Database (judgements, statutes)
  - External NLP/AI Model files
  - Cloud NLP API outputs
  - Vector Database

Total External Interface Files = 4

2) Value Adjustment Factor:

The 14 General System Characteristics (GSC) define the system's overall complexity. Each characteristic is evaluated on a scale of 0(no influence) to 5(strong influence) according to its impact on the project.

Table 1.4.1.2- Value Adjustment Factor

Sr.No	General System Characteristics	Effect on LegisCounsel	Influence (0-5)
1	Data communications	data communication between web interface, backend servers and AI APIs	3
2	Distributed data processing	Client-Server architecture	3
3	Performance	Fast response for each legal query	4
4	Heavily used configuration	Web-based app, moderate usage	2
5	Transaction Rate	limited concurrent users	2
6	Online Data Entry	Users enter legal queries in plain english	4
7	End-User Efficiency	Simple and intuitive interfaces for clients and lawyers	3

8	Online Update	Remote updation of Database and AI models	3
9	Complex Processing	AI/NLP tasks like semantic search and summarization using large models.	4
10	Reusability	Can be reused	2
11	Installation Ease	Web based and easy to deploy	4
12	Operational Ease	Automated and simple UI	3
13	Multiple Sites	Hosted centrally on a cloud server	1
14	Facilitate Change	Modular design allows adding new features (e.g., chat, mobile app) easily.	2
Total			**40**

Value Adjustment Factor Calculation:

VAF = 0.65 + (0.01*F*i*)
VAF = 0.65 + (0.01*40)
VAF = 0.65 + 0.4
VAF = 1.05

Adjusted Function Points (AFP):

AFP = UFP * VAF
AFP =146 * 1.05
AFP= 153.3 ~ 153 FP

Finally, Total Project Cost and Total Project Effort are calculated given the average productivity parameter for the system.

Let's assume labor rate for 3 students would be 1500/week/member and productivity parameter would be 10 .

The formulae are given as follows:

Cost / FP = labor rate / productivity parameter

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Cost / FP = 1500/10 = Rs. 150

Total Project Cost = FP est. * (cost / FP) = 153 *150 = **Rs. 22,950**

Total Estimated Effort = FP est. / productivity parameter = 153/10 = 15.3 **weeks.**

1.4.2. Project Cost Estimation by using COCOMO'81 (Constructive Cost Model)

Basic COCOMO Estimation

The Basic COCOMO (Constructive Cost Model) was applied to estimate the effort, duration, and staffing required for the development of the LegisCounsel System. The project falls under the **Organic mode** due to its relatively small size and well-understood functional requirements.

Table 1.4.2- Organic Mode Parameters

Project	a	b	c	d
Organic	2.4	1.05	2.5	0.38

Size Estimation
The system size was estimated using Function Point Analysis, resulting in **153 FP**.
Using a standard conversion of **50 LOC per FP**, the total size becomes:

- Total LOC = 153 × 50 = **7,650 LOC**
- KLOC = 7,650 / 1000 = **7.65 KLOC**

Basic COCOMO Effort Estimation
Using Organic Mode parameters (a = 2.4, b = 1.05):

Effort (Person-Months) = 2.4 × (7.65)^1.05
Effort ≈ **20.3 Person-Months**

Development Time Estimation
Using time parameters (c = 2.5, d = 0.38):

Development Time = 2.5 × (20.3)^0.38
Development Time ≈ **7.8 Months**

Step 4: Team Size Estimation

Team Size = Effort / Development Time
= 20.3 / 7.8 ≈ **2.6 ≈ 3 members**

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Since the project team includes three students, the COCOMO results align with the actual team size.

1.4.3. Activity Based Costing

Activity-Based Costing (ABC) is a method used to estimate project costs by identifying all major activities involved in the development process and assigning resources, time, and costs to each. Rather than estimating costs in general, this approach breaks the project into smaller, measurable tasks to determine how much each activity contributes to the total cost.
To calculate the entire cost of LegisCounsel's development, each activity was given a cost based on its duration, required resources and assigned labor rate of Rs. 1500.

ABC = Duration (weeks) * Number of Resources (Members) * Cost Rate per Resource

Table 1.4.3- Activity Based Costing

Activity Id	Activity Name	Duration (weeks)	No. of Members	Rate per week / per member	Total Cost (Rs.)
T1	Project Proposal and Approval	1	3	1500	4500
T2	Feasibility Study	1	2	1500	3000
T3	Requirement Gathering and Analysis	2	3	1500	9000
T4	System Design	1	2	1500	3000
T5	Frontend Development	2	1	1500	3000
T6	Database Design	1	2	1500	3000

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T7	Backend Development	2	2	1500	6000
T8	AI Model Integration	3	3	1500	13,500
T9	Testing and Debugging	2	3	1500	9000
T10	Deployment	1	2	1500	3000
T11	Final Submission and Viva	2	3	1500	9000
Total					**66000**

1.5. Task Dependency Table

Table 1.5- Task Dependency Table

Task Id	Task Name	Dependency on Predecessor
T1	Project Proposal and Approval	---
T2	Feasibility Study	T1
T3	Requirement Gathering and Analysis	T2

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T4	System Design	T3
T5	Frontend Development	T4
T6	Database Design	T4
T7	Backend Development	T4,T6
T8	AI Model Integration	T7
T9	Testing and Debugging	T5,T6,T7,T8
T10	Deployment	T9
T11	Final Submission and Viva	T9,T10

Tasks are scheduled such as after the approval of project proposal, Feasibility study is done followed by Requirement gathering and analysis phase. After that System Design is made. Then the development phase starts. Frontend and database is developed on the basis of system design. Then the backend is developed after completion of system design and database design. AI models are trained on the basis of system design and requirement analysis. After these development phases, System Components are integrated followed by testing and debugging. Finally the system is deployed and the final report is generated.

1.6. CPM - Critical Path Method

The **Critical Path Method (CPM)** is used to identify the sequence of dependent activities that directly affect the total duration of the LegisCounsel project. It helps the team determine which tasks must be completed on time to avoid project delays and ensures effective time management.

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Table 1.6.1- Task Dependency Table

Task Id	Task Name	Predecessor	Duration(weeks)
T1	Project Proposal and Approval	---	1
T2	Feasibility Study	T1	1
T3	Requirement Gathering and Analysis	T2	2
T4	System Design	T3	1
T5	Frontend Development	T4	2
T6	Database Design	T4	1
T7	Backend Development	T4,T6	2
T8	AI Model Integration	T7	3
T9	Testing and Debugging	T5,T6,T7,T8	2
T10	Deployment	T9	1
T11	Final Submission and Viva	T9,T10	2

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CPM Network Diagram for LegisCounsel System is constructed on the basis of activities and their dependencies.

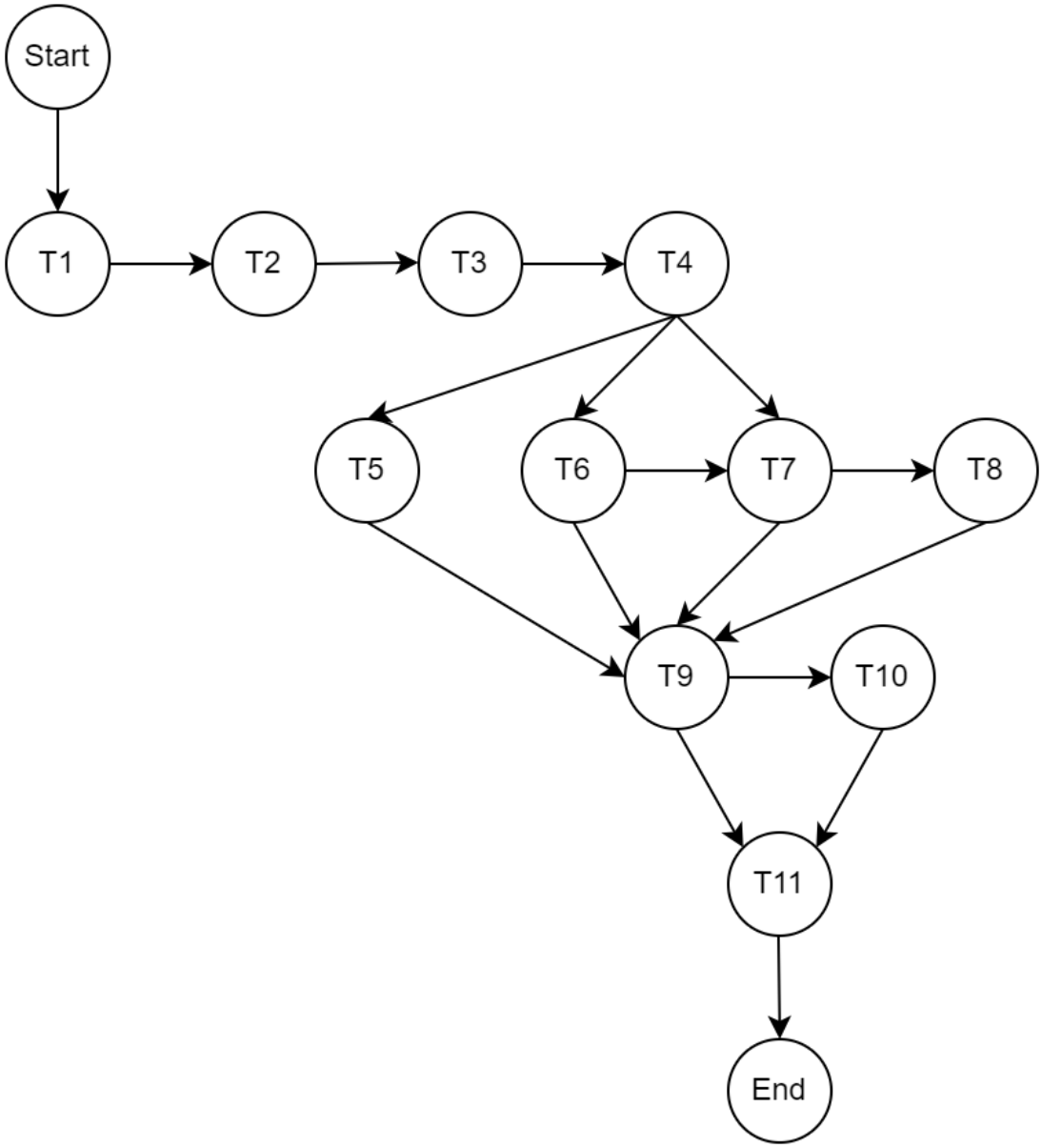


Figure 1.6.1- Network Diagram

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The CPM calculation table shows each activity's duration, earliest and latest start / finish times and slack to identify which tasks lie on the critical path.

Table 1.6.2- CPM Chart

Activity	Activity Name	Duration (Weeks)	ES	EF	LS	LF	TS
T1	Project Proposal and Approval	1	0	1	0	1	0
T2	Feasibility Study	1	1	2	1	2	0
T3	Requirement Gathering and Analysis	2	2	4	2	4	0
T4	System Design	1	4	5	4	5	0
T5	Frontend Development	2	5	7	9	11	4
T6	Database Design	1	5	6	5	6	0
T7	Backend Development	2	6	8	6	8	0
T8	AI Model Integration	3	8	11	8	11	0
T9	Testing and Debugging	2	11	13	11	13	0
T10	Deployment	1	13	14	13	14	0

T11	Final Submission and Viva	2	14	16	14	16	0

Following these formulas, we calculate:

$$\text{Total Slack (TS)} = LS - ES = LF - EF$$

Critical Path:

T1 → T2 → T3 → T4 → T6 → T7 → T8 → T9 → T10 → T11

There is only one non-critical path that is **T5(Frontend Development)** with slack= 4. That means if T5 delays by 4 weeks, it would not affect the overall project.

Sum of their duration is:

1+1+2+1+1+2+3+2+1+2= 16 weeks

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Update CPM diagram:

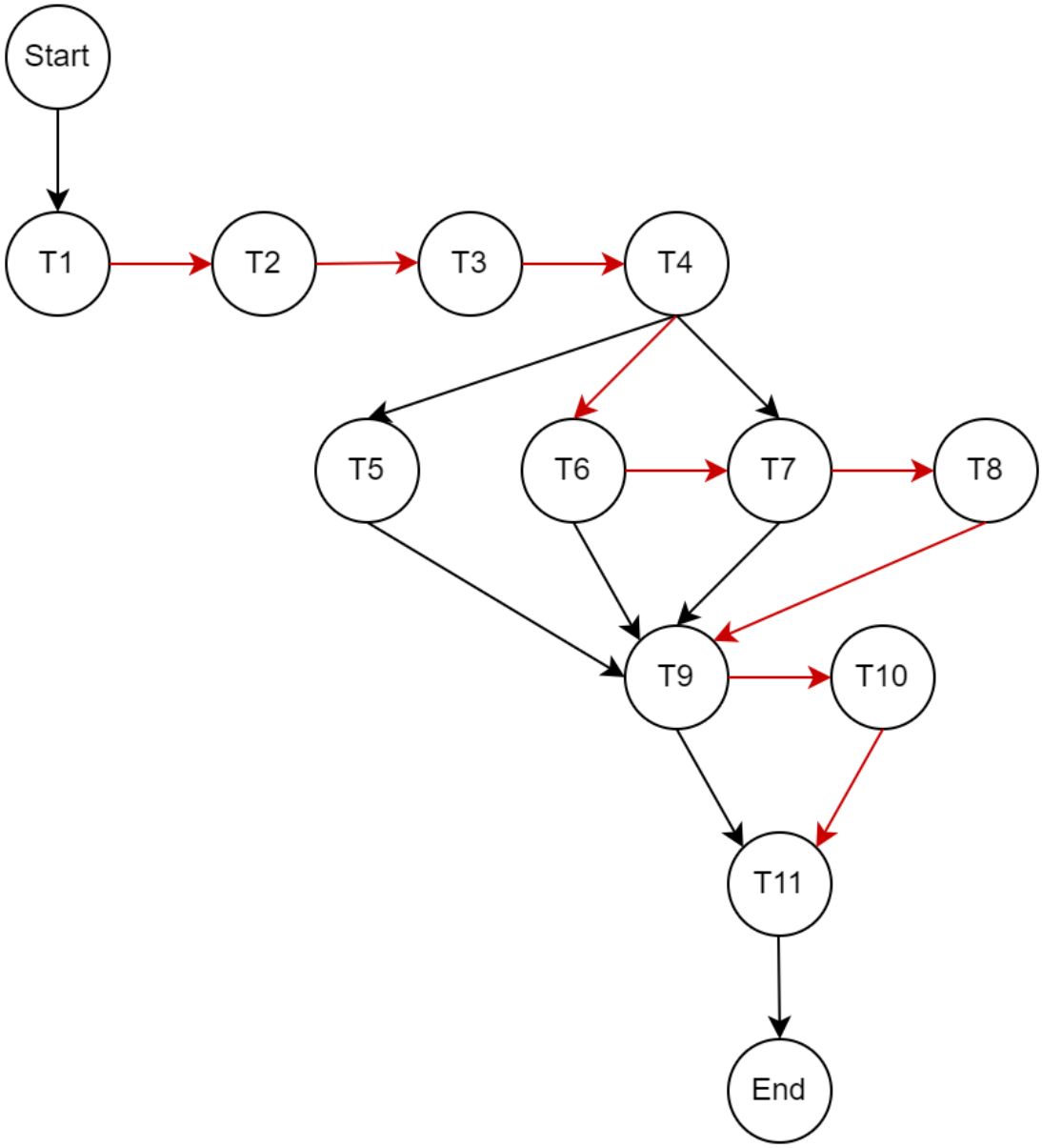


Figure 1.6.2- Updated Network Diagram

Critical Path is **T1 → T2 → T3 → T4 → T6 → T7 → T8 → T9 → T10 → T11**

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1.7. Gantt chart

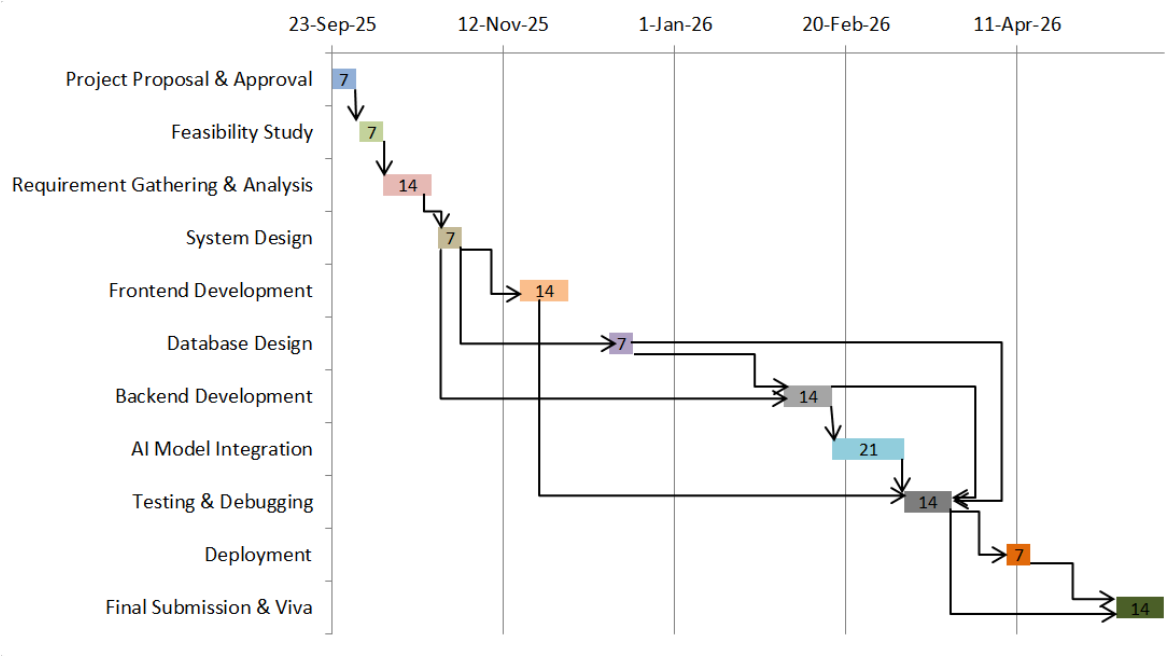


Figure 1.7- Gantt Chart

1.8. Introduction to Team member and their skill set

Table 1.8- Team Member & their Skill Set

Member ID	Member Name	Skill Set
M1	Ramiz Ilyas	Frontend developer, UI/UX Design, AI Models Integration, Documentation
M2	Nayab Gul	Designing UML Diagram, Database Management, Documentation, AI Models Integration
M3	Reham Batool	Software Design, Database Management, Documentation, Backend development

1.9. Task and Member Assignment Table

Table 1.9.1- Task and Member Assignment

Task Id	Task Name	Dependency	Duration (weeks)	Assigned Members
T1	Project Proposal and Approval	---	1	M1,M2,M3
T2	Feasibility Study	T1	1	M3
T3	Requirement Gathering and Analysis	T2	2	M1,M2
T4	System Design	T3	1	M2,M3
T5	Frontend Development	T4	2	M1
T6	Database Design	T4	1	M2,M3
T7	Backend Development	T4,T6	2	M1,M3
T8	AI Model Integration	T7	3	M1,M2
T9	Testing and Debugging	T8	2	M2,M3

T10	Deployment	T9	1	M1,M3
T11	Final Submission and Viva	T9,T10	2	M1,M2,M3

The table above outlines all major project activities along with their durations, dependencies, and assigned members.

Table 1.9.2- Assigned Roles

Task Id	Member Id	Assigned Roles
T1	M1,M2,M3	Prepare Proposal Document and Defence
T2	M3	Conduct Feasibility Study
T3	M1,M2	Gather and analyze requirements
T4	M2,M3	Design System Architecture and workflow diagrams
T5	M1	Develop frontend with React.js
T6	M2,M3	Design Database and setup connections
T7	M1,M3	Develop backend APIs and System Logic
T8	M1,M2	Integrate and train AI Model
T9	M2,M3	Perform unit and integration testing

T10	M1,M3	Deploy system
T11	M1,M2,M3	Prepare final report and viva

This table highlights the specific roles and responsibilities of each member for every project task, ensuring clear division of work, accountability, and smooth coordination throughout all development phases.

Table 1.9.3- Team Members and their Work Duration

Member ID	**Assigned Tasks**	**Total Weeks**	**Total Days**
M1	T1, T3, T5, T7, T8, T10, T11	13	91
M2	T1,T3, T4, T6, T8, T9, T11	12	84
M3	T1, T2, T4,T6, T7, T9, T10,T11	12	84

This table shows each member's assigned tasks and total working duration.

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Activity Bar Chart

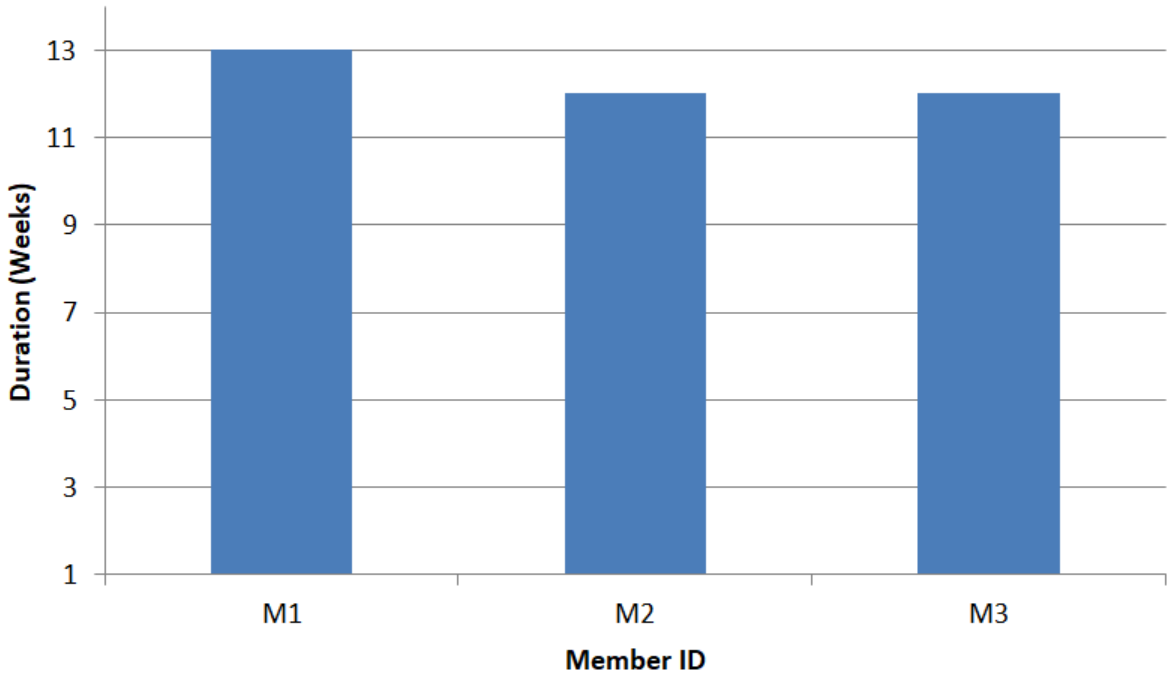


Figure 1.9- Activity Bar Chart

1.10. Tools and Technology with reasoning

Table 1.10- Tools and Technologies

Tool / Technology	**Reason for Selection**
React.js	Responsive, scalable frontend framework.
Node.js + Express.js	For robust and efficient backend APIs.
MongoDB	For user data and legal documents.
VectorSearch	For vector-based semantic search and retrieval.

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AI/NLP Models (Ollama, GPT)	To understand natural language queries and retrieve semantic results.
Summarization Models:	To convert complex judgments into client-friendly explanations.
Git/GitHub	Version control and team collaboration.
AWS/GCP	Cloud hosting with GPU support for AI inference.
GoogleDocs	Collaborative Documentation and Version Control
VS Code	Coding environment
Testing tools(Postman, Selenium, PyTest)	Testing tools for verifying AI model and backend functions, Frontend(UI) testing

1.11. Vision Document

Vision Statement:

To revolutionize legal research and access to justice by developing an AI-driven platform that delivers intelligent, context-aware, and simplified legal information for lawyers and the general public.

Key Features:

- Semantic search for legal arguments and judgments
- Summarization engine for client-friendly outputs
- Dual interface for lawyers and clients
- Optional modules: communication, appointment scheduling, and case tracking

Stakeholders:

- End Users (Lawyers, Clients, Students)
- Supervisor and Development Team

Constraints:

- Limited legal dataset availability in Pakistan
- Resource and time constraints for AI model training

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1.12. Risk List

The following risks are associated with the LegisCounsel System.

- There might not be enough open-source or structured legal data for training the AI model.
- The NLP model might not fully understand complex legal language or may return irrelevant results.
- Model training, system integration, and testing could take longer than expected due to technical complexity.
- AI model training may require high-end systems or GPU support, which might not always be available.
- Legal documents may contain sensitive information that must be handled carefully to avoid privacy issues.
- Difficulty in connecting AI components with backend APIs or frontend interfaces could delay the project.
- Since the system is web-based, poor internet connectivity or cloud service interruptions could affect performance.
- Lawyers or clients may find the AI-generated summaries inaccurate or hard to trust initially.
  Unauthorized access to stored legal data or user information could lead to misuse if security is not properly implemented.

1.13. Product Features/ Product Decomposition

Core Functional Modules:

1. **User Management Module** – registration, login, role assignment.
2. **Argument Searching Module** – core NLP-based semantic search.
3. **Summarization Module** – converts complex judgments into simple summaries.
4. **Knowledge Base Module** – stores case laws, statutes, and legal data.
5. **Result Presentation Module** – dual view for lawyers and clients.
6. **Chatbot Interface** – optional Q&A system for natural language queries.

Optional Modules:

- Lawyer–Client Communication
- Appointment & Scheduling
- Case Management and Notifications
- Admin Panel
- Future Mobile Application

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Chapter 2: Software Requirement Specification (For Object Oriented Approach)

2.1 Introduction:

Requirements engineering provides the mechanism for understanding user needs, analyzing them, assessing feasibility, and specifying the solution clearly. For LegisCounsel, this involves capturing requirements for an AI-powered legal research system that serves both lawyers and clients. The task involves capturing, structuring, and accurately representing user requirements so that they can be correctly embodied in a high-quality system.

- Requirements elicitation
- Requirements analysis and negotiation
- Requirements specification
- System modeling
- Requirements validation
- Requirements management

Here, we focus on requirements specification, which leads to the following steps:

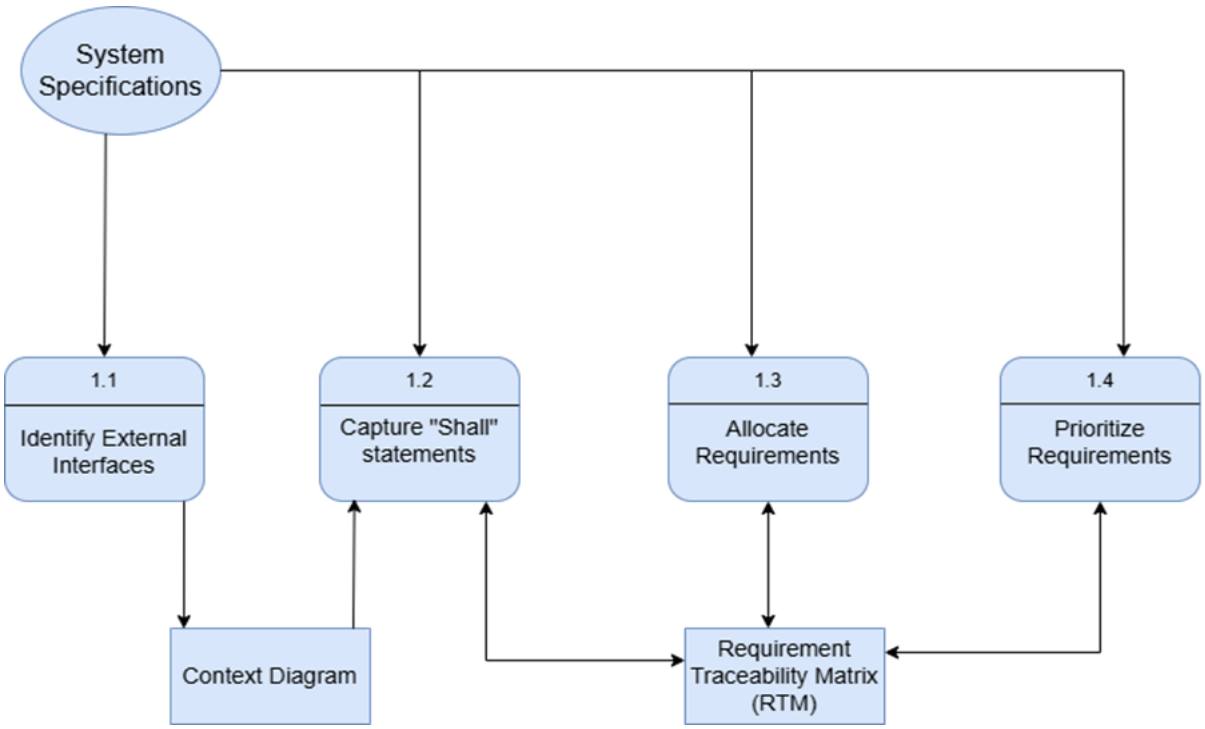


Figure 2.1 - System Specifications

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2.1.1 System Specifications

Introduction

LegisCounsel is an AI-powered intelligent legal argument searching system designed to revolutionize legal research in Pakistan. The system uses Natural Language Processing (NLP) and Semantic Search technologies to understand user queries and return the most relevant legal arguments, precedents, and judgments. Unlike traditional keyword-based search systems, LegisCounsel interprets the contextual meaning of legal queries and provides dual-view results - technical details for lawyers and simplified summaries for clients.

The platform addresses critical challenges in legal research:

- Time-consuming manual search through legal documents
- Complexity of legal language for non-lawyers
- Limited accessibility to legal precedents for general public
- High costs associated with legal research tools

Existing System

Currently, legal professionals in Pakistan rely on manual search methods through physical law journals, reporters, and basic digital databases with limited keyword search functionality. These traditional systems lack semantic understanding of legal context and cannot interpret the deep meaning behind complex legal queries. Existing platforms are either too expensive for individual practitioners or provide inadequate coverage of Pakistani case law and statutes.

Furthermore, they offer no client-friendly simplified summaries, making legal research inaccessible to the general public who cannot comprehend technical legal terminology. The absence of dual-view presentation forces lawyers to manually translate complex judgments for their clients, increasing time consumption and reducing efficiency in legal consultations.

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Organizational Chart

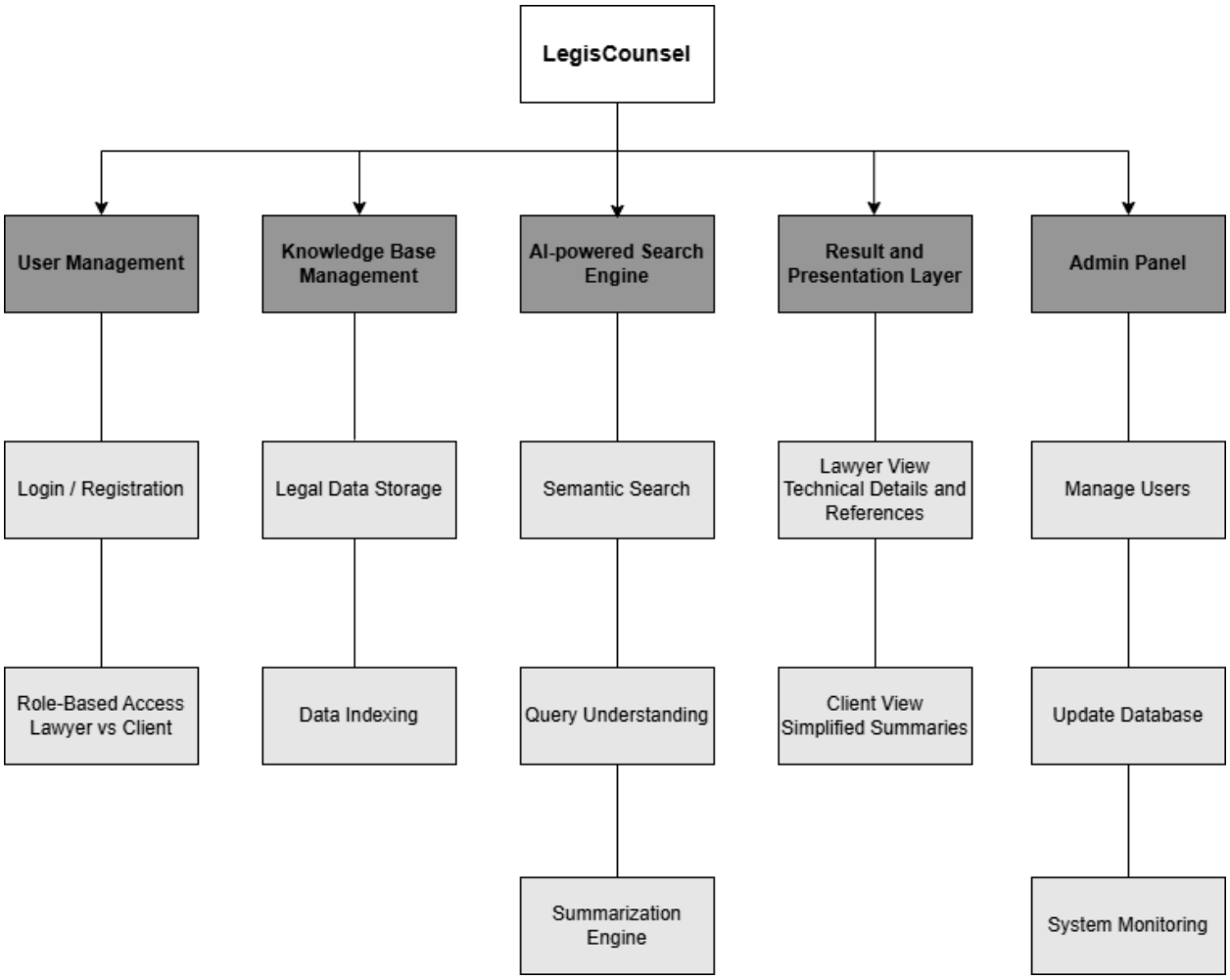


Figure 2.1.1 - Organizational Chart

Scope of the System

The scope of LegisCounsel is limited to providing an AI-powered intelligent legal argument searching system specifically for Criminal Law in Pakistan. The system focuses on simplifying the legal research process by allowing lawyers and clients to input natural language queries and receive relevant legal arguments, precedents, and judgments with semantic understanding.

The system's main functions include natural language processing, semantic search, dual-view result presentation, legal data summarization, and intelligent ranking of search results. It is primarily intended for criminal law professionals and the general public seeking criminal case understanding who need efficient and accurate legal research tools.

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Only the following domain areas are included in this project:

- AI-powered semantic search for criminal law arguments
- Natural language query processing in English
- Dual-view result presentation (Technical view for lawyers & Simplified view for clients)
- User authentication and role-based access control
- Currently limited to Criminal Law domain only

Out of scope are:

- Billing and payment systems
- Live video/audio consultations
- Case management and litigation support
- Court proceedings management systems
- Civil, Corporate, Family, and other legal domains

Summary of Requirements: (Initial Requirements)

The LegisCounsel system aims to simplify legal research using AI-driven automation. It enables lawyers and clients to input natural language queries and receive relevant legal arguments, precedents, and judgments with semantic understanding. The system ensures quick search results, dual-view presentation, and legal data summarization, supported by a secure and user-friendly interface.

Functional Requirements:

- User registration, login, and profile management for lawyers, clients, and administrators
- AI-based natural language processing and semantic search
- Legal argument searching and precedent retrieval
- Dual-view result presentation (technical for lawyers, simplified for clients)
- Legal data summarization and analysis
- Admin panel for user management and system monitoring

Non-Functional Requirements:

- User-friendly and responsive interface
- Fast performance and reliable uptime (5-10 seconds response time)

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- Secure authentication and data handling
- Scalable system architecture for future expansion

2.1.2. Identifying External Entities

For the LegisCounsel system, external entities are identified through the requirements analysis process. The identification involves two main phases to ensure all relevant entities are captured.

a. Over Specify Entities from Abstract:

Based on the LegisCounsel project requirements, the following entities have been identified:

- **Lawyers** - Legal professionals using the system for research
- **Clients** - General public seeking legal understanding
- **Admin** - System administrator managing content and users
- **Legal Database** - External source of laws and statutes
- **AI/NLP Model** - External NLP and machine learning model
- **Cloud Services** - Hosting and deployment platforms

b. Perform Refinement:

After comprehensive analysis, the following refined entities are most relevant to the LegisCounsel business logic:

- **Lawyers** (Primary users for technical legal research)
- **Clients** (Primary users for simplified legal summaries)
- **Admin** (System management and content curation)
- **AI/NLP Model** (Core processing for semantic search)
- **Legal Database** (External knowledge base source)

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2.1.3. Context Level Data Flow Diagram:

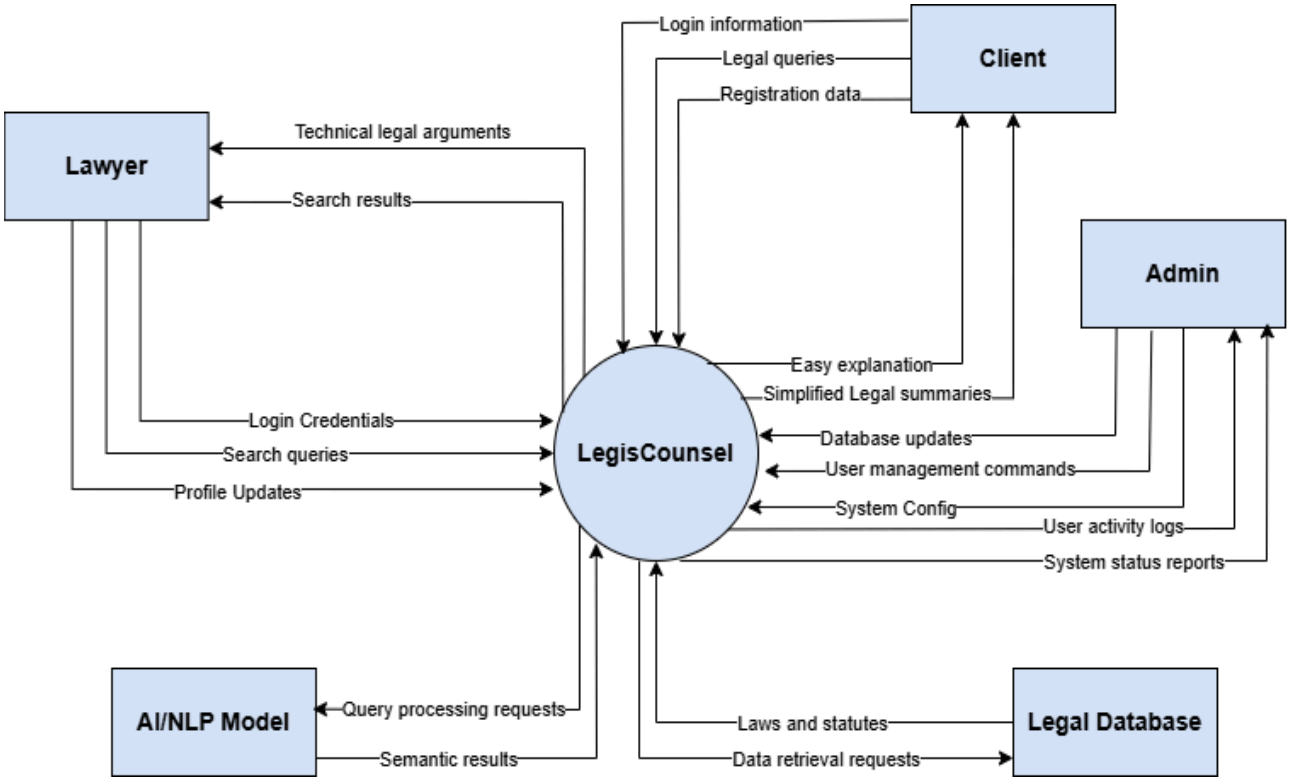


Figure 2.1.3 - Context Level Data Flow

2.1.4. Capture “shall” Statements:

Table 2.1.4 - Capture “shall” Statements

Para #	Initial Requirements
1	**Lawyer Requirements**
1.0	Lawyer “shall” register to the system by providing email and personal details.
1.1	Lawyer “shall” login to the system using valid credentials.
1.2	Lawyer “shall” update profile information.
1.3	Lawyer “shall” search for legal arguments using natural language queries.

1.4	Lawyer “shall” view detailed technical legal references and laws.
1.5	Lawyer “shall” bookmark important legal arguments for future reference.
1.6	Lawyer “shall” view search history of legal arguments.
1.7	Lawyer “shall” search by specific legal sections and acts (e.g., PPC Section 302, Contract Act).
1.8	Lawyer “shall” logout securely from the system.
2	**Client Requirements**
2.0	Client “shall” register to the system by providing email and personal details.
2.1	Client “shall” login to the system using valid credentials.
2.2	Client “shall” update profile information.
2.3	Client “shall” search for legal information using simple language queries.
2.4	Client “shall” view simplified summaries of legal concepts.
2.5	Client “shall” save helpful legal information for personal reference.
2.6	Client “shall” view previously searched legal topics.
2.7	Client “shall” logout securely from the system.
3	**Admin Requirements**
3.0	Admin “shall” login to the system using valid credentials.
3.1	Admin “shall” manage all user accounts (Lawyer, Client, Admin).
3.2	Admin “shall” update legal database with new laws and statutes.
3.3	Admin “shall” monitor system performance and user activities.
3.4	Admin “shall” logout securely from the system.
4	**System Requirements**

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4.0	System “shall” authenticate users during registration and login.
4.1	System “shall” validate all input data for correctness and format.
4.2	System “shall” process natural language queries using AI/NLP models.
4.3	System “shall” perform semantic search on legal database.
4.4	System “shall” display technical legal details to lawyers.
4.5	System “shall” display simplified summaries to clients.
4.6	System “shall” summarize complex legal judgments automatically.
4.7	System “shall” maintain user data and search history.
4.8	System “shall” provide role-based access control.
4.9	System “shall” log all user activities for audit purposes.

2.1.5. Allocate Requirements:

Table 2.1.5 - Allocate Requirements

Para #	Initial Requirements	Use Case Name
1.0	Lawyer “shall” register to the system by providing email and personal details.	UC-Register
1.1	Lawyer “shall” login to the system using valid credentials.	UC-Login
1.2	Lawyer “shall” update profile information.	UC-UpdateProfile
1.3	Lawyer “shall” search for legal arguments using natural language queries.	UC-SearchArguments
1.4	Lawyer “shall” view detailed technical legal references and laws.	UC-ViewLegalReferences

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1.5	Lawyer “shall” bookmark important legal arguments for future reference.	UC-BookmarkArguments
1.6	Lawyer “shall” view search history of legal arguments.	UC-ViewSearchHistory
1.7	Lawyer “shall” search by specific legal sections and acts (e.g., PPC Section 302, Contract Act).	UC-SearchBySections
1.8	Lawyer “shall” logout securely from the system.	UC-Logout
2.0	Client “shall” register to the system by providing email and personal details.	UC-Register
2.1	Client “shall” login to the system using valid credentials.	UC-Login
2.2	Client “shall” update profile information.	UC-UpdateProfile
2.3	Client “shall” search for legal information using simple language queries.	UC-SearchLegalInfo
2.4	Client “shall” view simplified summaries of legal concepts.	UC-ViewSimplifiedSummaries
2.5	Client “shall” save helpful legal information for personal reference.	UC-SaveLegalInfo
2.6	Client “shall” view previously searched legal topics.	UC-ViewPreviousSearches
2.7	Client “shall” logout securely from the system.	UC-Logout
3.0	Admin “shall” login to the system using valid credentials.	UC-Login
3.1	Admin “shall” manage all user accounts (Lawyer, Client, Admin).	UC-ManageUsers

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3.2	Admin “shall” update legal database with new laws and statutes.	UC-UpdateLaws
3.3	Admin “shall” monitor system performance and user activities.	UC-MonitorSystem
3.4	Admin “shall” logout securely from the system.	UC-Logout
4.0	System “shall” authenticate users during registration and login.	UC-AuthenticateUser
4.1	System “shall” validate all input data for correctness and format.	UC-ValidateInput
4.2	System “shall” process natural language queries using AI/NLP models.	UC-ProcessQueries
4.3	System “shall” perform semantic search on legal database.	UC-SemanticSearch
4.4	System “shall” display technical legal details to lawyers.	UC-DisplayTechnicalView
4.5	System “shall” display simplified summaries to clients.	UC-DisplaySimplifiedView
4.6	System “shall” summarize complex legal judgments automatically.	UC-SummarizeJudgments
4.7	System “shall” maintain user data and search history.	UC-ManageData
4.8	System “shall” provide role-based access control.	UC-ManageAccess
4.9	System “shall” log all user activities for audit purposes.	UC-LogActivities

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2.1.6. Prioritize Requirements:

Table 2.1.6 - Prioritize Requirements

Para #	Rank	Initial Requirements	Use Case ID	Use Case Name
1.3	Highest	Lawyer “shall” search for legal arguments using natural language queries.	UC-1	UC-SearchArguments
2.3	Highest	Client “shall” search for legal information using simple language queries.	UC-2	UC-SearchLegalInfo
1.0	Highest	Lawyer “shall” register to the system.	UC-3	UC-Register
2.0	Highest	Client “shall” register to the system.	UC-3	UC-Register
1.1	Highest	Lawyer “shall” login to the system.	UC-4	UC-Login
2.1	Highest	Client “shall” login to the system.	UC-4	UC-Login
3.0	Highest	Admin “shall” login to the system.	UC-4	UC-Login
4.0	Highest	System “shall” authenticate users.	UC-5	UC-AuthenticateUser
4.2	Highest	System “shall” process natural language queries using AI/NLP models.	UC-6	UC-ProcessQueries

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4.3	Highest	System “shall” perform semantic search on legal database.	UC-7	UC-SemanticSearch
1.4	Highest	Lawyer “shall” view detailed technical legal references and laws.	UC-8	UC-ViewLegalReferences
2.4	Highest	Client “shall” view simplified summaries of legal concepts.	UC-9	UC-ViewSimplifiedSummaries
4.4	Highest	System “shall” display technical legal details to lawyers.	UC-10	UC-DisplayTechnicalView
4.5	Highest	System “shall” display simplified summaries to clients.	UC-11	UC-DisplaySimplifiedView
1.7	Highest	Lawyer “shall” search by specific legal sections and acts.	UC-12	UC-SearchBySections
1.5	Medium	Lawyer “shall” bookmark important legal arguments.	UC-13	UC-BookmarkArguments
2.5	Medium	Client “shall” save helpful legal information.	UC-14	UC-SaveLegalInfo
1.6	Medium	Lawyer “shall” view search history of legal arguments.	UC-15	UC-ViewSearchHistory
2.6	Medium	Client “shall” view previously searched legal topics.	UC-16	UC-ViewPreviousSearches

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4.6	Medium	System “shall” summarize complex legal judgments automatically.	UC-17	UC-SummarizeJudgments
1.2	Lowest	Lawyer “shall” update profile information.	UC-18	UC-UpdateProfile
2.2	Lowest	Client “shall” update profile information.	UC-18	UC-UpdateProfile
1.8	Lowest	Lawyer “shall” logout securely from the system.	UC-19	UC-Logout
2.7	Lowest	Client “shall” logout securely from the system.	UC-19	UC-Logout
3.4	Lowest	Admin “shall” logout securely from the system.	UC-19	UC-Logout
3.1	Lowest	Admin “shall” manage all user accounts.	UC-20	UC-ManageUsers
3.2	Lowest	Admin “shall” update legal database with new laws and statutes.	UC-21	UC-UpdateLaws
3.3	Lowest	Admin “shall” monitor system performance and user activities.	UC-22	UC-MonitorSystem
4.1	Lowest	System “shall” validate all input data for correctness and format.	UC-23	UC-ValidateInput

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4.7	Lowest	System “shall” maintain user data and search history.	UC-24	UC-ManageData
4.8	Lowest	System “shall” provide role-based access control.	UC-25	UC-ManageAccess
4.9	Lowest	System “shall” log all user activities for audit purposes.	UC-26	UC-LogActivities

2.1.7. Requirements Trace-ability Matrix:

Table 2.1.7 - Requirements Trace-ability Matrix

Sr #	Para #	System Specification Text	Build	Use Case Name	Category
1	1.0	Lawyer “shall” register to the system by providing email and personal details.	B1.0	UC-Register	Account Management
2	1.1	Lawyer “shall” login to the system using valid credentials.	B1.0	UC-Login	Account Management
3	1.2	Lawyer “shall” update profile information.	B1.2	UC-UpdateProfile	Account Management
4	1.3	Lawyer “shall” search for legal arguments using natural language queries.	B1.1	UC-SearchArguments	Legal Search
5	1.4	Lawyer “shall” view detailed technical legal references and laws.	B1.1	UC-ViewLegalReferences	Legal Search

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6	1.5	Lawyer “shall” bookmark important legal arguments for future reference.	B1.2	UC-BookmarkArguments	User Features
7	1.6	Lawyer “shall” view search history of legal arguments.	B1.2	UC-ViewSearchHistory	User Features
8	1.7	Lawyer “shall” search by specific legal sections and acts.	B1.1	UC-SearchBySections	Legal Search
9	1.8	Lawyer “shall” logout securely from the system.	B1.0	UC-Logout	Account Management
10	2.0	Client “shall” register to the system by providing email and personal details.	B1.0	UC-Register	Account Management
11	2.1	Client “shall” login to the system using valid credentials.	B1.0	UC-Login	Account Management
12	2.2	Client “shall” update profile information.	B1.2	UC-UpdateProfile	Account Management
13	2.3	Client “shall” search for legal information using simple language queries.	B1.1	UC-SearchLegalInfo	Legal Search
14	2.4	Client “shall” view simplified summaries of legal concepts.	B1.1	UC-ViewSimplifiedSummaries	Legal Search
15	2.5	Client “shall” save helpful legal information for personal reference.	B1.2	UC-SaveLegalInfo	User Features

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16	2.6	Client “shall” view previously searched legal topics.	B1.2	UC-ViewPreviousSearches	User Features
17	2.7	Client “shall” logout securely from the system.	B1.0	UC-Logout	Account Management
18	3.0	Admin “shall” login to the system using valid credentials.	B1.0	UC-Login	Account Management
19	3.1	Admin “shall” manage all user accounts (Lawyer, Client, Admin).	B1.3	UC-ManageUsers	Admin Functions
20	3.2	Admin “shall” update legal database with new laws and statutes.	B1.3	UC-UpdateLaws	Admin Functions
21	3.3	Admin “shall” monitor system performance and user activities.	B1.3	UC-MonitorSystem	Admin Functions
22	3.4	Admin “shall” logout securely from the system.	B1.0	UC-Logout	Account Management
23	4.0	System “shall” authenticate users during registration and login.	B1.0	UC-AuthenticateUser	System Security
24	4.1	System “shall” validate all input data for correctness and format.	B1.1	UC-ValidateInput	System Validation
25	4.2	System “shall” process natural language queries using AI/NLP models.	B1.1	UC-ProcessQueries	AI Processing

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26	4.3	System “shall” perform semantic search on legal database.	B1.1	UC-SemanticSearch	AI Processing
27	4.4	System “shall” display technical legal details to lawyers.	B1.1	UC-DisplayTechnicalView	Result Presentation
28	4.5	System “shall” display simplified summaries to clients.	B1.1	UC-DisplaySimplifiedView	Result Presentation
29	4.6	System “shall” summarize complex legal judgments automatically.	B1.1	UC-SummarizeJudgments	AI Processing
30	4.7	System “shall” maintain user data and search history.	B1.2	UC-ManageData	Data Management
31	4.8	System “shall” provide role-based access control.	B1.0	UC-ManageAccess	System Security
32	4.9	System “shall” log all user activities for audit purposes.	B1.3	UC-LogActivities	System Monitoring

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2.1.8. High Level Use Case Diagram:

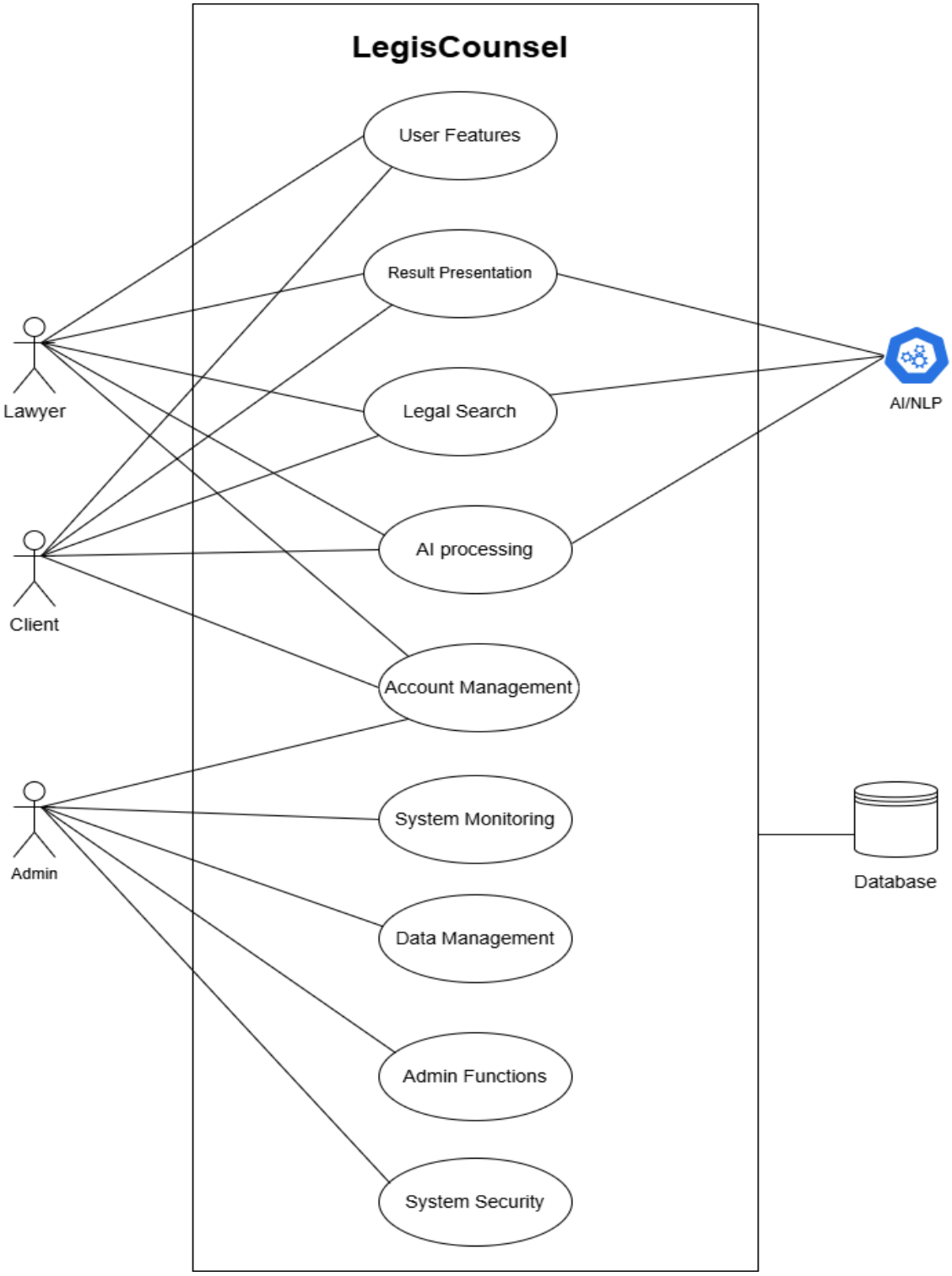


Figure 2.1.8 - High Level Use Case Diagram

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2.1.9. Analysis Level Use Case Diagram:

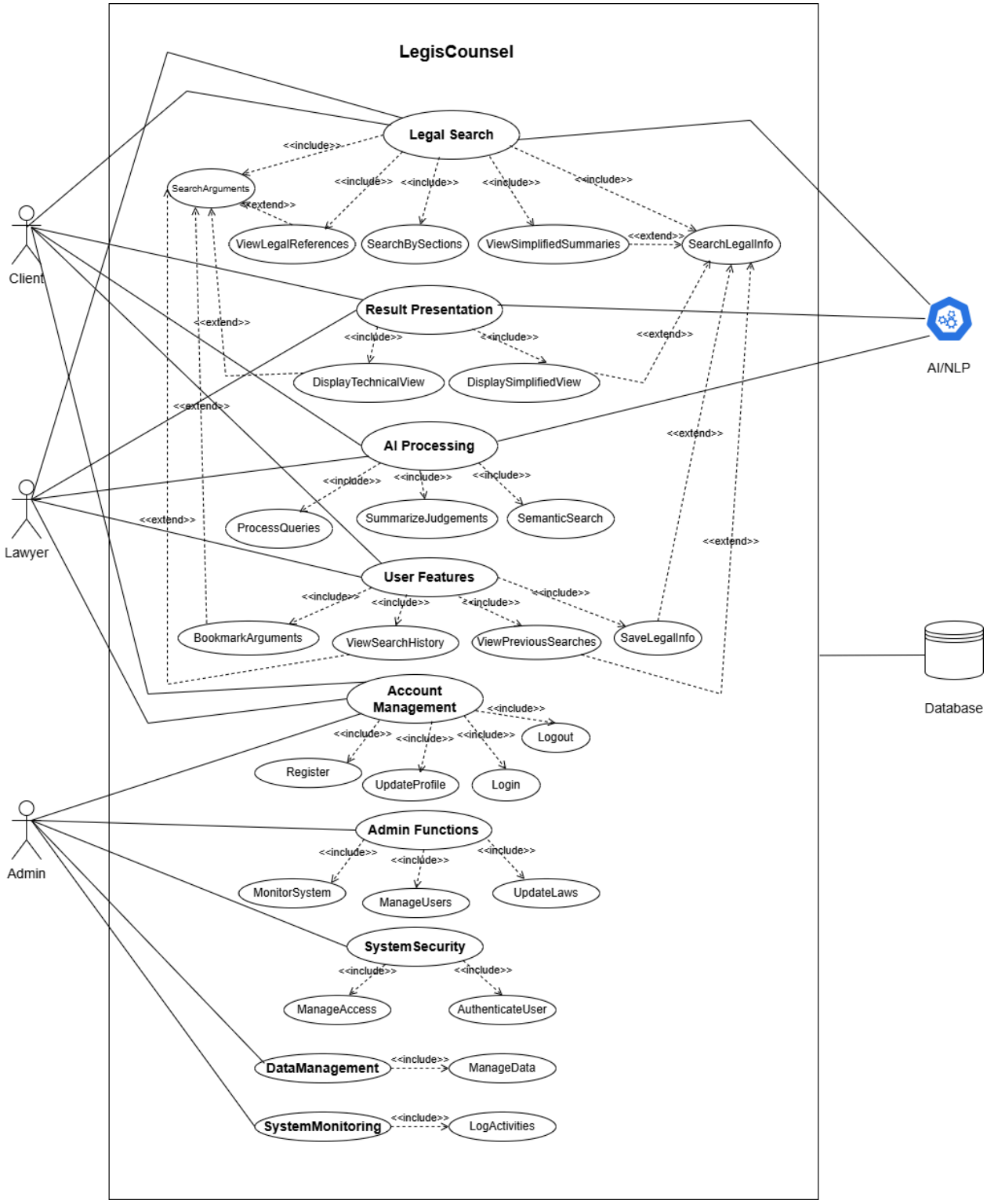


Figure 2.1.9 - Analysis Use Case Diagram

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2.1.10. Use Case Description

Search Arguments :

Table 2.1.10.1 - Search Arguments

Use case Name: UC-SearchArguments **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-1 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Lawyer must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer enters query.	2. System processes query using AI/NLP.
3. Lawyer clicks search button.	4. System performs semantic search on legal database.
5. Lawyer views search results.	6. System displays technical legal arguments.
Success Scenario: Relevant legal arguments displayed successfully.	
Alternate Scenario: ● If no results found, system shows "No matching arguments found".	
Post condition: Search results available for viewing and bookmarking.	

Search Legal Information :

Table 2.1.10.2 - Search Legal Information

Use case Name: UC-SearchLegalInfo **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-2 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Client must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client enters query.	2. System processes query using AI/NLP.
3. Client clicks search button.	4. System performs semantic search on legal database.
5. Client views search results.	6. System displays simplified summaries.

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Success Scenario: Simplified legal information displayed successfully.
Alternate Scenario: ● If no results found, system shows "No relevant information found".
Post condition: Client can save helpful information for future reference.

Register :

Table 2.1.10.3 - Register

Use case Name: UC-Register **Actors:** Lawyer, Client **Date:** 05/11/2025	**Use case ID:** UC-3 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User must have a valid email address and internet connection.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User selects registration option.	2. System displays registration form.
3. User enters email and personal details.	4. System validates input data.
5. User submits registration form.	6. System creates account and sends confirmation.
Success Scenario: User's account created successfully.	
Alternate Scenario: ● If email already exists, system displays "Email already registered" error.	
Post condition: User can login to the system.	

Login :

Table 2.1.10.4 - Login

Use case Name: UC-Login **Actors:** Lawyer, Client, Admin **Date:** 05/11/2025	**Use case ID:** UC-4 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User must have registered account.	
Main Success Scenario	
Actor's Action	**System's Response**

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1. User enters email and password.	2. System validates credentials.
3. User clicks login button.	4. System authenticates and grants access.
5. User accesses dashboard.	6. System redirects to appropriate interface.
Success Scenario: User logged in successfully.	
Alternate Scenario: ● If invalid credentials, system shows "Invalid email/password".	
Post condition: User gains role-based access to system.	

Authenticate User :

Table 2.1.10.5 - Authenticate User

Use case Name: UC-AuthenticateUser **Actors:** Lawyer, Client, Admin **Date:** 05/11/2025	**Use case ID:** UC-5 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User attempts to login or access secure features.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User enters email and password.	2. System validates credentials.
3. User requests access to feature.	4. System checks user role and permissions.
5. User accesses authorized features.	6. System grants appropriate access.
Success Scenario: User authenticated successfully.	
Alternate Scenario: ● If invalid credentials, system denies access.	
Post condition: User granted secure access to system.	

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Process Queries :

Table 2.1.10.6 - Process Queries

Use case Name: UC-ProcessQueries **Actors:** Lawyer, Client **Date:** 05/11/2025	**Use case ID:** UC-6 **Created by:** Nayab, Reham, Ramiz
Pre-condition: System receives user search query.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User enters search query.	2. System sends query to AI/NLP Model.
3. User waits for results.	4. System extracts key legal concepts.
5. User receives structured results.	6. System prepares structured search parameters.
Success Scenario: Query processed successfully with accurate intent understanding.	
Alternate Scenario: ● If query unclear, system shows Please clarify your query".	
Post condition: Structured search parameters ready for semantic search.	

Semantic Search :

Table 2.1.10.7 - Semantic Search

Use case Name: UC-SemanticSearch **Actors:** Lawyer, Client **Date:** 05/11/2025	**Use case ID:** UC-7 **Created by:** Nayab, Reham, Ramiz
Pre-condition: System has processed user query.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User waits for search results	2. System searches legal database using semantic matching.
3. User views search progress.	4. System retrieves relevant legal data.
5. User receives search results.	6. System ranks results by relevance.
Success Scenario: Semantic search completed successfully.	
Alternate Scenario: ● If no matches, system shows "No relevant results found".	

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Post condition: Relevant legal arguments retrieved and ranked.

View Legal References :

Table 2.1.10.8 - View Legal References

Use case Name: UC-ViewLegalReferences **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-8 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Lawyer must have search results.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer selects legal argument from results.	2. System displays detailed technical references.
3. Lawyer views laws and statutes.	4. System shows citation details.
5. Lawyer analyzes legal precedents.	6. System maintains access log.
Success Scenario: Technical legal references displayed successfully.	
Alternate Scenario: ● If reference unavailable, system shows "Reference not found".	
Post condition: Lawyer can use references for legal research.	

View Simplified Summaries :

Table 2.1.10.9 - View Simplified Summaries

Use case Name: UC-ViewSimplifiedSummaries **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-9 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Client must have search results.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client selects legal topic from results.	2. System displays simplified summary.

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3. Client reads easy-to-understand explanation.	4. System shows key points in simple language.
5. Client comprehends legal concept.	6. System tracks client understanding.
Success Scenario: Simplified legal summary displayed successfully.	
Alternate Scenario: ● If summary generation fails, system shows "Summary unavailable".	
Post condition: Client understands legal concept easily.	

Display Technical View :

Table 2.1.10.10 - Display Technical View

Use case Name: UC-DisplayTechnicalView **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-10 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Search results available for lawyer.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer requests detailed view of results.	2. System formats technical legal details.
3. Lawyer views technical references.	4. System displays laws and statutes.
5. Lawyer analyzes legal data.	6. System provides detailed analysis options.
Success Scenario: Technical view displayed successfully.	
Alternate Scenario: ● If formatting error, system shows basic results.	
Post condition: Lawyer accesses detailed legal references.	

Display Simplified View :

Table 2.1.10.11 - Display Simplified View

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Use case Name: UC-DisplaySimplifiedView **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-11 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Search results available for client.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client requests simple explanation.	2. System generates simplified summaries.
3. Client reads easy-to-understand content.	4. System converts legal jargon to simple language.
5. Client comprehends information.	6. System highlights key points.
Success Scenario: Simplified view displayed successfully.	
Alternate Scenario: ● If summarization fails, system shows original content.	
Post condition: Client understands legal concepts easily.	

Search By Sections :

Table 2.1.10.12 - Search By Sections

Use case Name: UC-SearchBySections **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-12 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Lawyer must be logged in.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer selects "Search by Sections" option.	2. System shows legal section/acts list.
3. Lawyer chooses specific section (e.g., PPC 302).	4. System searches relevant laws.
5. Lawyer views section-specific arguments.	6. System displays targeted legal content.
Success Scenario: Section-specific legal arguments found successfully.	
Alternate Scenario: ● If section not found, system shows "Section not in database".	

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Post condition: Lawyer gets precise legal arguments for specific sections.

Bookmark Arguments :

Table 2.1.10.13 - Bookmark Arguments

Use case Name: UC-BookmarkArguments **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-13 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Lawyer must have search results displayed.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer clicks bookmark icon on legal argument.	2. System saves argument to lawyer's bookmark.
3. Lawyer views bookmarks section.	4. System shows "Bookmarked" confirmation.
5. Lawyer accesses bookmarked arguments later.	6. System maintains bookmark collection.
Success Scenario: Legal argument bookmarked successfully.	
Alternate Scenario: ● If already bookmarked, system shows "Already Bookmarked".	
Post condition: Bookmarked argument available in lawyer's personal collection.	

Save Legal Information :

Table 2.1.10.14 - Save Legal Information

Use case Name: UC-SaveLegalInfo **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-14 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Client must have viewed legal information.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client clicks save button on legal information.	2. System saves information to client's collection.

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3. Client views saved items section.	4. System shows "Saved" confirmation.
5. Client organizes saved information.	6. System maintains a personal library.
Success Scenario: Legal information saved successfully.	
Alternate Scenario: ● If already saved, system shows "Already saved".	
Post condition: Legal information available in client's personal library.	

View Search History :

Table 2.1.10.15 - View Search History

Use case Name: UC-ViewSearchHistory **Actors:** Lawyer **Date:** 05/11/2025	**Use case ID:** UC-15 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Lawyer must be logged in and have previous search activity.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Lawyer navigates to search history section.	2. System displays search list.
3. Lawyer selects previous search query.	4. System executes the search again.
5. Lawyer views updated results.	6. System logs this access.
Success Scenario: Search history accessed successfully.	
Alternate Scenario: ● If no history exists, system shows "No search history available".	
Post condition: Lawyer can reuse previous searches efficiently.	

View Previous Searches :

Table 2.1.10.16 - View Previous Searches

Use case Name: UC-ViewPreviousSearches **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-16 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Client must be logged in and have previous search activity.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client navigates to search history.	2. System displays list of previously searched topics.
3. Client selects a previous search.	4. System shows the saved results.
5. Client reviews the information.	6. System updates access timestamp.
Success Scenario: Previous searches accessed successfully.	
Alternate Scenario: ● If no history, system shows "No previous searches found".	
Post condition: Client can revisit previously searched legal topics.	

Summarize Judgements :

Table 2.1.10.17 - Summarize Judgements

Use case Name: UC-SummarizeJudgements **Actors:** Client **Date:** 05/11/2025	**Use case ID:** UC-17 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Client must have selected a legal judgment to view.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Client selects "Get Summary" option on a judgment.	2. System processes judgment text through AI.
3. Client waits for summary generation.	4. System extracts key points and rulings.
5. Client reads the simplified summary.	6. System displays concise, easy-to-understand version.
Success Scenario: Complex judgment summarized successfully.	
Alternate Scenario: ● If summarization fails, system shows original judgment.	

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Post condition: Client understands the key points of the judgment.

Update Profile :

Table 2.1.10.18 - Update Profile

Use case Name: UC-UpdateProfile **Actors:** Client, Lawyer **Date:** 05/11/2025	**Use case ID:** UC-18 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User navigates to profile settings.	2. System displays current profile information.
3. User edits personal details.	4. System validates new information.
5. User saves changes.	6. System updates profile and confirms "Profile updated successfully".
Success Scenario: Profile information updated successfully.	
Alternate Scenario: ● If invalid data, system shows "Please check entered information".	
Post condition: Updated profile information stored in system.	

Logout :

Table 2.1.10.19 - Logout

Use case Name: UC-Logout **Actors:** Client, Lawyer, Admin **Date:** 05/11/2025	**Use case ID:** UC-19 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User clicks logout button.	2. System terminates user session.

3. User confirms logout.	4. System clears authentication tokens.
5. User redirected to login page.	6. System logs logout activity for audit.
Success Scenario: User logged out securely from system.	
Alternate Scenario: ● If network issue, system forces logout after timeout.	
Post condition: User session ended and secure access revoked.	

Manage Users :

Table 2.1.10.20 - Manage Users

Use case Name: UC-ManageUsers **Actors:** Admin **Date:** 05/11/2025	**Use case ID:** UC-20 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Admin must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Admin navigates to user management.	2. System displays all user accounts.
3. Admin selects user for action.	4. System shows user details and options.
5. Admin performs action (enable/disable/delete).	6. System updates user status and confirms.
Success Scenario: User account managed successfully.	
Alternate Scenario: ● If user not found, system shows "User account does not exist".	
Post condition: User accounts properly maintained in system.	

Update Laws :

Table 2.1.10.21 - Update Laws

Use case Name: UC-UpdateLaws **Actors:** Admin **Date:** 05/11/2025	**Use case ID:** UC-21 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Admin must be logged into the system.	

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Main Success Scenario	
Actor's Action	**System's Response**
1. Admin navigates to laws management section.	2. System shows current laws and statutes database.
3. Admin uploads new laws/statutes.	4. System validates data format.
5. Admin confirms update.	6. System adds new legal data and shows "Update successful".
Success Scenario: Legal database updated successfully with new laws.	
Alternate Scenario: ● If data format wrong, system shows "Invalid data format".	
Post condition: New legal data available for searching.	

Monitor System :

Table 2.1.10.22 - Monitor System

Use case Name: UC-MonitorSystem **Actors:** Admin **Date:** 05/11/2025	**Use case ID:** UC-22 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Admin must be logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Admin checks user accounts list.	2. System shows all registered users.
3. Admin reviews user search activities.	4. System displays search logs and history.
5. Admin monitors system performance.	6. System shows basic usage statistics.
Success Scenario: User activities and system status checked successfully.	
Alternate Scenario: ● If no activities, system shows "No recent activities found".	
Post condition: Admin has overview of user activities and system usage.	

Validate Input :

Table 2.1.10.23 - Validate Input

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Use case Name: UC-ValidateInput **Actors:** Admin, Lawyer, Client **Date:** 05/11/2025	**Use case ID:** UC-23 **Created by:** Nayab, Reham, Ramiz
Pre-condition: User enters data in any form.	
Main Success Scenario	
Actor's Action	**System's Response**
1. User submits form/data.	2. System checks data format and rules.
3. User corrects errors if any.	4. System accepts valid data.
Success Scenario: All input data validated successfully.	
Alternate Scenario: ● If invalid data, system shows "Please correct errors".	
Post condition: Only correct data stored in system.	

Manage Access :

Table 2.1.10.24 - Manage Access

Use case Name: UC-ManageAccess **Actors:** Admin **Date:** 05/11/2025	**Use case ID:** UC-25 **Created by:** Nayab, Reham, Ramiz
Pre-condition: Admin is logged into the system.	
Main Success Scenario	
Actor's Action	**System's Response**
1. Admin sets user roles.	2. System saves role permissions.
3. Admin assigns access levels.	4. System implements access control.
Success Scenario: Role-based access control implemented successfully.	
Alternate Scenario: ● If unauthorized access, system shows "Access denied".	
Post condition: Users have appropriate access levels.	

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Chapter 3: Design Document (For Object Oriented Approach)

3.1. Introduction:

The third deliverable is all about the software design. In the previous deliverable, analysis of the system is completed. So we understand the current situation of the problem domain. Now we are ready to strive for a solution for the problem domain by using an object-oriented approach. Following artifacts are included in this deliverable for the LegisCounsel system:

1. Domain Model
2. System Sequence Diagram
3. Sequence Diagram
4. Collaboration Diagram
5. Operation Contracts
6. Design Class Diagram
7. State Transition Diagram
8. Data Model

Now we discuss these artifacts one by one as follows:

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3.2. Domain Model

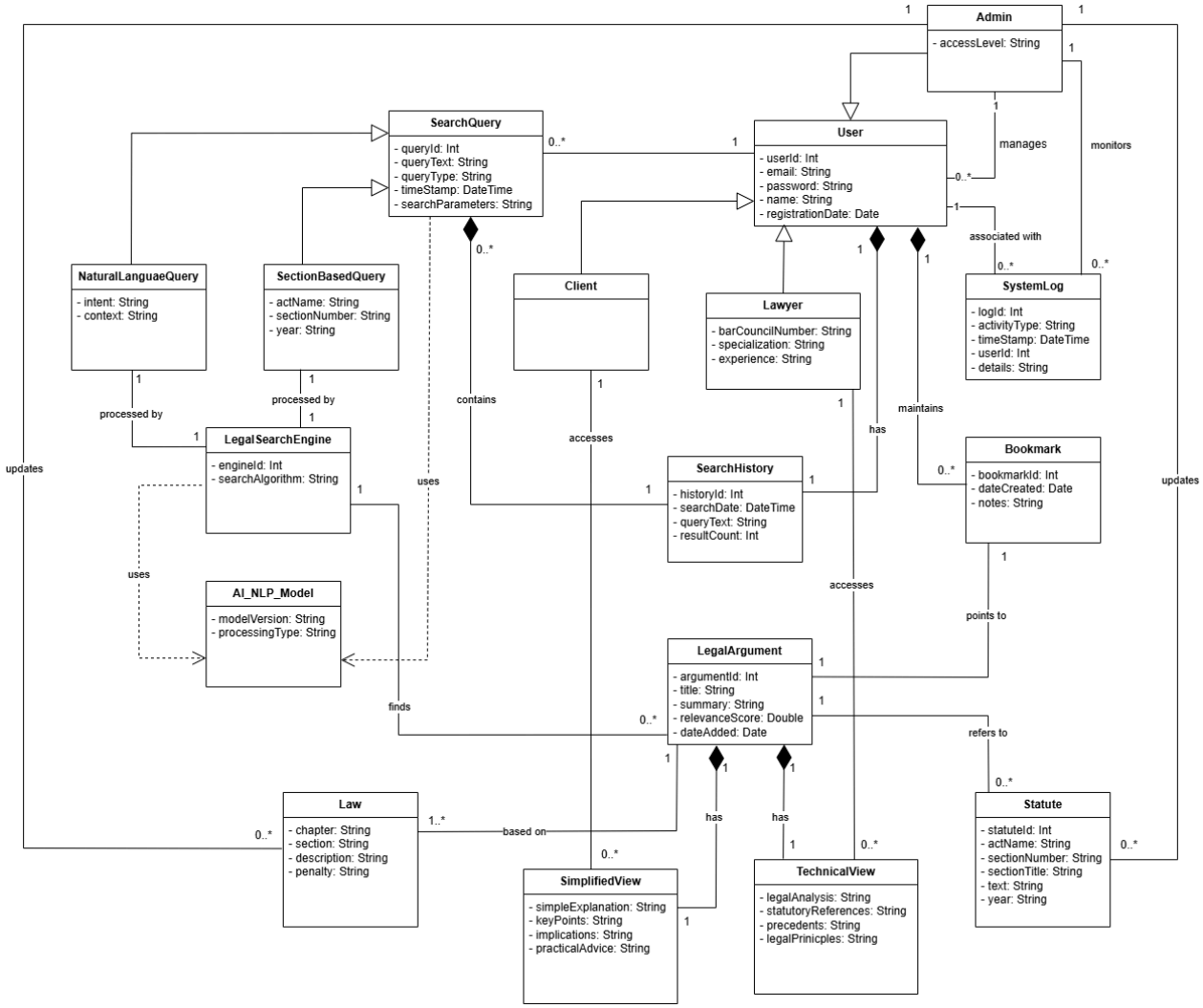


Figure 3.2 - Domain Model Diagram

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3.3. System Sequence Diagram

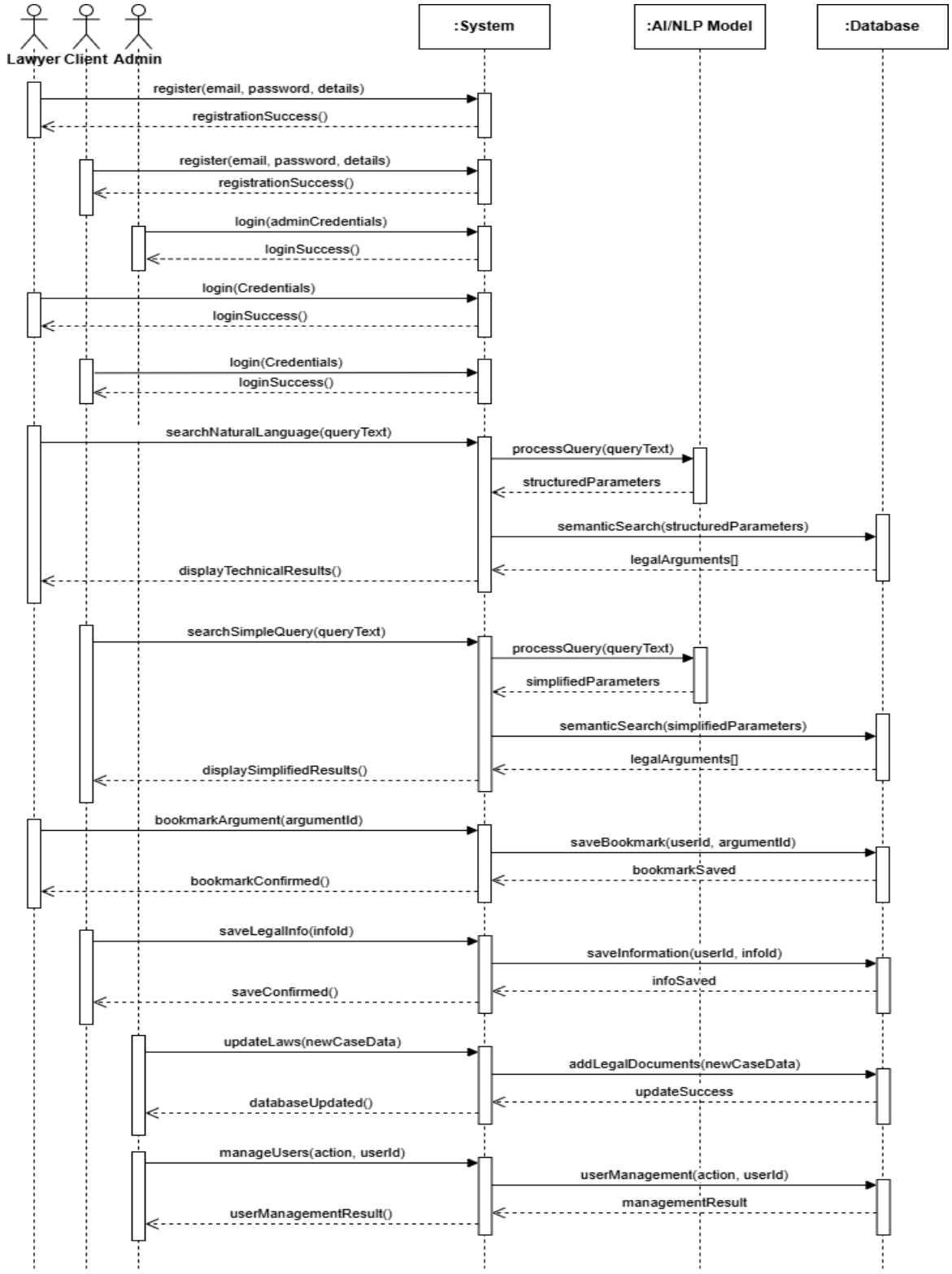


Figure 3.3 - System Sequence Diagram

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3.4. Sequence Diagram

Legal Argument Search (Lawyer) Sequence Diagram:

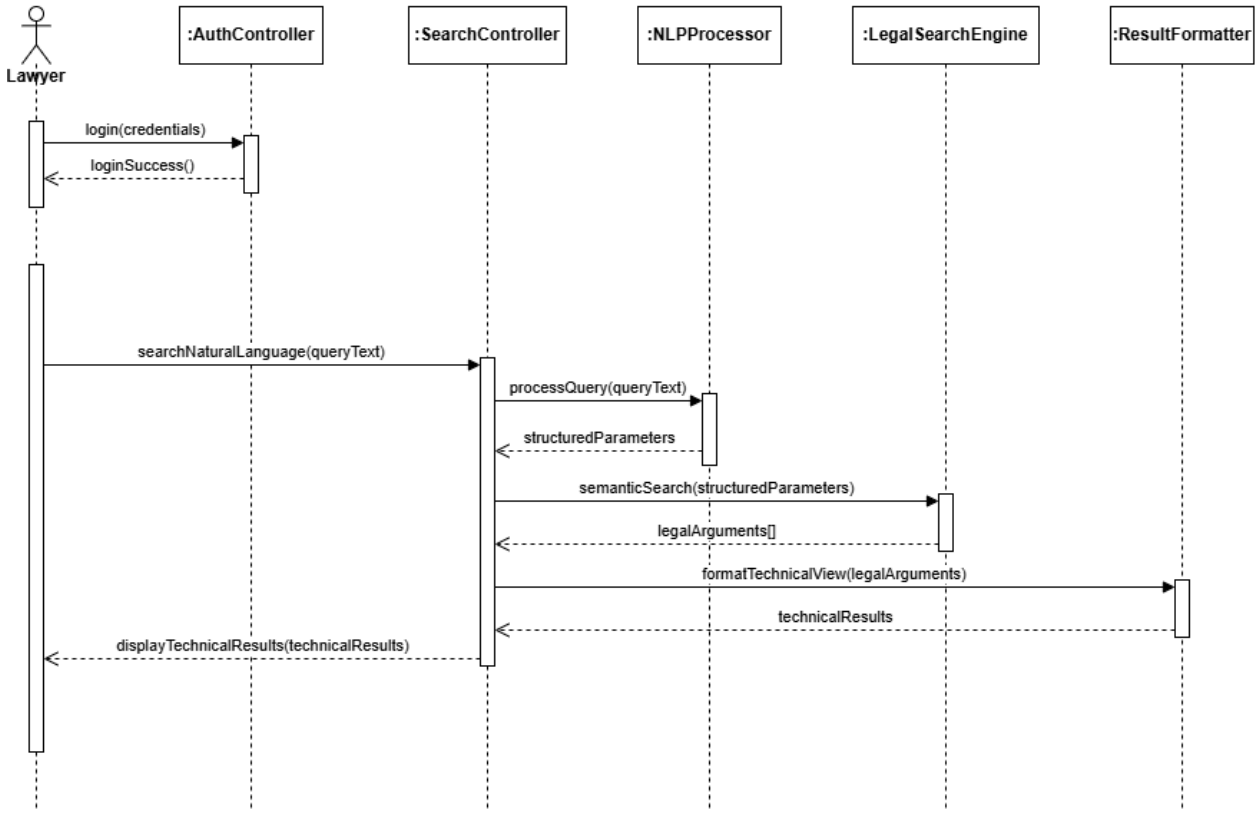


Figure 3.4.1 - Legal Argument Search Sequence Diagram

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Legal Information Search (Client) Sequence Diagram:

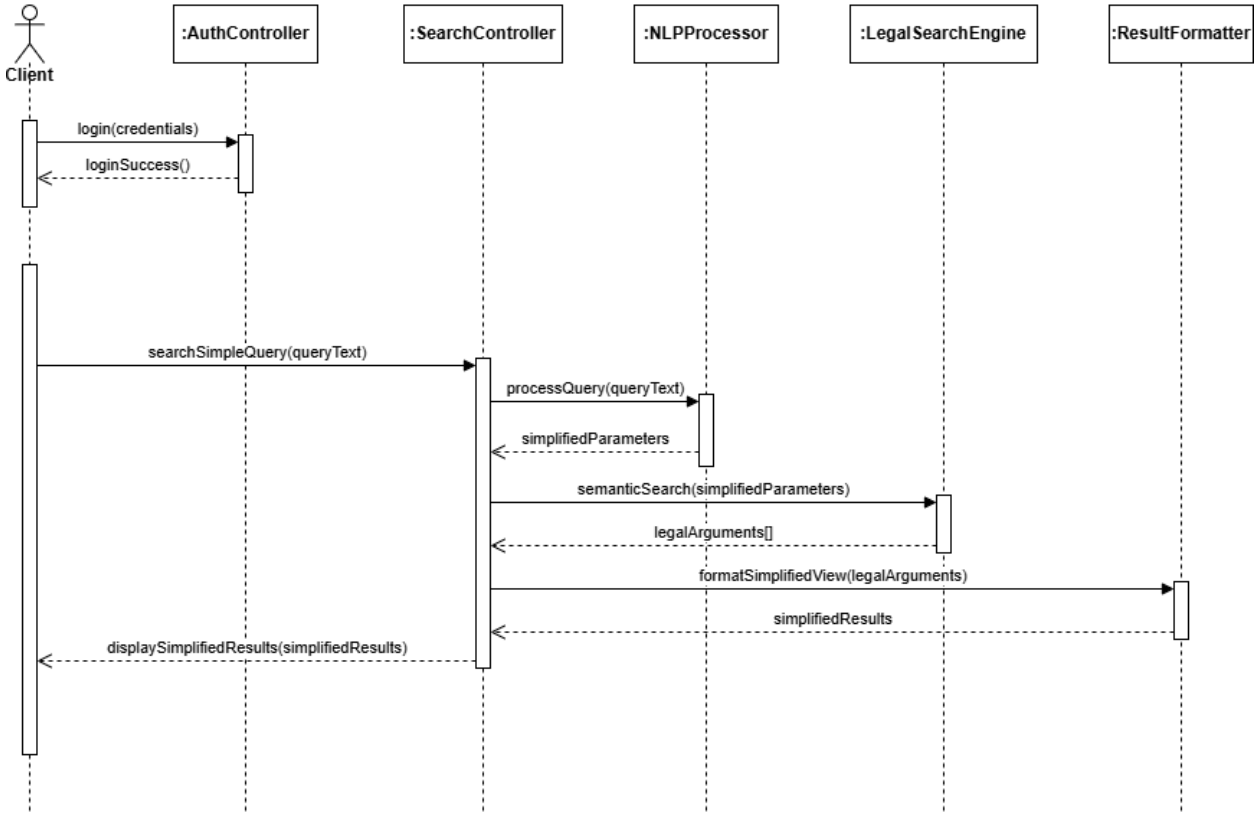


Figure 3.4.2 - Legal Information Search Sequence Diagram

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Admin Update Laws Sequence Diagram:

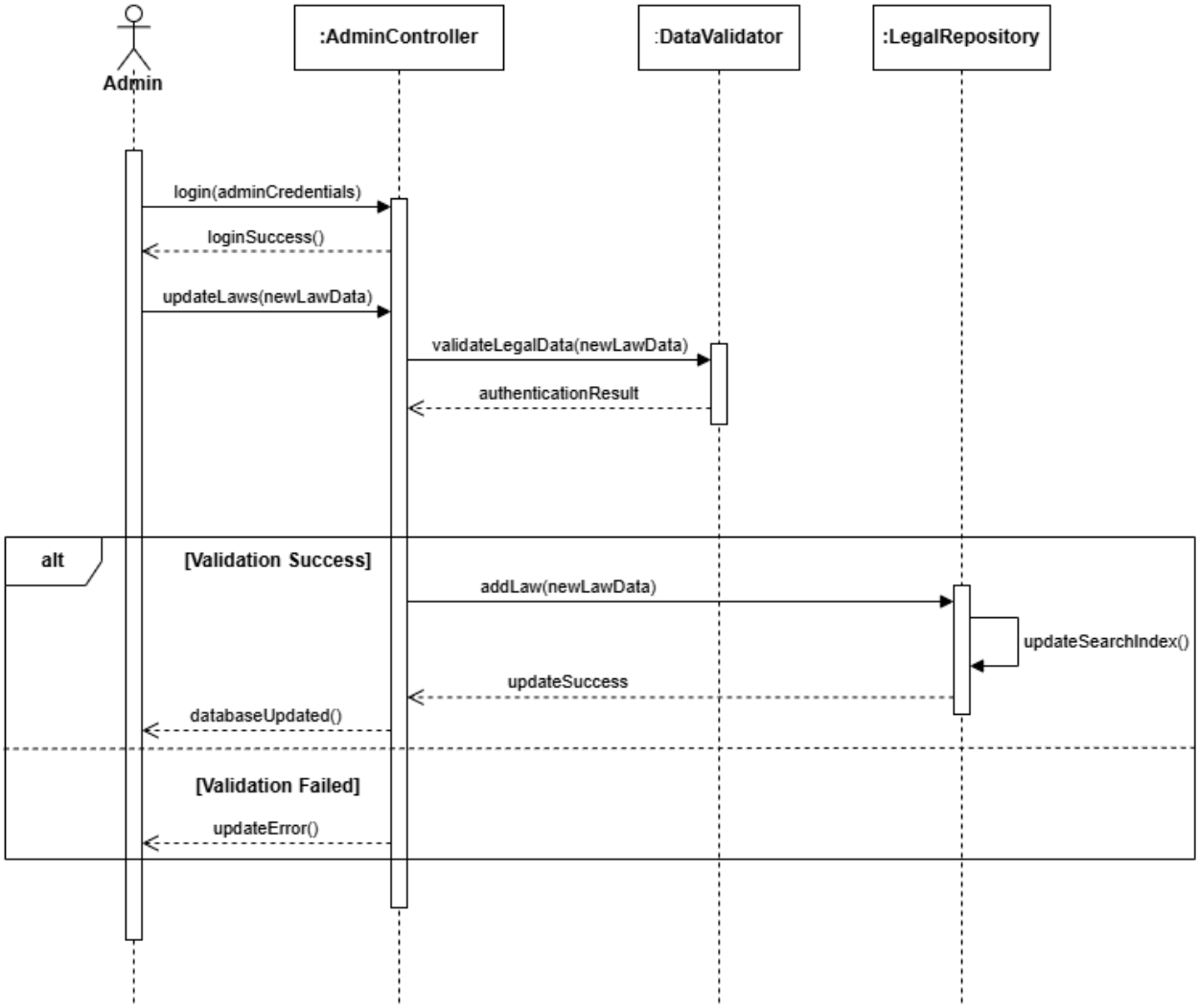


Figure 3.4.3 - Admin Update Laws Sequence Diagram

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User Login Sequence Diagram:

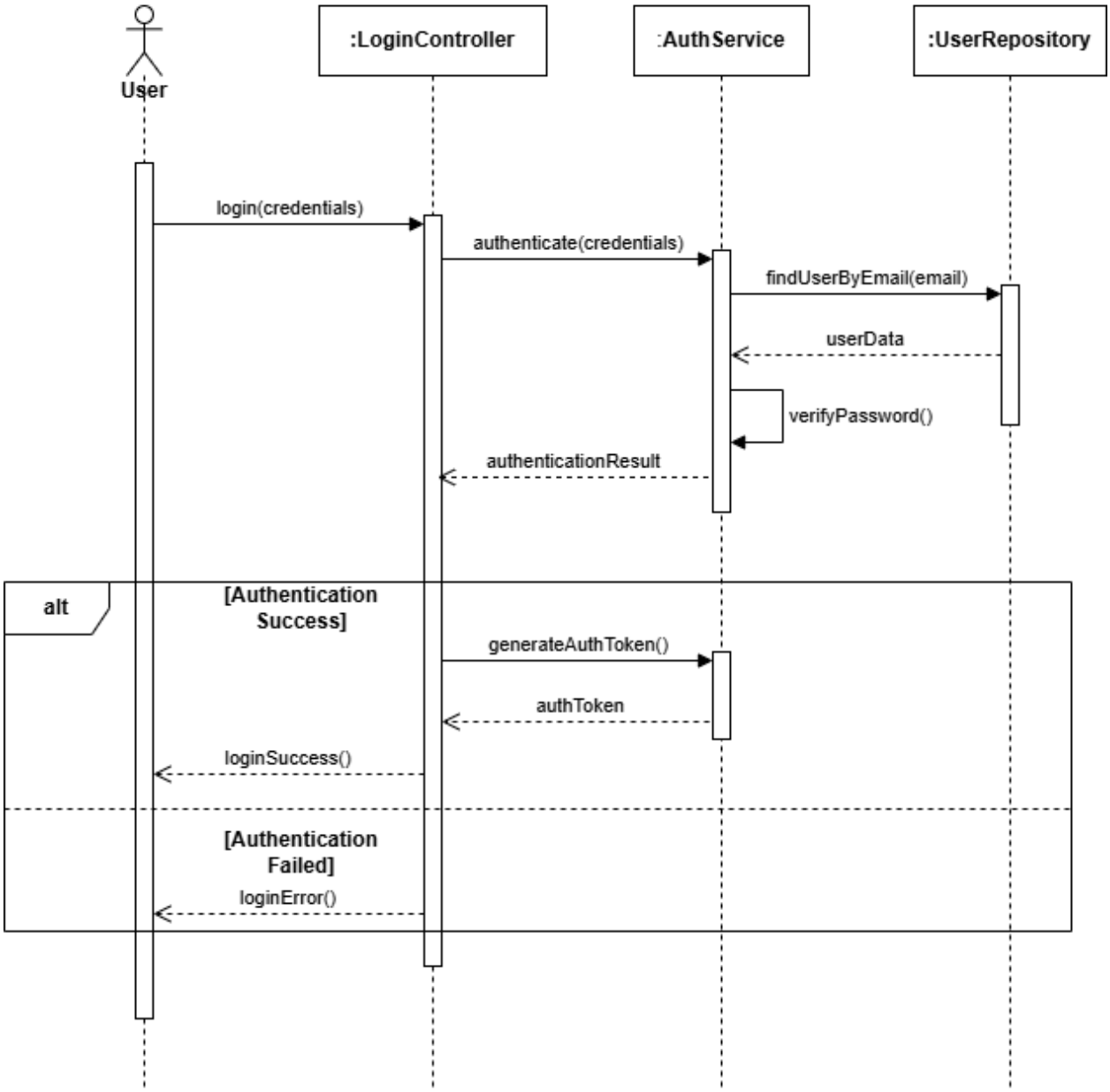


Figure 3.4.4 - User Login Sequence Diagram

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User Registration Sequence Diagram:

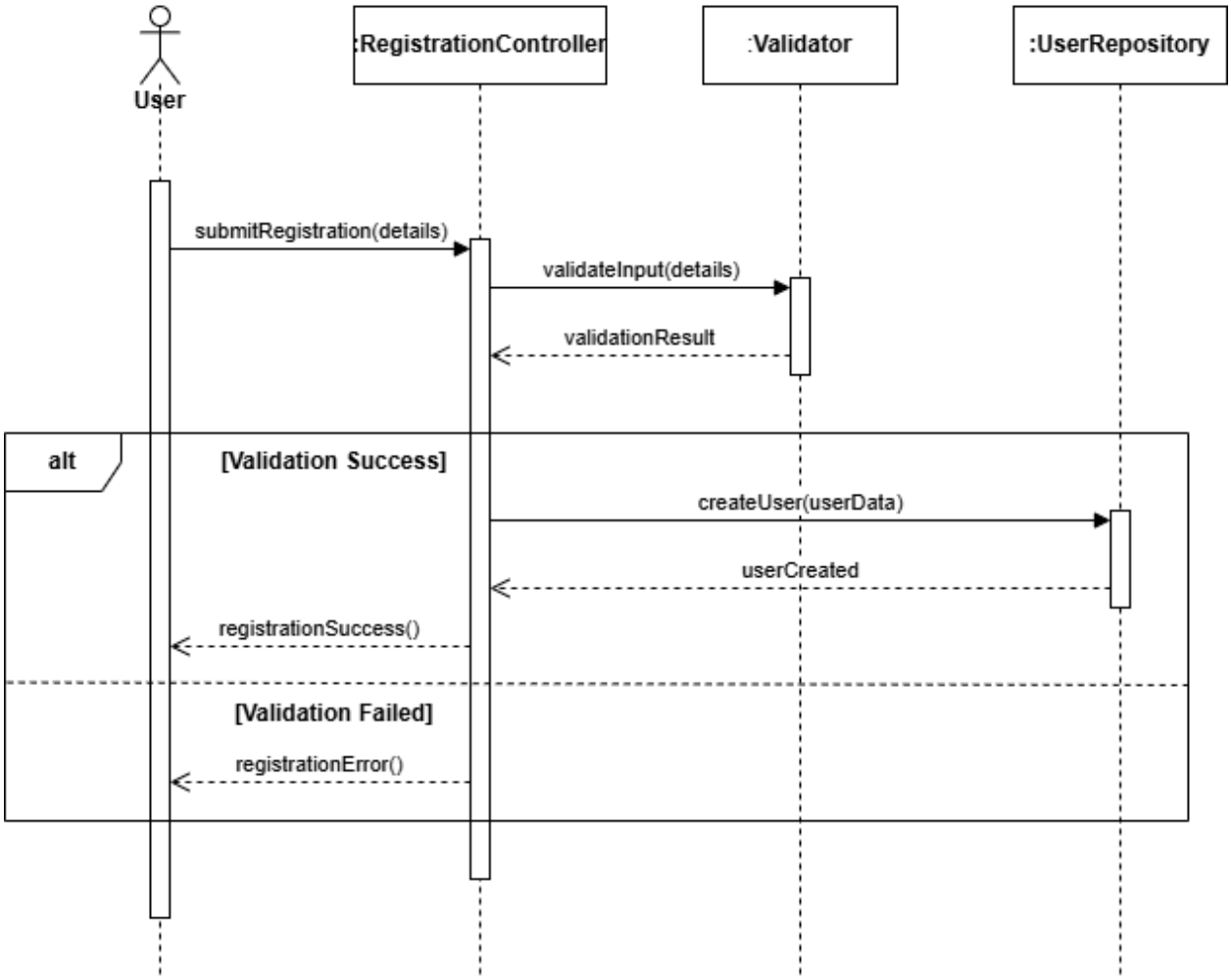


Figure 3.4.5 - User Registration Sequence Diagram

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Section-based Search Sequence Diagram:

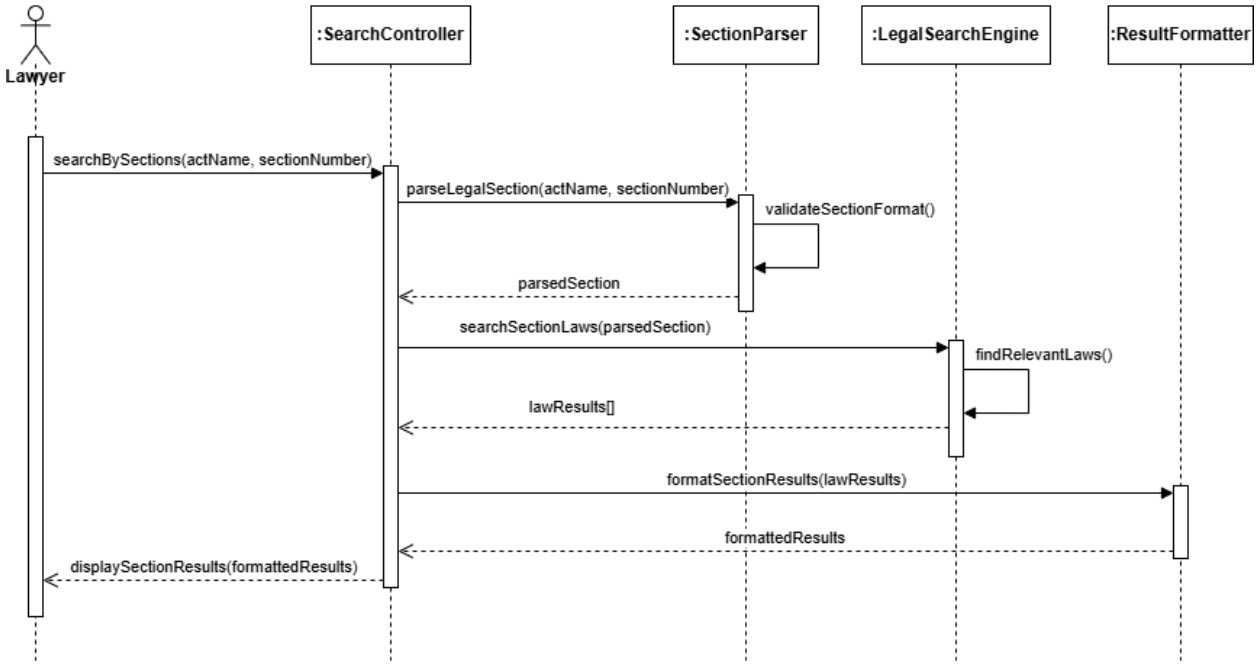


Figure 3.4.6 - Section-based Search Sequence Diagram

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Bookmark Legal Argument Sequence Diagram:

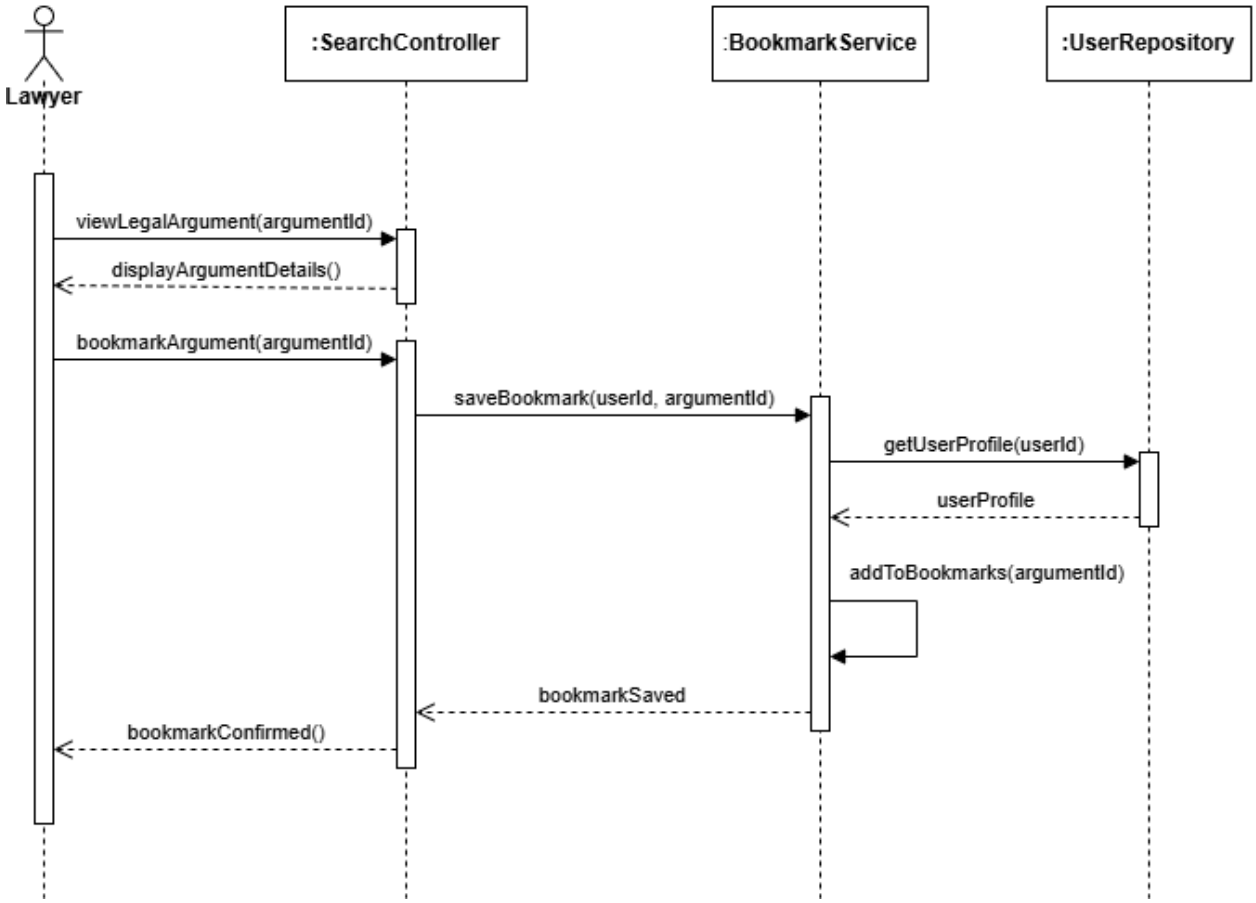


Figure 3.4.7 - Bookmark Legal Argument Sequence Diagram

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View Search History Sequence Diagram:

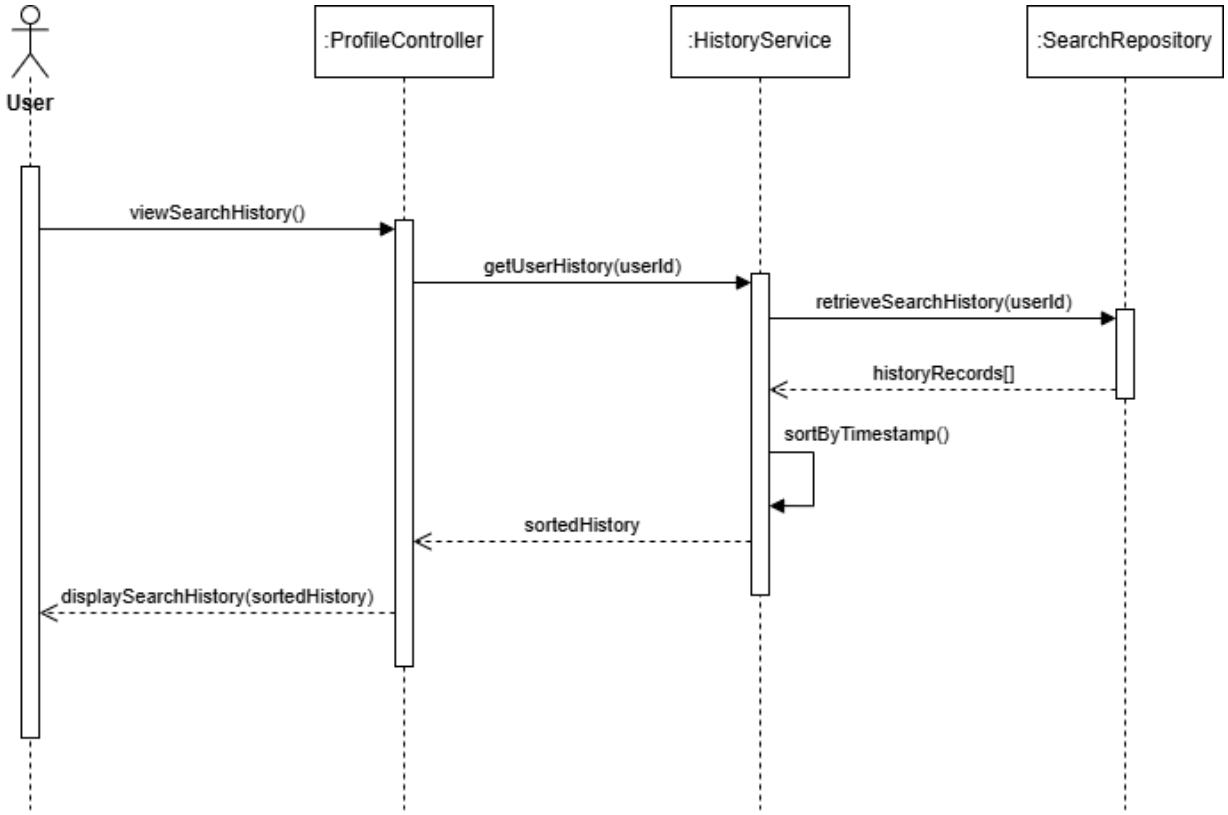


Figure 3.4.8 - View Search History Sequence Diagram

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Admin User Management Sequence Diagram:

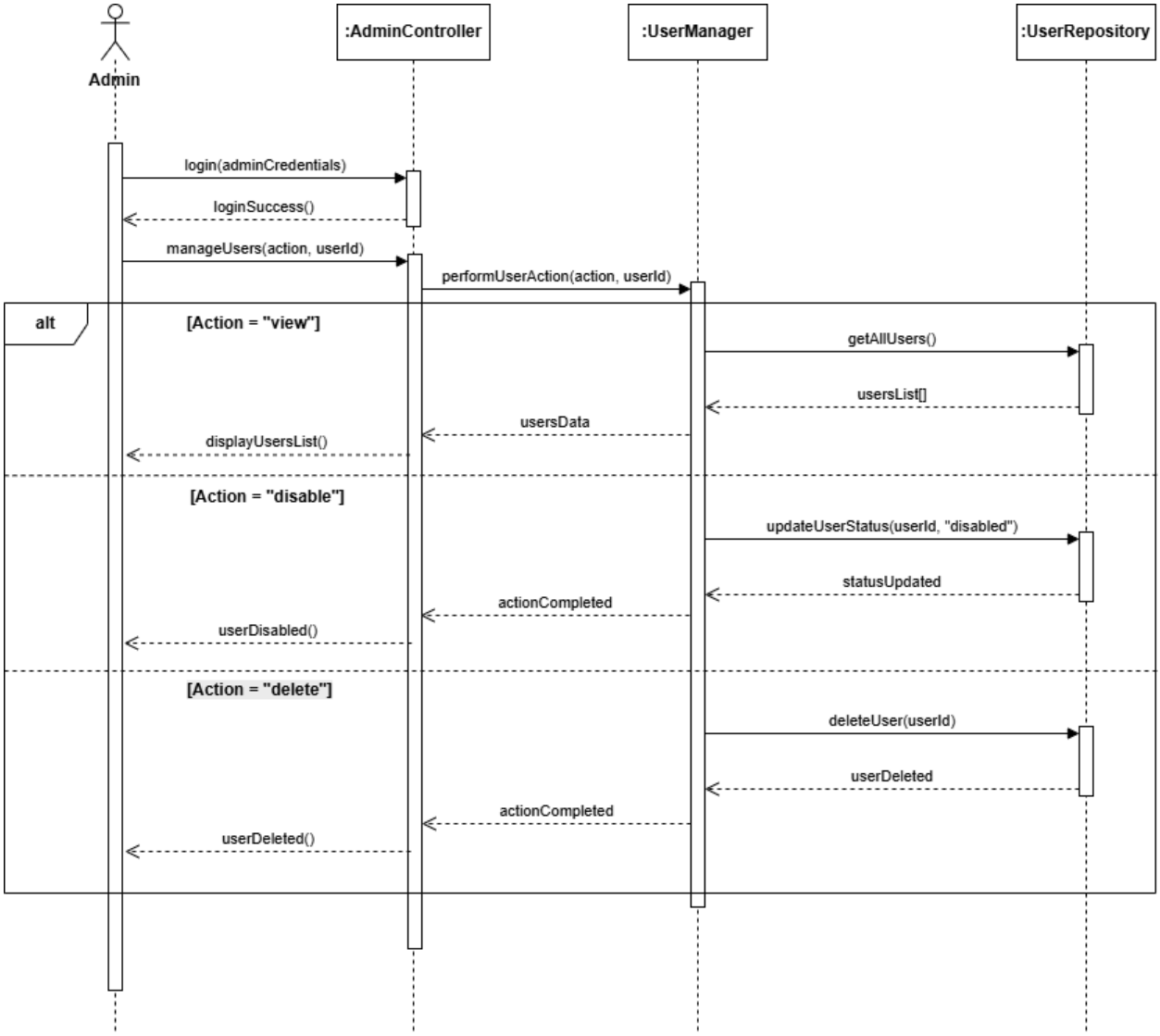


Figure 3.4.9 - Admin User Management Sequence Diagram

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Save Legal Information (Client) Sequence Diagram:

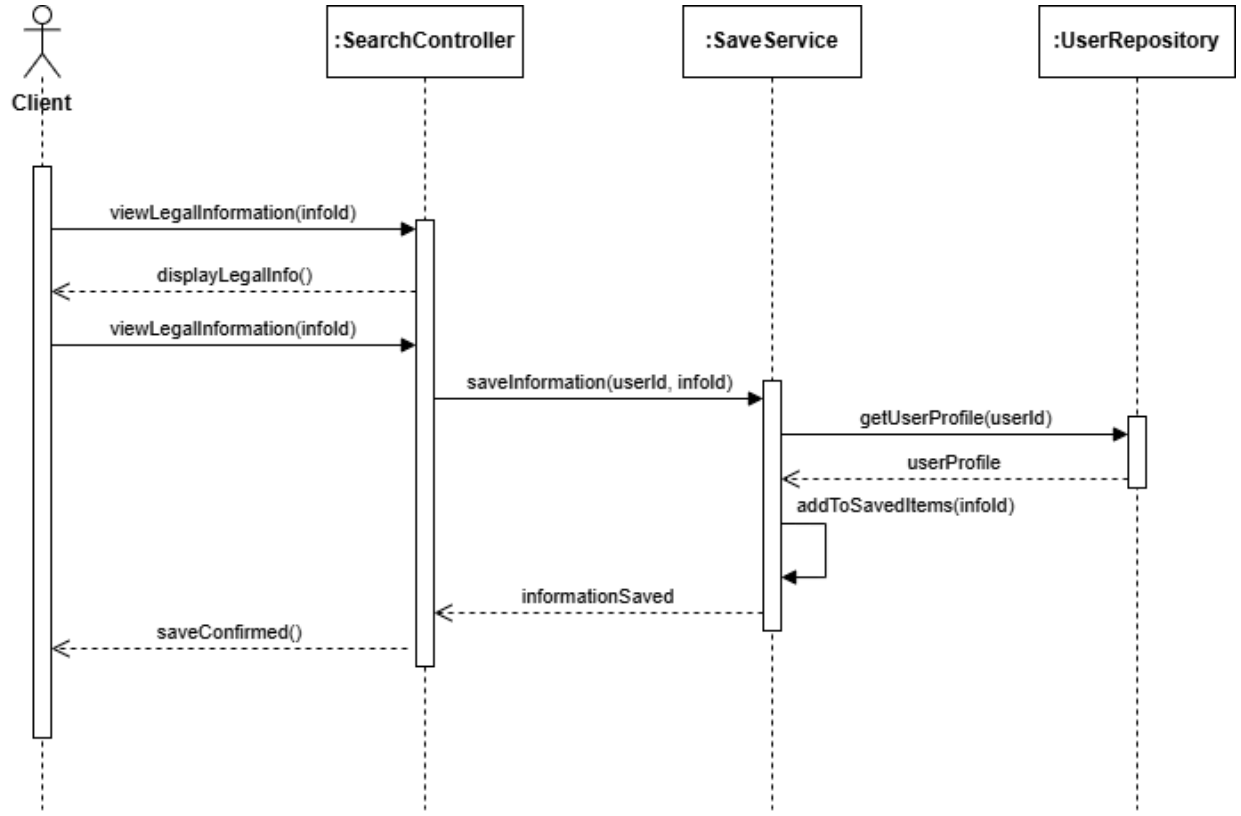


Figure 3.4.10 - Save Legal Information Sequence Diagram

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User Logout Sequence Diagram:

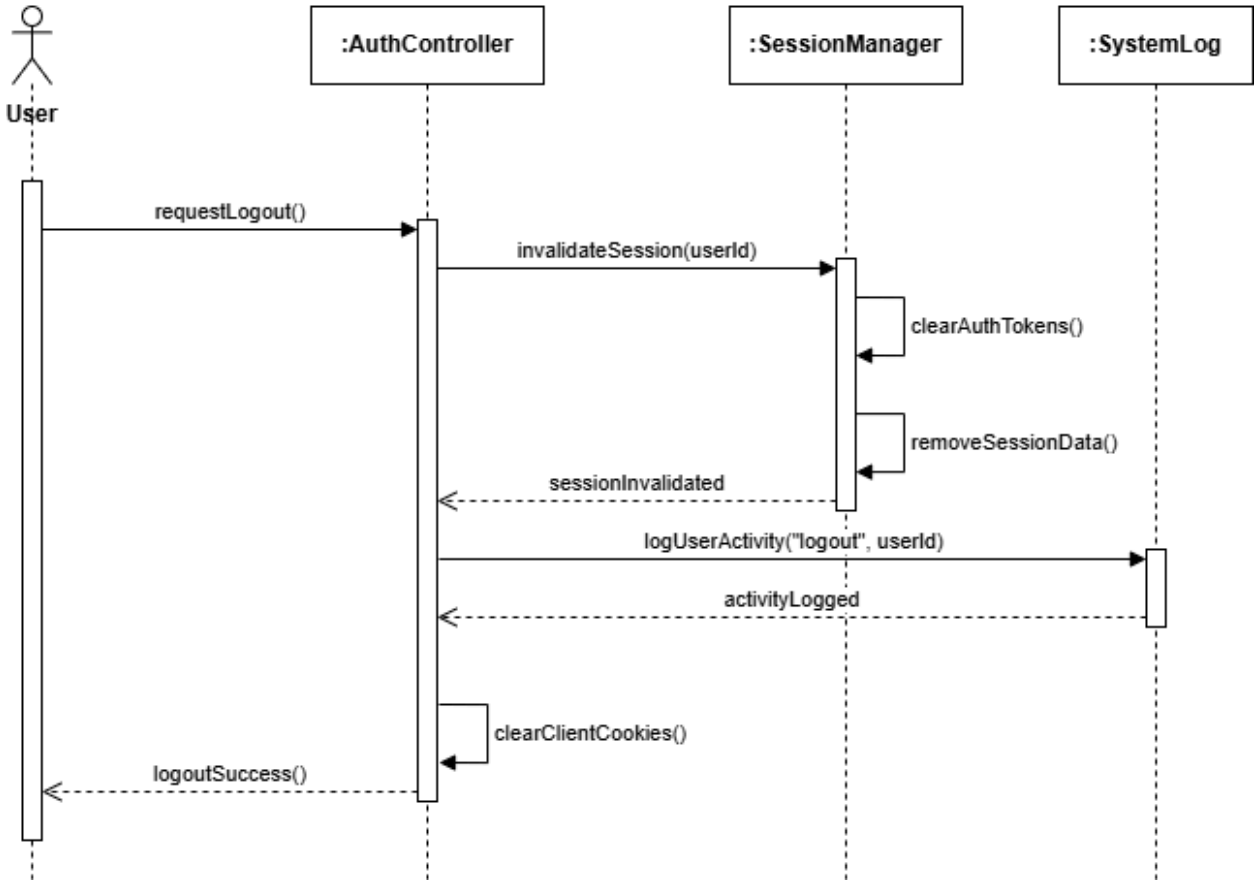


Figure 3.4.11 - User Logout Sequence Diagram

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User Update Profile Sequence Diagram:

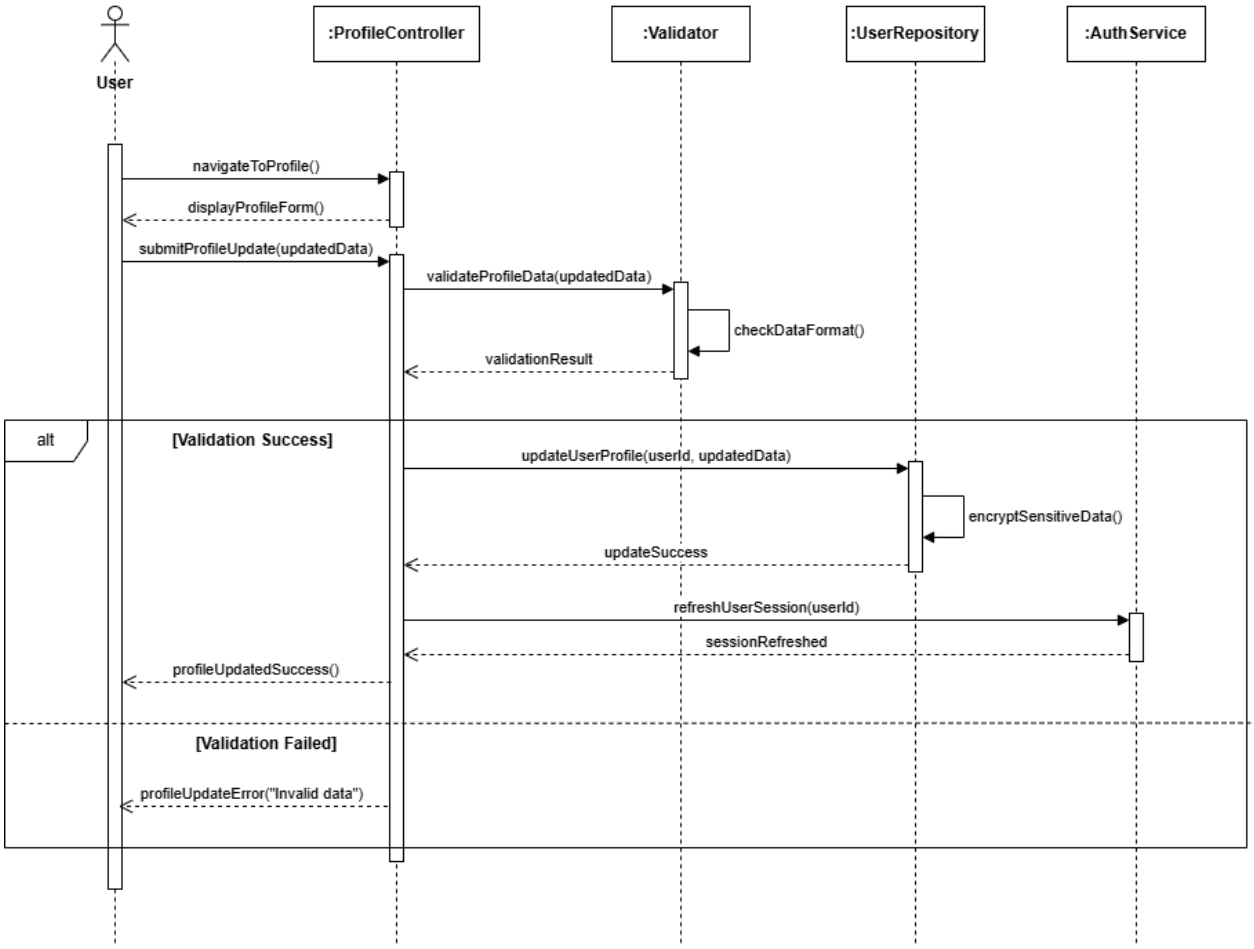


Figure 3.4.12 - User Update Profile Sequence Diagram

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View Simplified Summaries Sequence Diagram:

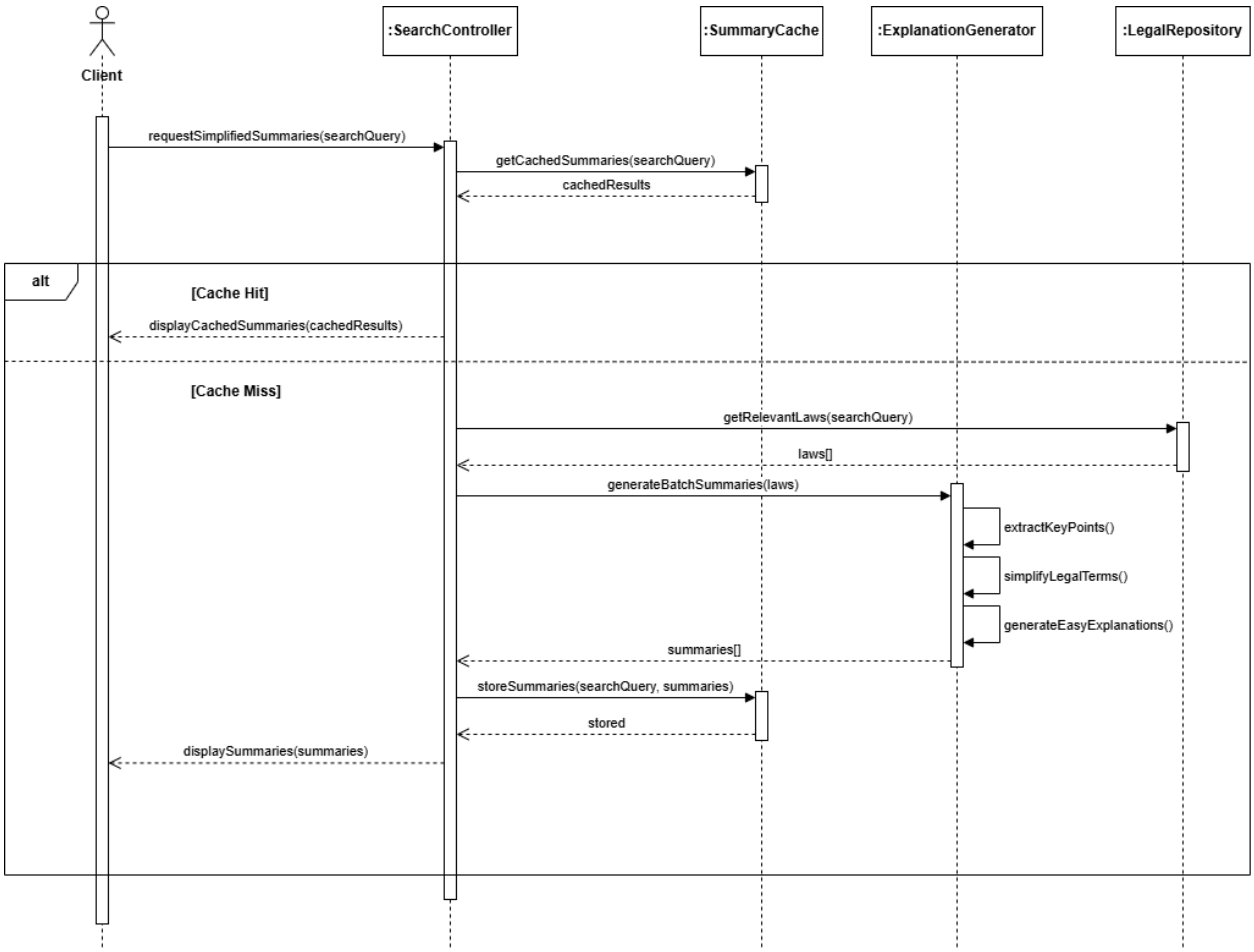


Figure 3.4.13 - View Simplified Summaries Sequence Diagram

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3.5. Collaboration Diagram

Lawyer - Legal Argument Search Collaboration Diagram:

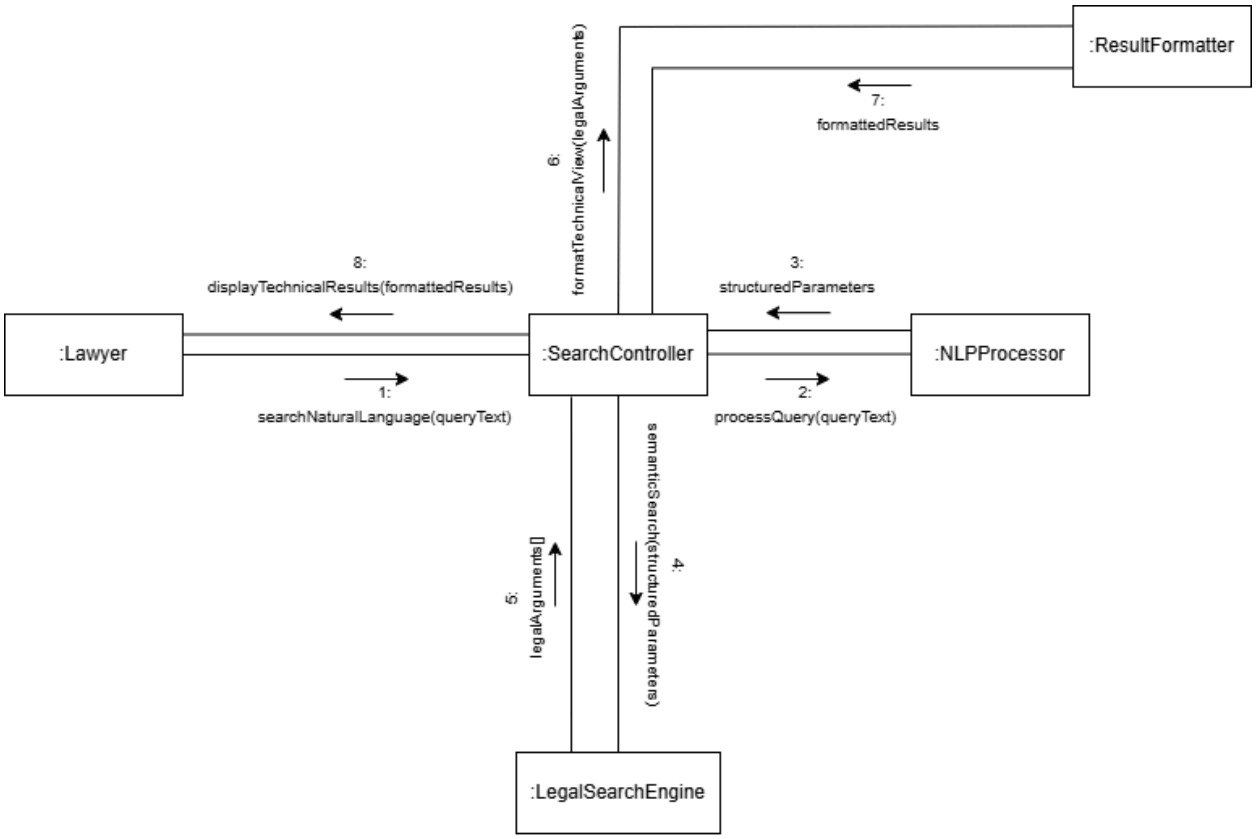


Figure 3.5.1 - Lawyer - Legal Argument Search Collaboration Diagram

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Client - Legal Information Search Collaboration Diagram:

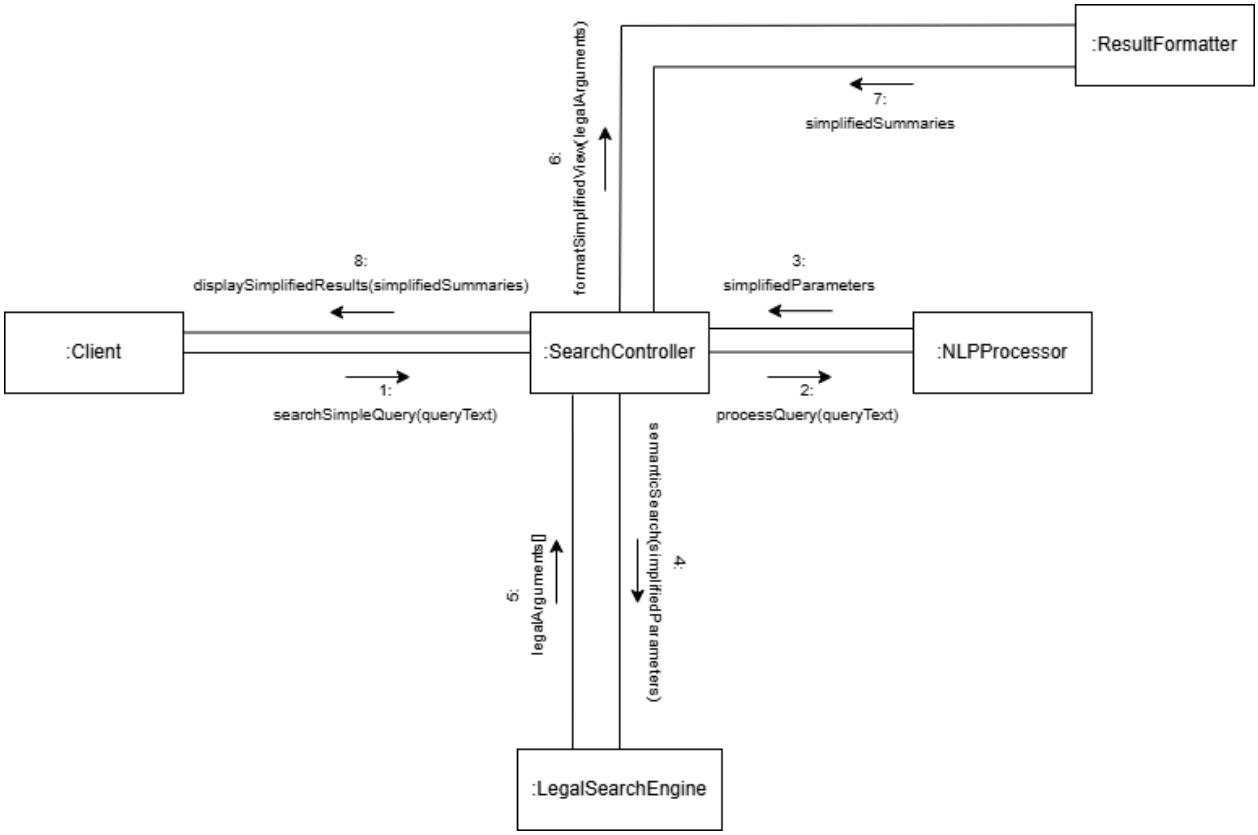


Figure 3.5.2 - Client - Legal Information Search Collaboration Diagram

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User Registration Collaboration Diagram:

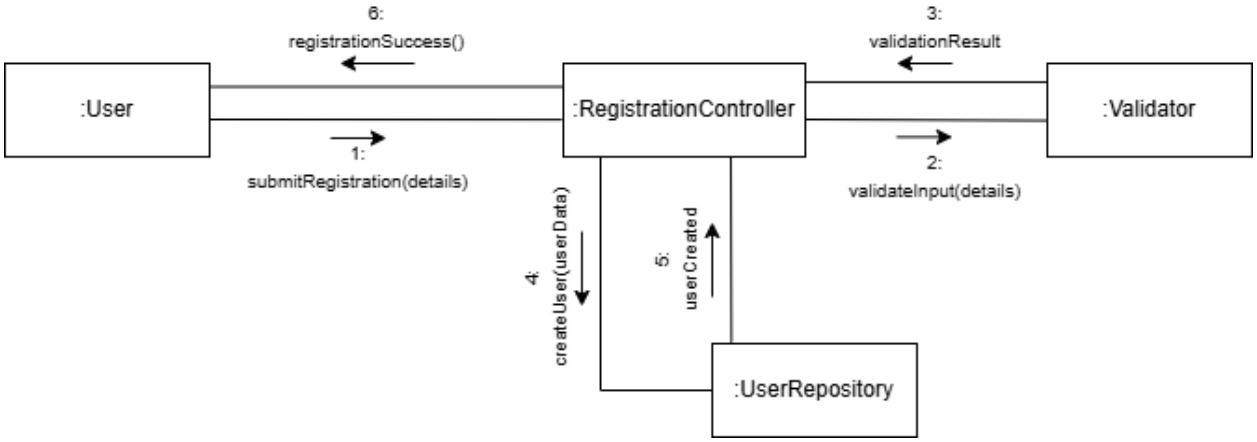


Figure 3.5.3 - User Registration Collaboration Diagram

Admin - Update Laws Collaboration Diagram:

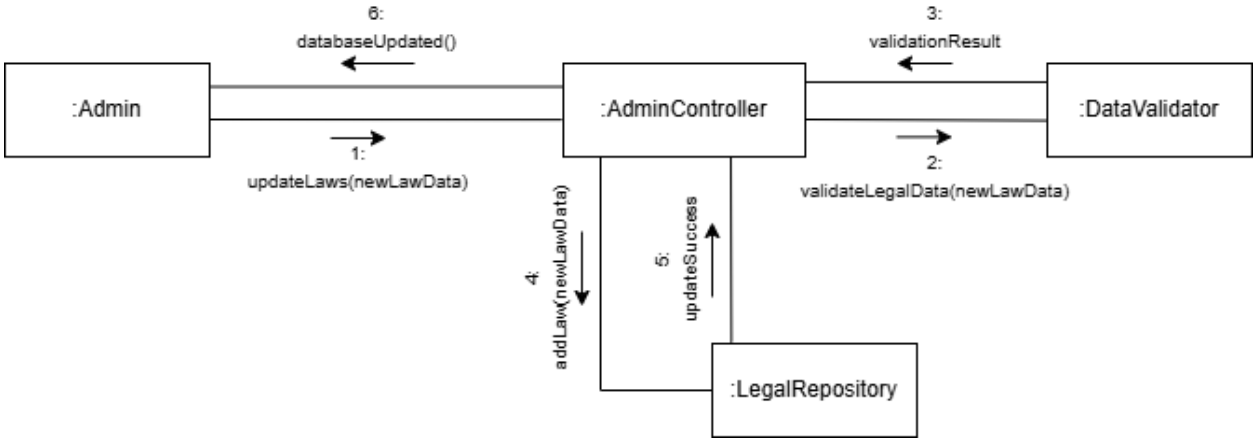


Figure 3.5.4 - Admin - Update Laws Collaboration Diagram

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Admin - Manage Users Collaboration Diagram:

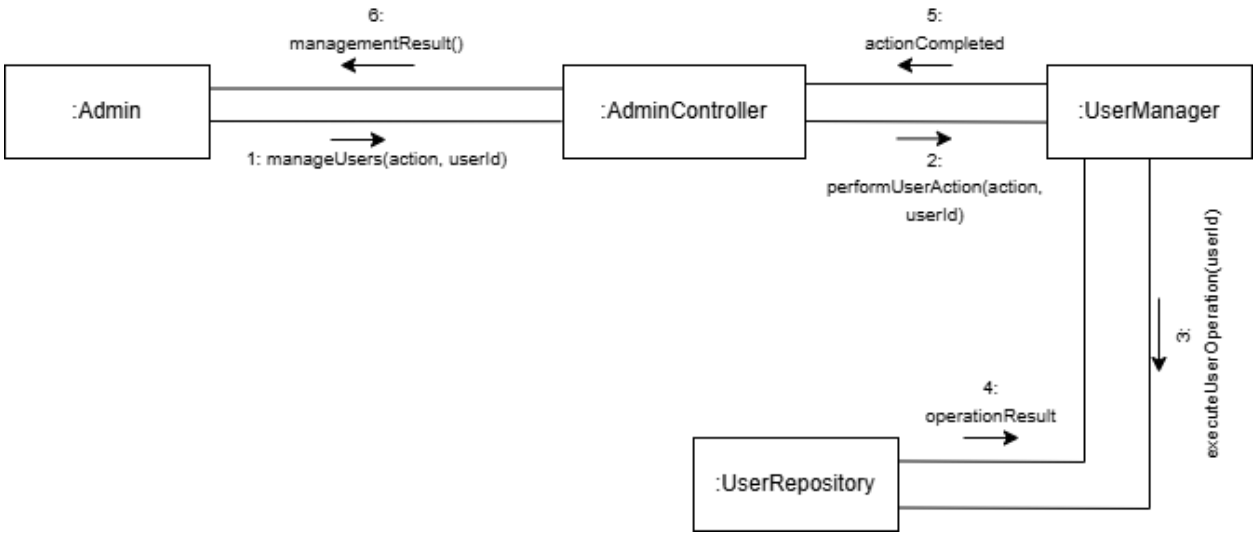


Figure 3.5.5 - Admin - Manage Users Collaboration Diagram

3.6. Operation Contracts

Search Natural Language Operation Contract:

Table 3.6.1 - Search Arguments

Name	searchNaturalLanguage
Responsibilities	Process natural language query and return legal arguments.
Cross References	UC-SearchArguments
Preconditions	● Lawyer is logged into the system. ● Legal database contains laws and statutes. ● AI/NLP model is available.
Postconditions	● A new SearchQuery object was created. ● SearchQuery was associated with Lawyer. ● LegalArguments were retrieved from LegalRepository.

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	• SearchHistory was updated with the query. • SystemLog was generated for the activity.
Exceptions	Invalid query format, No results found.

Search Simple Query Operation Contract :

Table 3.6.2 - Search Legal Info

Name	searchSimpleQuery
Responsibilities	Process simple legal query and return simplified summaries.
Cross References	UC-SearchLegalInfo
Preconditions	• Client is logged into the system. • Legal database contains laws and statutes. • ExplanationGenerator is available.
Postconditions	• A new SearchQuery object was created. • SearchQuery was associated with Client. • Relevant laws were retrieved from LegalRepository. • Simplified summaries were generated. • SearchHistory was updated. • SystemLog was generated.
Exceptions	Invalid query, No relevant laws found.

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Register Operation Contract :

Table 3.6.3 - Register

Name	register
Responsibilities	Create new user account in the system.
Cross References	UC-Register
Preconditions	● Email is not already registered. ● Internet connection is available.
Postconditions	● A new User object was created. ● User credentials were stored securely. ● SystemLog was generated for registration.
Exceptions	Email already exists, Invalid data format.

Login Operation Contract :

Table 3.6.4 - Login

Name	login
Responsibilities	Authenticate user and create session.
Cross References	UC-Login
Preconditions	● User account exists in the system. ● User is not already logged in.
Postconditions	● User session was created. ● Authentication token was generated. ● User role permissions were loaded. ● Last login timestamp was updated. ● SystemLog was generated for login activity.
Exceptions	Invalid credentials, Account disabled.

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Update Laws Operation Contract :

Table 3.6.5 - Update Laws

Name	updateLaws
Responsibilities	Add new laws to the legal database.
Cross References	UC-UpdateLaws
Preconditions	● Admin is logged into the system. ● Admin has appropriate privileges. ● Legal database is accessible.
Postconditions	● New Law objects were created. ● Laws were added to LegalRepository. ● Vector embeddings were updated. ● SystemLog was generated for the update.
Exceptions	Invalid data format, Duplicate entry.

Bookmark Operation Contract :

Table 3.6.6 - Bookmark Argument

Name	bookmarkArgument
Responsibilities	Save legal argument for future reference.
Cross References	UC-BookmarkArguments
Preconditions	● Lawyer is logged into the system. ● Legal argument exists in the system.
Postconditions	● A new Bookmark object was created. ● Bookmark was associated with Lawyer. ● Bookmark was associated with LegalArgument. ● User's bookmark collection was updated.

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Exceptions	Argument not found, Already bookmarked.

Update Profile Operation Contract :

Table 3.6.7 - Update Profile

Name	updateProfile
Responsibilities	Update user profile information.
Cross References	UC-UpdateProfile
Preconditions	● User is logged into the system. ● User account exists in the system.
Postconditions	● User object attributes were updated. ● Profile modification timestamp was updated. ● SystemLog was generated for profile update. ● User session was refreshed with updated data.
Exceptions	Invalid data format

Logout Operation Contract :

Table 3.6.8 - Logout

Name	logout
Responsibilities	Terminate user session securely.
Cross References	UC-Logout
Preconditions	● User is currently logged into the system. ● Active session exists.
Postconditions	● User session was invalidated. ● Authentication tokens were cleared. ● SystemLog was generated for logout

	activity.
Exceptions	Session already expired, Network error.

Save Legal Information Operation Contract :

Table 3.6.9 - Save Legal Information

Name	saveLegalInfo
Responsibilities	Save legal information for personal reference.
Cross References	UC-SaveLegalInfo
Preconditions	● Client is logged into the system. ● Legal information exists in the system.
Postconditions	● A new SavedItem object was created. ● SavedItem was associated with the client. ● SavedItem was associated with LegalInformation. ● Client's saved items collection was updated. ● SystemLog was generated for save activity.
Exceptions	Information not found, Already saved

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3.7. Design Class Diagram

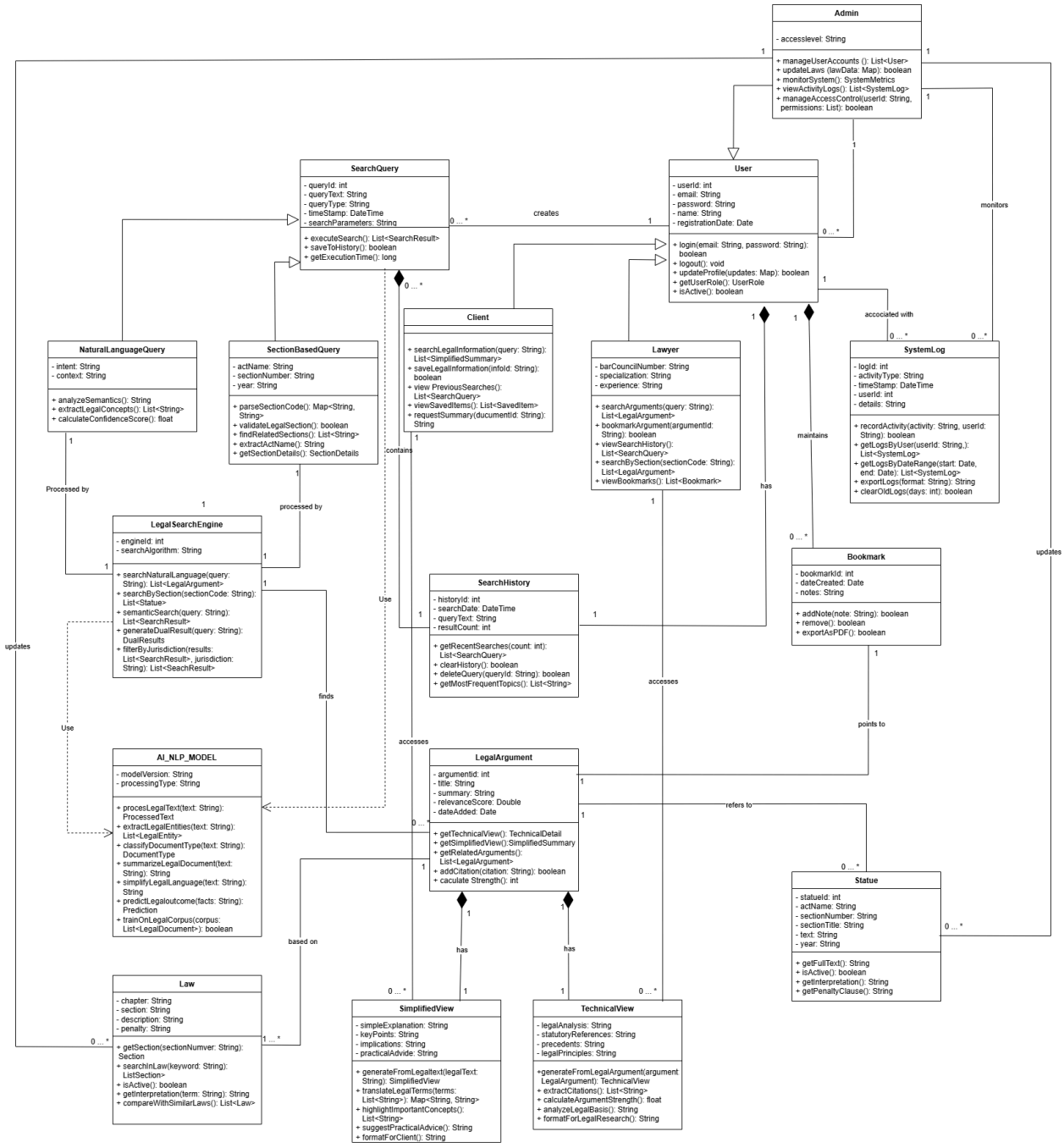


Figure 3.7 -System Class Diagram

3.8. State Chart Diagram

3.8.1 For Admin

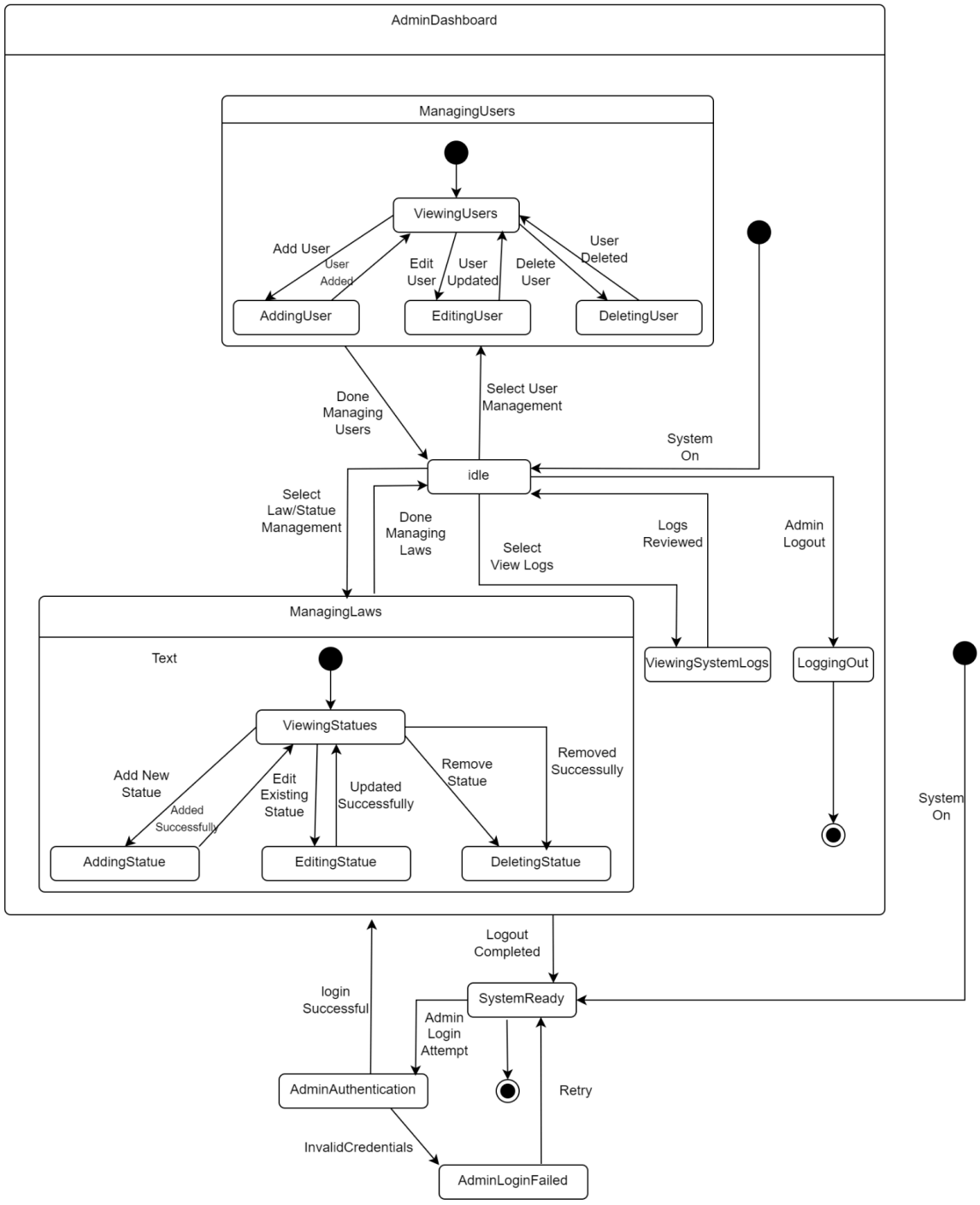


Figure 3.8.1 - Admin State Chart Diagram

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3.8.2 For Lawyer/Client

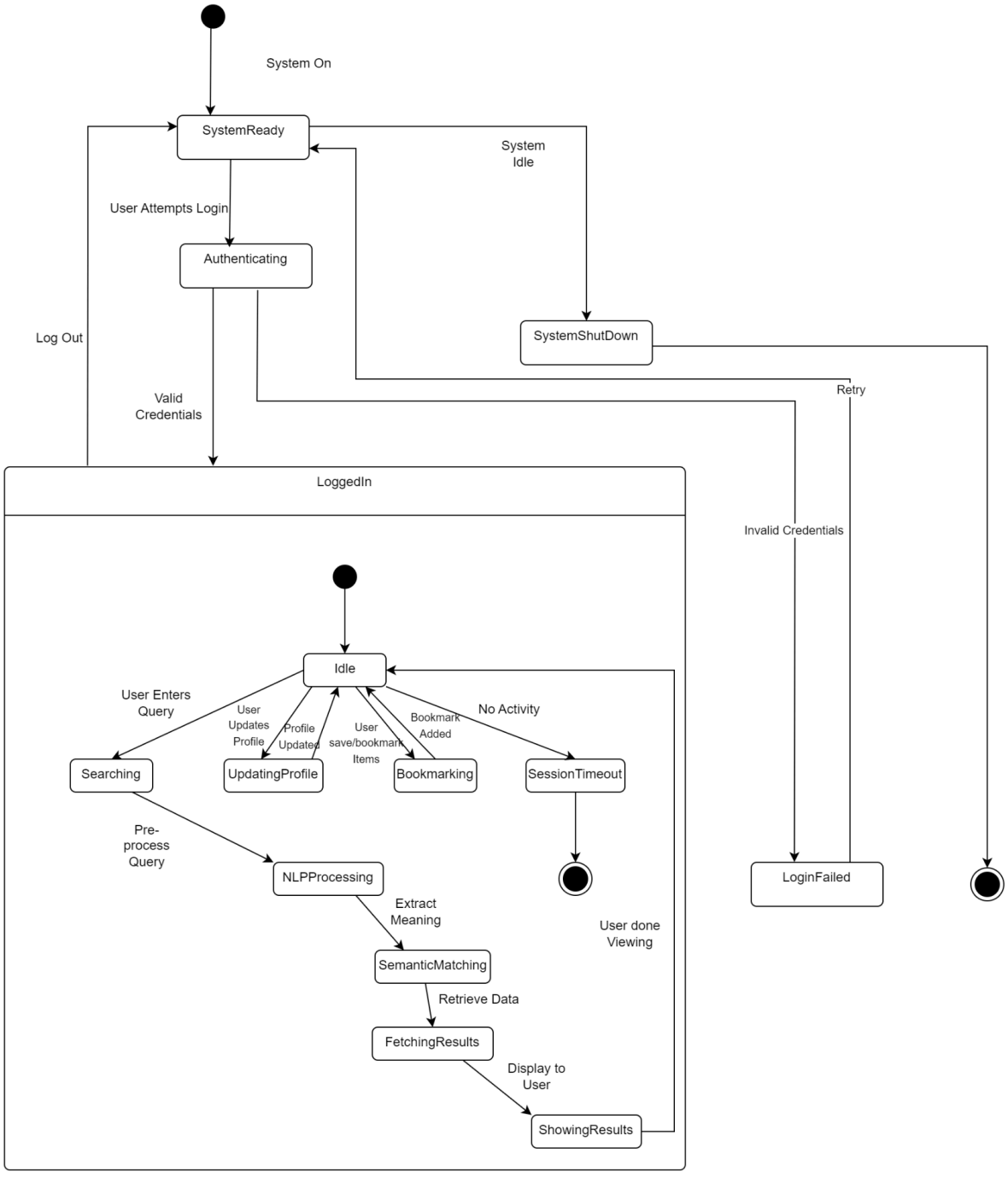


Figure 3.8.2 - Lawyer/Client State Chart Diagram

3.9. Data Model

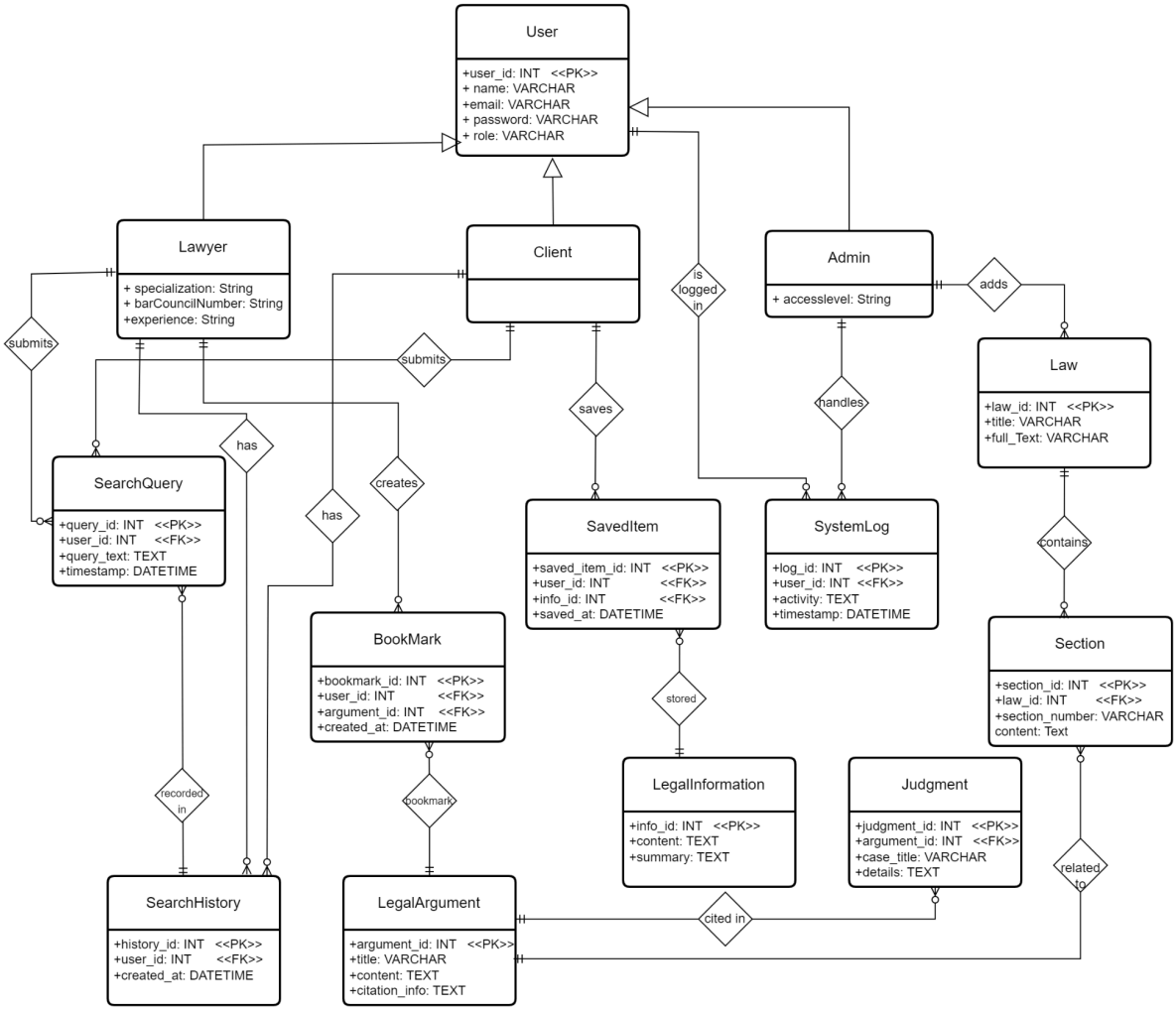


Figure 3.9 - System Data Model

Chapter 4: User Interface Design

4.1. Introduction

A user interface design consists of three main parts:

Page elements should be visualized on paper before building them in the computer. Just as you draw a site map to plan the site, use cartoons and storyboards to begin blocking out the site's appearance and navigational scheme.The design ensures smooth interaction between users and the AI-powered legal system.

1. Site maps
2. Storyboards
3. Navigational maps
4. Traceability Matrix

4.2. Site Maps

4.2.1 Landing Page

- Acts as the entry point of the system.
- Provides overview of the system, features, and navigation to login or registration.

4.2.2 Login Page

- Allows users (Admin, Lawyer, Client) to log in using credentials.
- Validates user and redirects to respective dashboards.

4.2.3 Admin Panel

- Provides administrative control over the system.
- Admin can manage users, update legal data, and monitor system activities.

4.2.4 Client Dashboard

- Main interface for clients.
- Allows searching legal queries in simple language.
- Displays simplified summaries and saved information.

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4.2.5 Lawyer Dashboard

- Main interface for lawyers.
- Allows advanced legal search with technical results, case references, and statutes.

4.2.6 Settings Page

- Central hub for managing user profile and account settings.

Sub-pages include:

a) Profile Page

- Allows users to update personal information.

b) Security Page

- Allows password change and account security settings.

4.2.7 Case Management Page (Optional Module)

- **For Lawyers**
  - Displays Case lists.
  - Lawyers can add, update, assign or delete cases.
- **For Clients**
  - Clients can see and track a list of personal cases with status labels.
  - Status can be in progress, pending or closed with results displayed.

4.2.8 Communication Page (Optional Module)

- Enables communication between lawyers and clients.
- Provides messaging or query-based interaction.

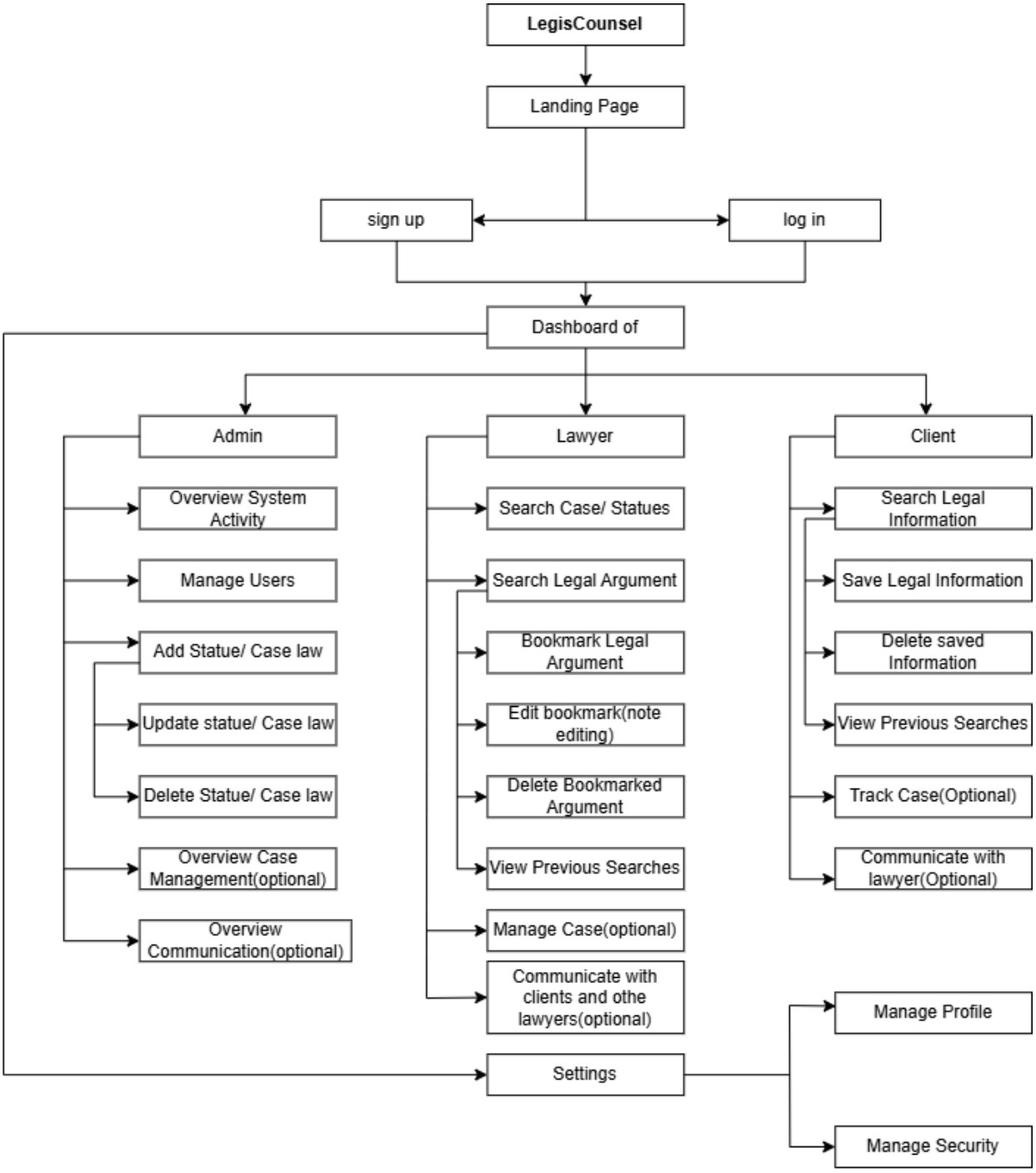


Figure 4.2.8 - System Specifications

4.3. Story boards

Storyboards are used to visually represent the interaction between the user and the system in a step-by-step manner. They illustrate how users navigate through different interfaces and how the system responds to user actions. In LegisCounsel, storyboards help in understanding the flow of activities such as login, legal search, result viewing, and profile management. These visual and descriptive sequences provide a clear view of user experience, system behavior, and overall interaction design.

4.3.1 Login

Table 4.3.1 - Login_Storyboard_Table

Element	Description
Environment	User accesses the system through a web browser on a laptop or mobile device.
Visual Cues	Login form with email, password fields, role and login button.
Audible Cues	No sound (optional notification sound on error/success).
Tactile Cues	User clicks input fields and login button.
User Input	User enters email, password, select role and clicks login.
Machine Output	System validates credentials and redirects to respective dashboard or shows error message.
User's Emotions	User expects quick and secure login; frustration if login fails.
Technology	React.js frontend, backend authentication API, database validation.
Quality of Experience	Fast login process improves usability and user satisfaction.

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4.3.2 Register

Table 4.3.2 - Register_Storyboard_Table

Element	Description
Environment	User accesses the system through a web browser on a laptop or mobile device.
Visual Cues	Registration form with fields like name, email, password, role and submit button.
Audible Cues	Optional alert sound for success or error messages.
Tactile Cues	User types into input fields and clicks the register button.
User Input	User enters personal details and submits the registration form.
Machine Output	System validates input, creates account, and shows success or error message.
User's Emotions	User expects easy and quick registration; satisfaction on success.
Technology	Frontend forms (React), backend validation, database storage.
Quality of Experience	Simple and fast registration improves user onboarding experience.

4.3.3 Admin Dashboard

Table 4.3.3 - Admin_Dashboard_Storyboard_Table

Element	Description
Environment	Admin accesses the system through a web browser after logging into the admin panel.
Visual Cues	Dashboard with system overview, user list, activity logs, and options to manage users and legal data (statutes/laws).

Audible Cues	Optional alert sounds for system notifications or error messages.
Tactile Cues	Clicking buttons, selecting options, and entering data in forms.
User Input	Admin views system performance, manages users, updates legal data, and monitors activity logs.
Machine Output	System displays system statistics, updates records, and confirms actions such as adding laws or managing users.
User's Emotions	Admin feels in control and confident while monitoring and managing the system.
Technology	Backend APIs, database operations, admin services, and dashboard UI components.
Quality of Experience	Provides centralized control and efficient system management.

4.3.4 Lawyer Legal Search(Technical View)

Table 4.3.4 - Lawyer_LegalSearch_Storyboard_Table

Element	Description
Environment	Lawyer accesses the system via dashboard using a web browser.
Visual Cues	Advanced search interface with detailed results including case references, statutes, and legal arguments.
Audible Cues	None (optional system alerts).
Tactile Cues	Typing query, clicking search button, navigating results.
User Input	Lawyer enters detailed legal query or searches specific laws/sections.
Machine Output	System retrieves and displays technical legal arguments, case laws, and references.

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User's Emotions	Lawyer feels confident and efficient while accessing accurate legal data.
Technology	VectorSearch, NLP models, backend APIs, database queries.
Quality of Experience	Enables fast, accurate, and professional legal research.

4.3.5 Client Legal Search(Simplified View)

Table 4.3.5 - Client_LegalSearch_Storyboard_Table

Element	Description
Environment	Client accesses the system through a web browser after logging into the dashboard.
Visual Cues	Chat interface with message bubbles, input field, and AI responses displayed like a conversation.
Audible Cues	Optional alert sound for errors or successful search.
Tactile Cues	Typing in input field and clicking the send button.
User Input	Client enters a simple query (e.g., "theft law in Pakistan").
Machine Output	System processes query using AI and displays simplified summaries of relevant legal information.
User's Emotions	Client feels relieved and satisfied when complex legal information is easy to understand.
Technology	NLP models, backend APIs, frontend UI rendering.
Quality of Experience	Provides easy access to legal knowledge for non-technical users.

4.3.6 Settings Management(Profile, Security)

Table 4.3.6 - Setting_Storyboard_Table

Element	Description
Environment	User accesses the system through a web browser after logging into the dashboard.
Visual Cues	Settings page with multiple sections/tabs such as Profile and Security. Each section contains editable fields and save buttons.
Audible Cues	Optional notification sound for successful update or error alerts.
Tactile Cues	User clicks tabs, types in input fields, toggles options, and presses save/update buttons.
User Input	User selects a settings category (e.g., Profile, Security), updates information (e.g., name, password, preferences), and submits changes.
Machine Output	System validates input, updates data in database, and displays confirmation or error message.
User's Emotions	User expects smooth and secure updates; satisfaction when changes are saved successfully.
Technology	Frontend UI (React), backend APIs for updating data, database storage, authentication for secure actions.
Quality of Experience	Centralized and easy-to-use settings improve user control, personalization, and overall system usability.

4.3.7 Lawyer Case Management(Optional)

Table 4.3.7 - LawyerCase_Storyboard_Table

Element	Description
Environment	Lawyer accesses system through dashboard using a web browser.

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Visual Cues	Case list with options to add, update, assign, or delete cases.
Audible Cues	Optional confirmation alerts for actions.
Tactile Cues	Clicking buttons, typing case details, selecting options.
User Input	Lawyer creates or updates case information and assigns clients.
Machine Output	System saves changes, updates case records, and displays confirmation message.
User's Emotions	Lawyer feels in control and efficient while managing cases.
Technology	Backend APIs, database operations, frontend forms.
Quality of Experience	Enables efficient management of multiple cases with ease.

4.3.8 Client Case Tracking(Optional)

Table 4.3.8 - ClientCase_Storyboard_Table

Element	Description
Environment	Client accesses the system through a web browser after logging in.
Visual Cues	List of personal cases with status labels (e.g., Pending, In Progress, Closed).
Audible Cues	Optional notification alerts for updates.
Tactile Cues	Scrolling and clicking on case items.
User Input	Client selects a case to view details.
Machine Output	System displays case status, updates, and assigned lawyer information.

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User's Emotions	Client feels informed and reassured about case progress.
Technology	Frontend UI, backend APIs, database retrieval.
Quality of Experience	Provides transparency and easy tracking of legal cases.

4.3.9 Communication(Optional)

Table 4.3.9 - Communication_Storyboard_Table

Element	Description
Environment	User uses communication module within system.
Visual Cues	Chat interface or message panel.
Audible Cues	Notification sounds (optional).
Tactile Cues	Typing and sending messages.
User Input	Sending message/query.
Machine Output	Displays message and response.
User's Emotions	Engagement and interaction satisfaction.
Technology	Real-time communication APIs (Socket/Web).
Quality of Experience	Enhances collaboration between users.

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4.4. Navigational maps:

- **Landing Page:**

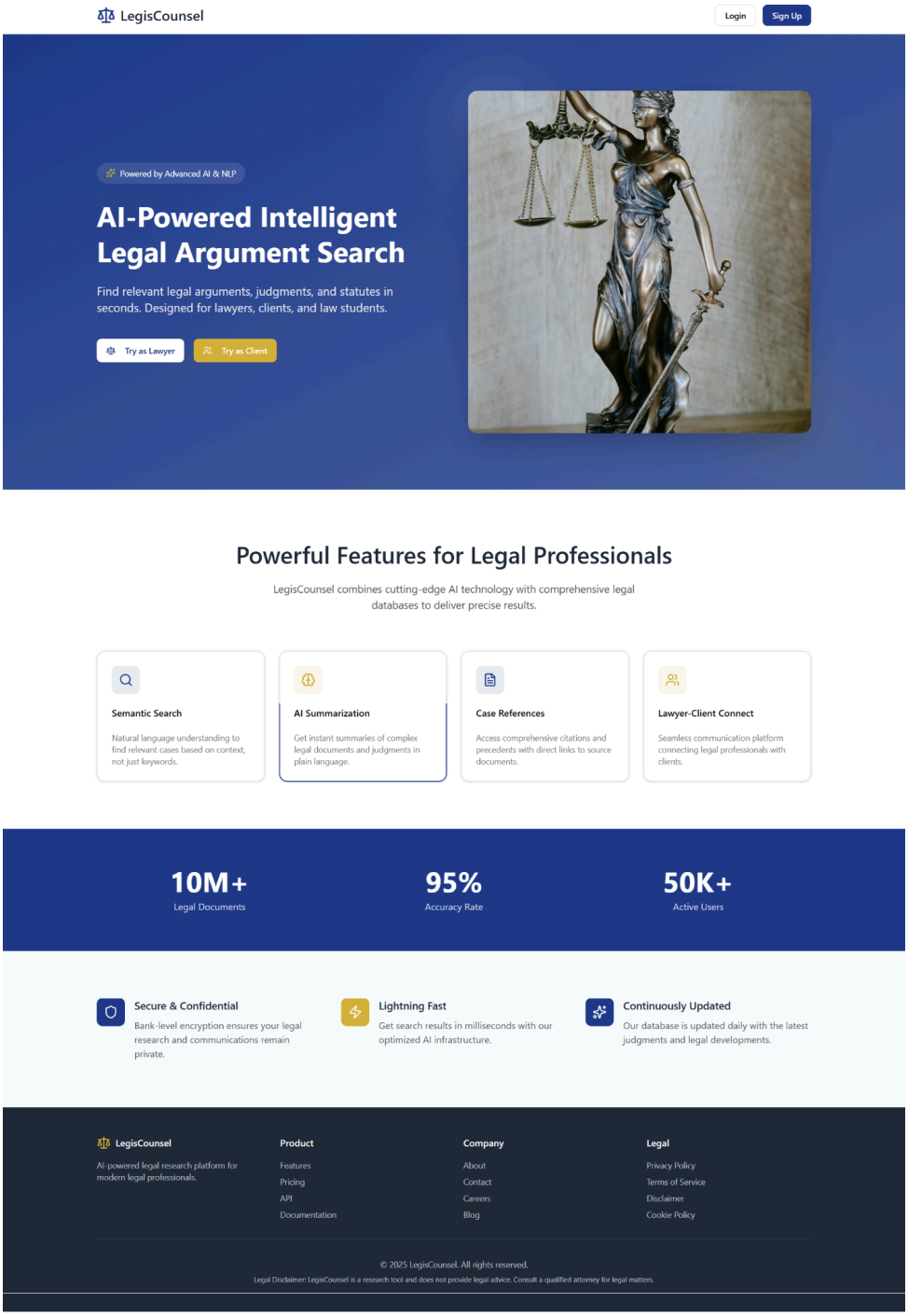


Figure 4.4.1 - Landing Page

- **Sign Up Page**

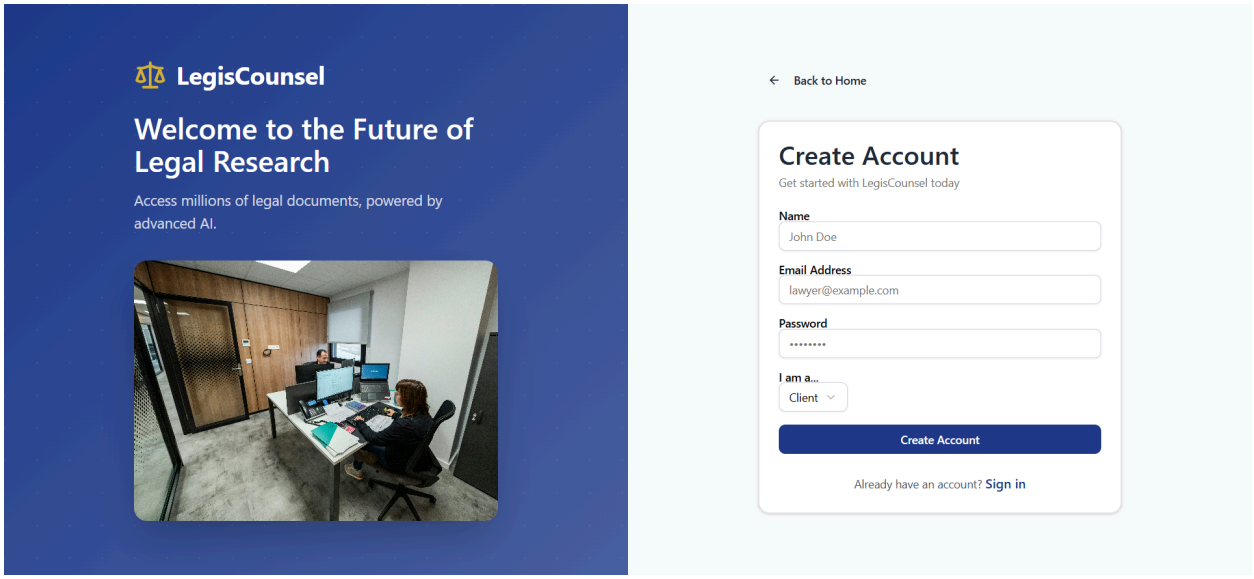


Figure 4.4.2 - Sign Up Page

- **Log In Page:**

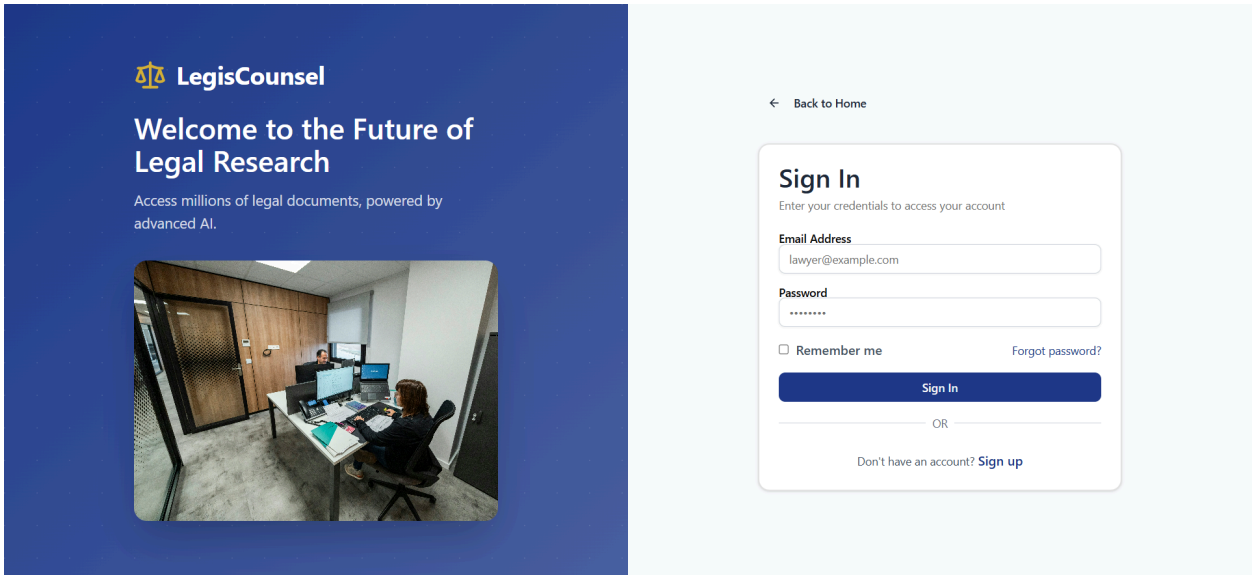


Figure 4.4.3 - Log In Page

- **Lawyer Dashboard**

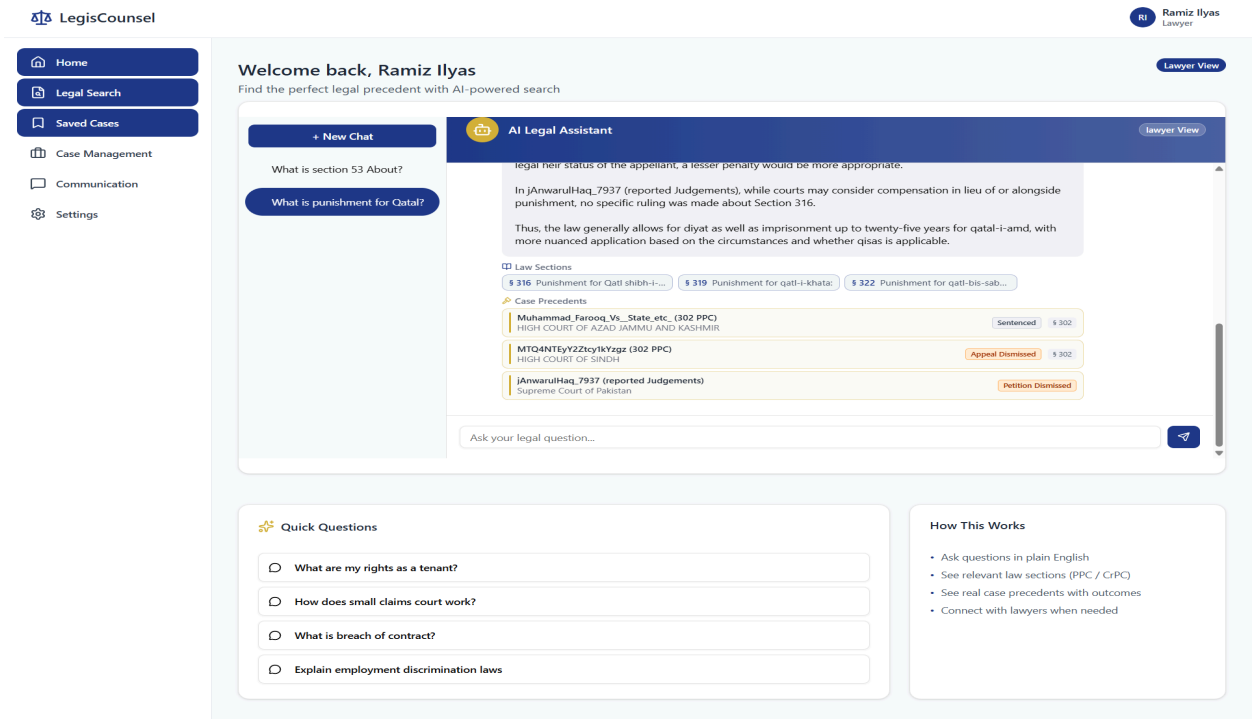


Figure 4.4.4 - Lawyer Dashboard

- **Client Dashboard:**

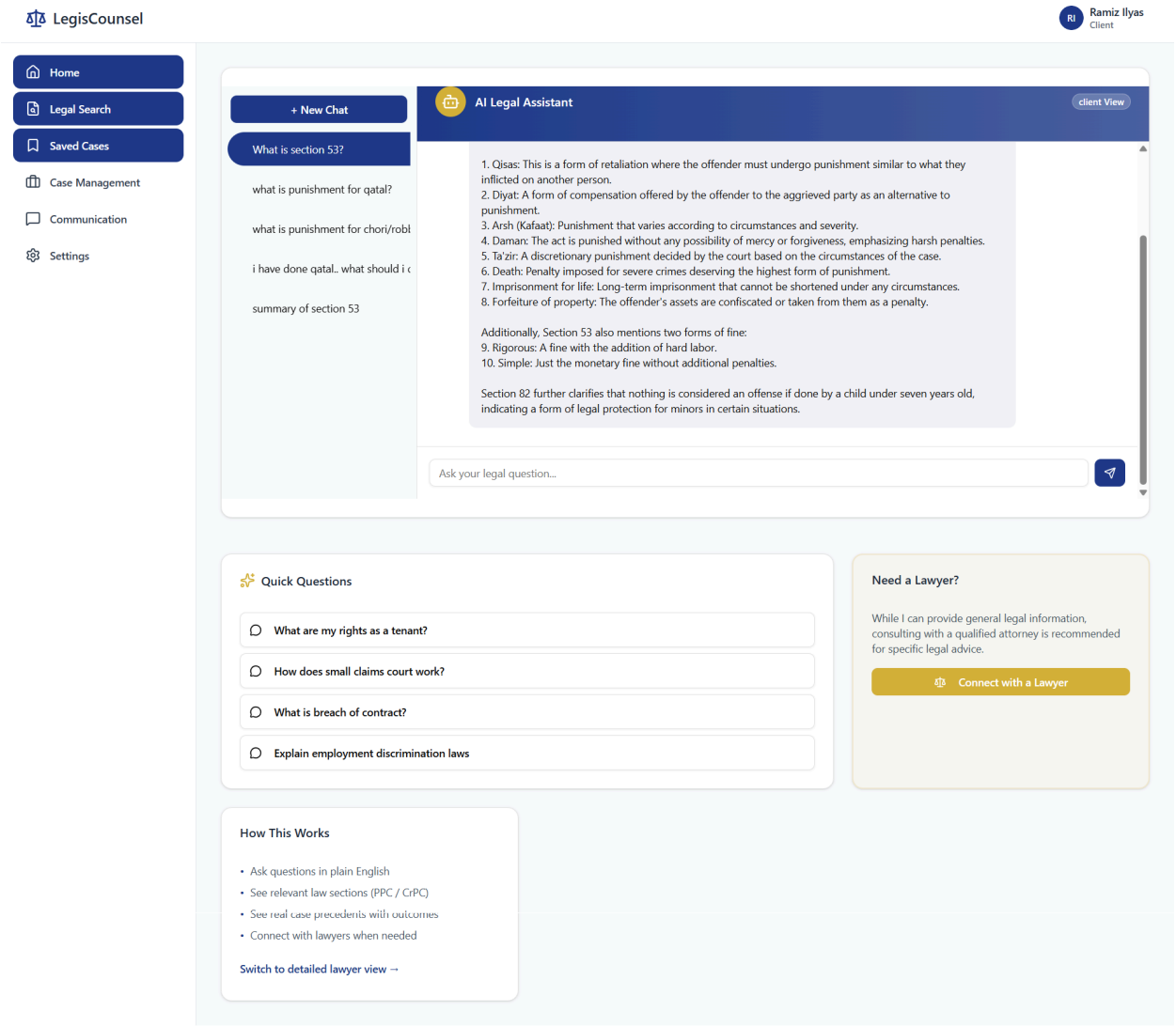


Figure 4.4.5 - Client Dashboard

- **Case Management:**

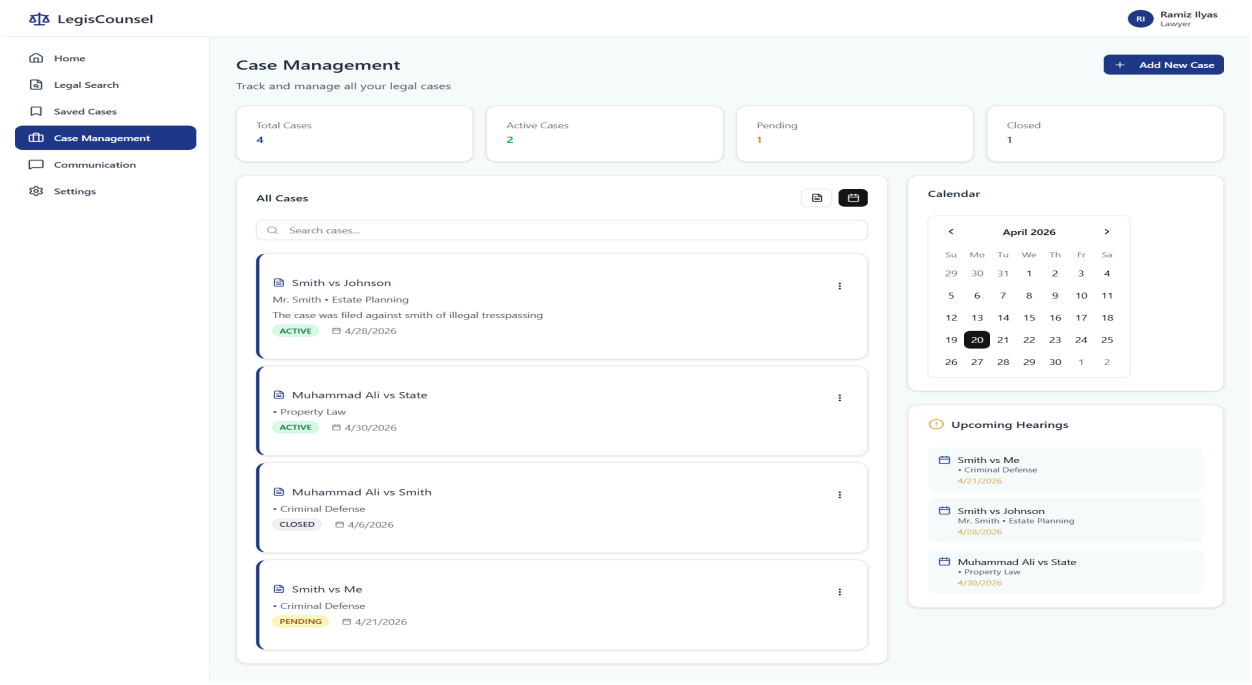


Figure 4.4.6 - Case Management Page

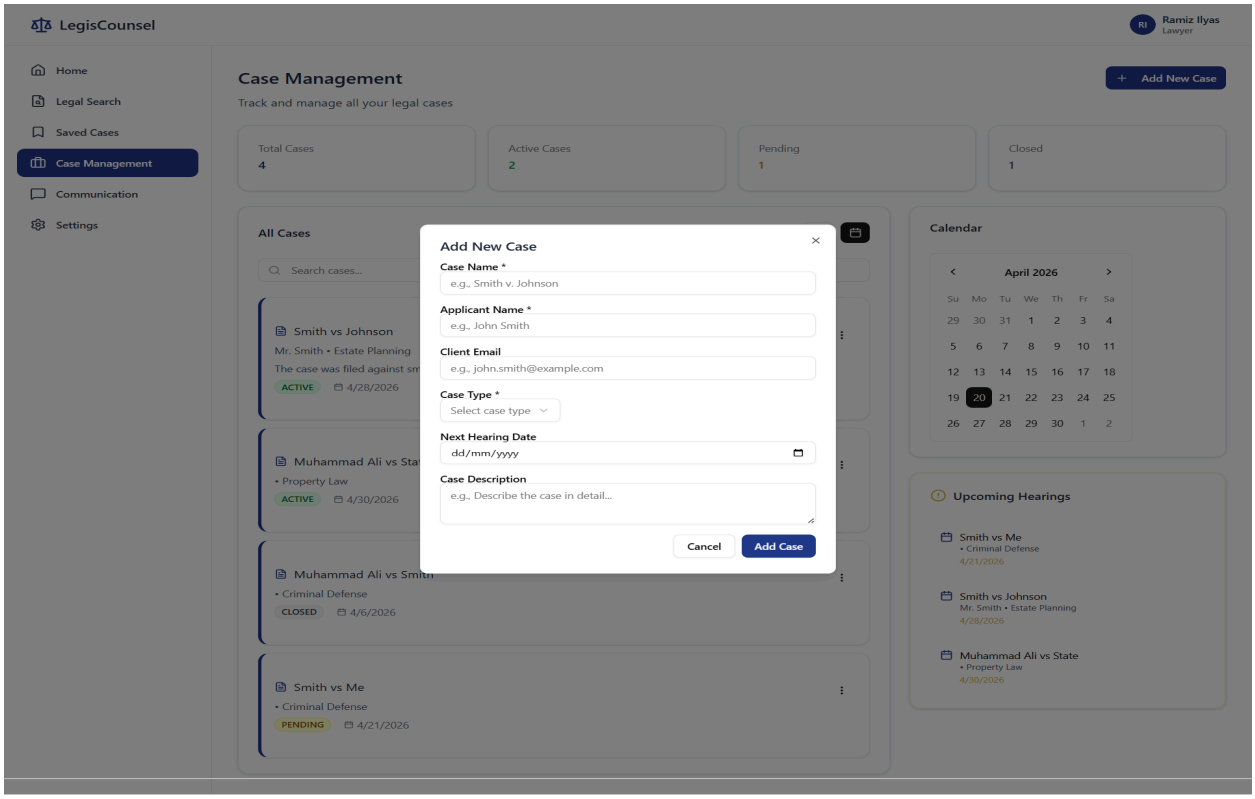


Figure 4.4.7 - Add Case Page

- **Communication Page:**

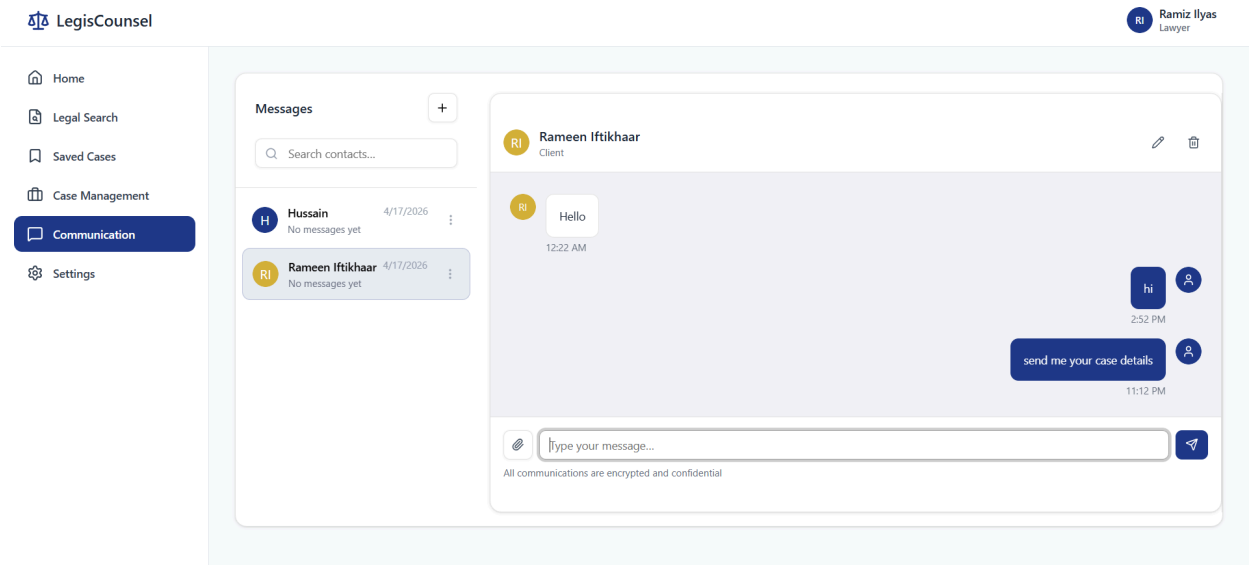


Figure 4.4.8 - Communication Page

- **Admin Dashboard:**

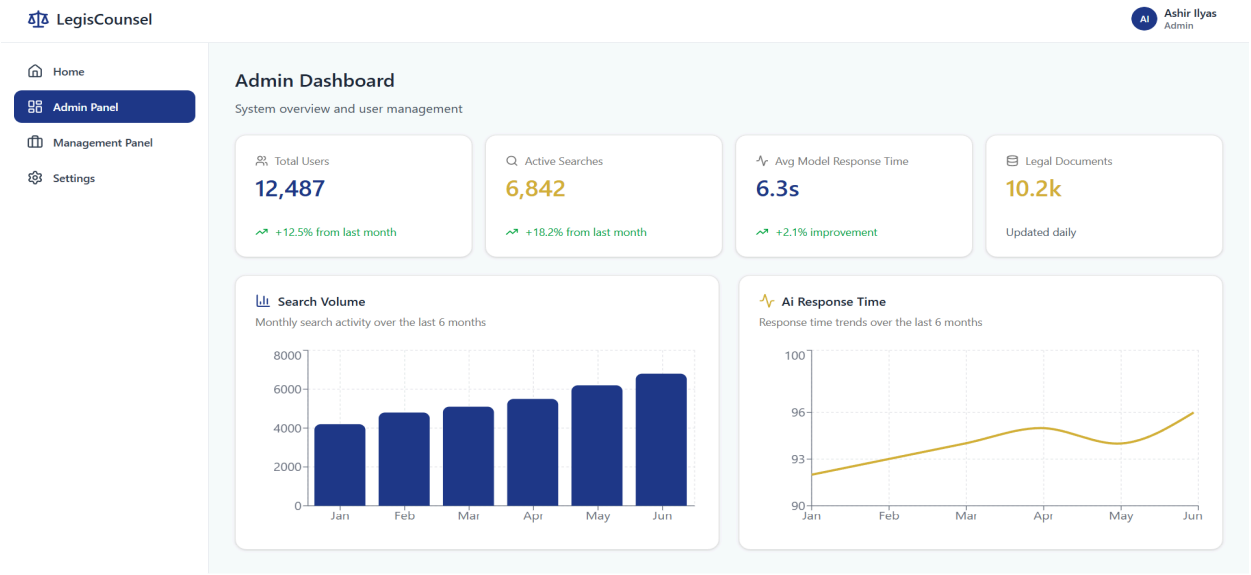


Figure 4.4.9 - Admin Dashboard

- **Management Panel:**

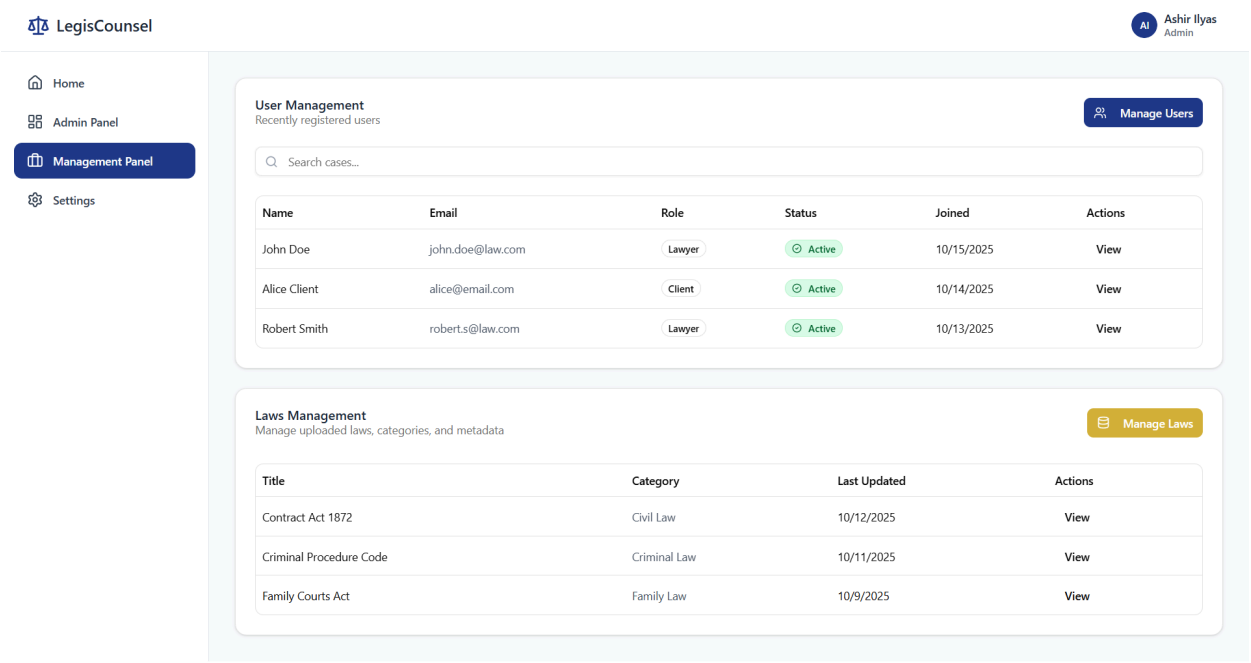


Figure 4.4.10- User & Law Management

- **Settings Page:**

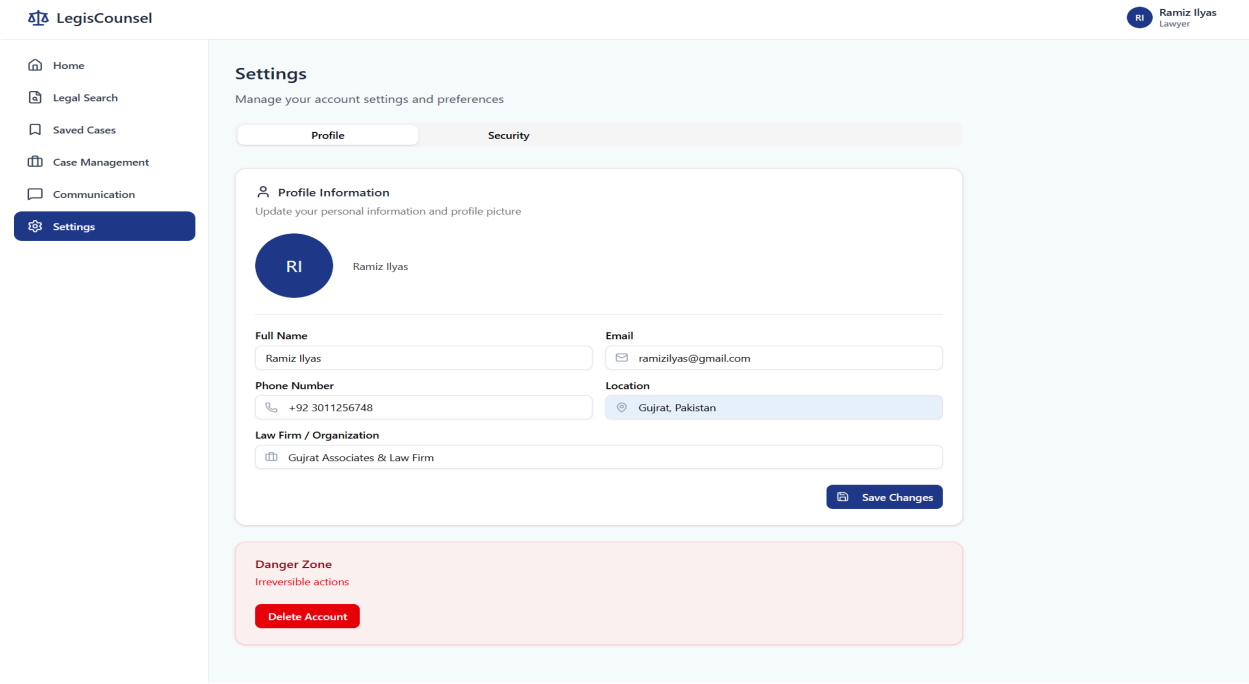


Figure 4.4.11 - Profile Management

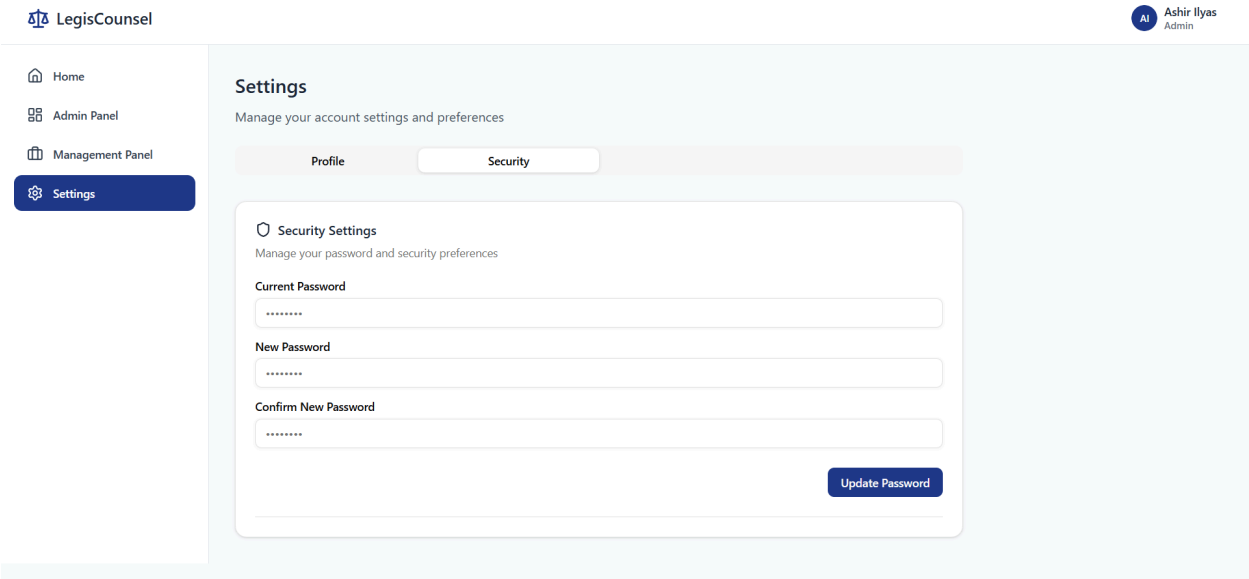


Figure 4.4.12 - Security

4.5 Trace-ability Matrix

4.5.1 Login

Table 4.5.1 - Login_Matrix

Field	Description
Use Case Id	UC-4
UI Id	UI-001
Priority	High
Use Case Cross Ref	UC-3, UC-10, UC-11, UC-22
DB Collection Id	users
Elaborated Use Case ID	UC-3
Dependent Classes	User

4.5.2 User Registration

Table 4.5.2 - User_Registration_Matrix

Field	Description
Use Case Id	UC-3
UI Id	UI-002
Priority	High
Use Case Cross Ref	UC-4
DB Collection Id	users
Elaborated Use Case ID	UC-3
Dependent Classes	User

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4.5.3 Lawyer Dashboard (Technical View)

Table 4.5.3 - Lawyer_Dashboard_Matrix

Field	Description
Use Case Id	UC-10
UI Id	UI-003-A
Priority	High
Use Case Cross Ref	UC-4, UC-26
DB Collection Id	messages, conversations, laws, judgements
Elaborated Use Case ID	UC-21
Dependent Classes	SearchQuery, Bookmark, TechnicalView, SearchHistory, AI_NLP_MODEL, Lawyer

4.5.4 Client Dashboard (Simplified View)

Table 4.5.4 - Client_Dashboard_Matrix

Field	Description
Use Case Id	UC-11
UI Id	UI-003-B
Priority	High
Use Case Cross Ref	UC-4, UC-26
DB Collection Id	conversations, messages
Elaborated Use Case ID	UC-21
Dependent Classes	SearchQuery, SimplifiedView, AI_NLP_MODEL, SearchHistory, Client

4.5.5 Settings

Table 4.5.5 - Settings_Matrix

Field	Description
Use Case Id	UC-18
UI Id	UI-004
Priority	Medium
Use Case Cross Ref	UC-10, UC-11, UC-22
DB Collection Id	users
Elaborated Use Case ID	UC-18
Dependent Classes	User

4.5.6Admin Dashboard

Table 4.5.6 - Admin_Dashboard_Matrix

Field	Description
Use Case Id	UC-22
UI Id	UI-005
Priority	Medium
Use Case Cross Ref	UC-4
DB Collection Id	users, laws, judgements
Elaborated Use Case ID	UC-20, UC-21, UC-25
Dependent Classes	Admin, User, SystemLog, Law, Statue

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4.5.7 Lawyer Case Management (Optional)

Table 4.5.7 - Lawyer_Case_Matrix

Field	Description
Use Case Id	UC-ManageCases
UI Id	UI-006-A
Priority	Low
Use Case Cross Ref	UC-AssignCase, UC-UpdateCase
DB Collection Id	cases
Elaborated Use Case ID	UC-ManageCases
Dependent Classes	Case, Lawyer, CaseService

4.5.8 Client Case Tracking (Optional)

Table 4.5.8 - Client_Case_Matrix

Field	Description
Use Case Id	UC-ViewCaseStatus
UI Id	UI-006-B
Priority	Low
Use Case Cross Ref	UC-SearchLegalInfo
DB Collection Id	cases
Elaborated Use Case ID	UC-ViewCaseStatus
Dependent Classes	Case, Client, CaseService

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4.5.9 Communication (Optional)

Table 4.5.9 - Communication_Matrix

Field	Description
Use Case Id	UC-Communication
UI Id	UI-007
Priority	Low
Use Case Cross Ref	UC-Message
DB Collection Id	chatmessages, chatconversations
Elaborated Use Case ID	UC-Communication
Dependent Classes	ChatService, Message

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5.1 Introduction:

This deliverable is based on the IEEE Standard of Software Testing i.e., IEEE SOFTWARE TEST DOCUMENTATION Std 829-1998. This standard describes a set of basic test documents associated with the dynamic aspects of software testing, specifically the execution of procedures and code. The standard defines the purpose, outline, and content of each basic document.

In this deliverable we will make test cases and test the LegisCounsel system according to those defined test cases. It includes the complete testing report with all test items, test logs, and summary reports.

LegisCounsel is an AI-powered legal argument search system developed for legal professionals and clients in Pakistan. The system integrates React.js (frontend), Node.js (backend), MongoDB (database), VectorSearch (semantic search engine), and Python-based NLP models to deliver relevant legal arguments, case law, and precedents through a dual-view interface — Lawyer View and Client View.

Following are standard artifacts, which must be included in this deliverable for the **LegisCounsel (AI-Powered Intelligent Legal Argument Searching System)**:

1. Test Plan
2. Test Design Specification
3. Test Case Specification
4. Test Procedure Specification
5. Test Item Transmittal Report
6. Test Log
7. Test Incident Report
8. Test Summary Report

5.2. Test plan

5.2.1. Purpose

The purpose of the test plan is to prescribe the scope, approach, resources, and schedule of the testing activities for the LegisCounsel system. It aims to identify the items being tested, the features to be tested, the testing tasks to be performed, the personnel responsible for each task, and the risks associated with this plan. The primary objective is to ensure that the AI-powered legal argument searching system functions correctly, delivers accurate semantic search results,

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provides appropriate dual-view outputs for lawyers and clients, and meets all functional and non-functional requirements specified in Chapters 1-3.

5.2.2. Outline

A test plan shall have the following structure:

a. Test plan identifier
b. Introduction
c. Test items
d. Features to be tested
e. Features not to be tested
f. Approach
g. Item pass/fail criteria
h. Suspension criteria and resumption requirements
i. Test deliverables
j. Testing tasks
k. Environmental needs
l. Responsibilities
m. Staffing and training needs
n. Schedule
o. Risks and contingencies
p. Approvals

The sections shall be ordered in the specified sequence. Additional sections may be included immediately prior to Approvals. If some or all of the content of a section is in another document, then a reference to that material may be listed in place of the corresponding content. The referenced material must be attached to the test plan or available to users of the plan.

Details on the content of each section are contained in the following sub-clauses.

5.2.2.1. Test Plan Identifier

The unique identifier assigned to this test plan is **TP-LGS-001**.

5.2.2.2. Introduction

This test plan is a comprehensive blueprint that aims at conducting the testing activities for the system **"LegisCounsel: AI-Powered Intelligent Legal Argument Searching System"**. The motivation behind this test plan is to ensure all required features are tested for their working and reliability. The purpose of this test plan is to verify that features this system provides are

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compatible with each other and to evaluate how the system reacts to various inputs provided by users including lawyers, clients, and administrators. The testing will also verify whether expected and actual outcomes are being met, particularly focusing on the accuracy of NLP-based semantic search, the quality of AI-generated summaries, and the proper functioning of dual-view result presentation.

Testing verifies all requirements defined in SRS and design documents.

5.2.2.3. Test Items

The following test items including their version/revision level are identified for testing in the LegisCounsel system:

Table 5.2.2.3 - Test Items

Test Item	Version/Revision	Description
User Registration	v1.0	Registration functionality for Lawyers, Clients, and Admin
User Login	v1.0	Authentication system with role-based access
Profile Management	v1.0	Update profile information for all user types
AI/NLP Query Processing	v1.0	Natural language processing of legal queries using transformer models
Semantic Search Engine	v1.0	Context-aware search on legal database using VectorSearch
Dual-View Result Presentation	v1.0	Technical view for lawyers, simplified view for clients
Legal Summarization	v1.0	Automatic summarization of complex legal judgments
Bookmark Management	v1.0	Save and manage bookmarked legal arguments for lawyers

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Save Legal Information	v1.0	Save helpful legal information for clients
Search History	v1.0	View and manage search history for both user types
Section-Based Search Module	v1.0	Search by specific legal sections (e.g., PPC 302)
Admin Management Module	v1.0	User management, legal database updates, system monitoring
Legal Database	v1.0	Repository of laws, statutes, and judgments
Vector Database	v1.0	Vector database with legal document embeddings for semantic search

Supply references to the following test item documentation:

- Requirements Specification
- Design Specification
- Users Guide

Items that are specifically excluded from testing are identified in Section 5.2.2.5.

5.2.2.4. Features to be Tested

All software features and combinations of software features to be tested are identified below:

Table 5.2.2.4.1 - Features to be Tested

Feature ID	**Feature Name**	**Associated Use Case**	**Priority**	**Test Design Specification Reference**
F-01	User Registration	UC-Register	Highest	TDS-001
F-02	User Login	UC-Login	Highest	TDS-001
F-03	Lawyer Natural Language Search	UC-SearchArguments	Highest	TDS-002
F-04	Client Simple Language Search	UC-SearchLegalInfo	Highest	TDS-002
F-05	Semantic Search Processing	UC-SemanticSearch	Highest	TDS-002
F-06	AI/NLP Query Processing	UC-ProcessQueries	Highest	TDS-002
F-07	Technical View Display for Lawyers	UC-DisplayTechnicalView	Highest	TDS-003
F-08	Simplified View Display for Clients	UC-DisplaySimplifiedView	Highest	TDS-003
F-09	View Legal References	UC-ViewLegalReferences	Highest	TDS-003

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F-10	View Simplified Summaries	UC-ViewSimplifiedSummaries	Highest	TDS-003
F-11	Legal Judgment Summarization	UC-SummarizeJudgments	Medium	TDS-003
F-12	Section-Based Legal Search	UC-SearchBySections	Highest	TDS-002
F-13	Bookmark Legal Arguments	UC-BookmarkArguments	Medium	TDS-004
F-14	Save Legal Information	UC-SaveLegalInfo	Medium	TDS-004
F-15	View Search History (Lawyer)	UC-ViewSearchHistory	Medium	TDS-004
F-16	View Previous Searches (Client)	UC-ViewPreviousSearches	Medium	TDS-004
F-17	Update Profile	UC-UpdateProfile	Lowest	TDS-001
F-18	Secure Logout	UC-Logout	Lowest	TDS-001
F-19	Admin User Management	UC-ManageUsers	Lowest	TDS-005
F-20	Admin Update Legal Database	UC-UpdateLaws	Lowest	TDS-005

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F-21	Admin System Monitoring	UC-MonitorSystem	Lowest	TDS-005
F-22	User Authentication	UC-AuthenticateUser	Highest	TDS-001
F-23	Input Validation	UC-ValidateInput	Lowest	TDS-001
F-24	Role-Based Access Control	UC-ManageAccess	Lowest	TDS-001

Explanation of Each TDS Reference:

Table 5.2.2.4.2 - TDS Reference

Reference	Features Covered	Section in Document
TDS-001	**Test Design Specification for User Management & Authentication Features:** User Registration, Login, Authentication, Profile Update, Logout, Input Validation, Role-Based Access Control	Section 5.3
TDS-002	**Test Design Specification for Search & AI Processing Features:** Lawyer Search, Client Search, Semantic Search, AI/NLP Processing, Section-Based Search	Section 5.3

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TDS-003	**Test Design Specification for Result Presentation & Summarization Features:** Technical View, Simplified View, Legal References, Simplified Summaries, Summarization	Section 5.3
TDS-004	**Test Design Specification for Bookmark, Save & History Features:** Bookmark, Save, Search History (Lawyer), Previous Searches (Client)	Section 5.3
TDS-005	**Test Design Specification for Admin Management Features:** Admin User Management, Update Legal Database, System Monitoring	Section 5.3

5.2.2.5. Features not to be Tested

The following features and significant combinations of features will not be tested and the reasons are provided:

Table 5.2.2.5 - Features not to be Tested

Feature	**Reason for Exclusion**
Billing and Payment Systems	Out of scope as per project scope definition in Chapter 1, Section 1.3
Live Video/Audio Consultations	Out of scope – identified as optional future enhancement
Case Management and Litigation Support	Out of scope – not part of core system functionality
Mobile Application	Out of scope – future enhancement

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Civil, Corporate, Family Law Domains	Out of scope – system initially limited to Criminal Law domain only

5.2.2.6. Approach

The overall approach to testing LegisCounsel follows a comprehensive strategy combining both manual and automated testing techniques. The testing approach is described in sufficient detail to permit identification of major testing tasks and estimation of time required for each.

Testing Levels:

1. **Unit Testing:** Individual components such as AI/NLP processing functions, search algorithms, and summarization modules will be tested in isolation using Jest and Python unit test frameworks.
2. **Integration Testing:** Interaction between frontend (React.js) and backend (Node.js) APIs, database connectivity, and VectorSearch integration will be tested to ensure seamless data flow.
3. **System Testing:** End-to-end testing of complete workflows including user registration, natural language query processing, semantic search execution, and result presentation will be conducted.
4. **Acceptance Testing:** Real-world legal queries will be tested by sample users (lawyers and clients) to validate the accuracy and usefulness of search results.

Testing Techniques:

- **Black Box Testing:** For user-facing features such as registration, login, search interfaces, and result display
- **White Box Testing:** For AI/NLP model integration and semantic search algorithms
- **Regression Testing:** After each bug fix or feature addition to ensure existing functionality remains intact
- **Performance Testing:** To verify search response times meet the requirement of 5-10 seconds

Comprehensiveness Criteria:

- All functional requirements identified in Chapter 2, Section 2.1.4 will be tested
- Minimum 90% code coverage for critical modules (AI processing, search engine)
- All use cases from UC-1 to UC-26 will be validated

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Traceability:

Requirements will be traced to test cases using the Requirements Traceability Matrix defined in Chapter 2, Section 2.1.7.

5.2.2.7. Item Pass/Fail Criteria

For the LegisCounsel system to be categorized as complete and working, it must perform according to the specified requirements. The pass/fail criteria for each test item are as follows:

Table 5.2.2.7 - Item Pass/Fail Criteria

Category	Pass Criteria	Fail Criteria
Functional Requirements	All required features produce expected output matching specifications; expected output equals actual output	Any required feature produces unexpected output, crashes, or returns errors not handled gracefully
AI/NLP Semantic Search	Search results are contextually relevant to the query; top 5 results have relevance score >70%	Search results are irrelevant or unrelated to query; no results returned for valid queries
Dual-View Presentation	Lawyers see technical references with citations; Clients see simplified language summaries	Wrong view presented to user type; missing citations; overly complex language for clients
Summarization	Summaries accurately capture key points of judgments; length appropriate (100-200 words)	Summaries miss key points; contain errors; too long or too short
Response Time	Search results returned within 5-10 seconds	Search takes more than 10 seconds
Security	Unauthorized access attempts blocked; role-based access enforced	Unauthorized users can access restricted features

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User Experience	Interface is responsive; error messages clear and helpful	Interface crashes; error messages are unclear or absent

5.2.2.8. Suspension Criteria and Resumption Requirements

Testing will be suspended under the following conditions:

Table 5.2.2.8 - Suspension Criteria and Resumption Requirements

Suspension Criteria	**Description**
Critical System Failure	Database or backend server becomes unavailable
Network Connectivity Issues	Internet connection lost, affecting cloud-based AI model access
High Failure Rate	More than 30% of test cases in a test suite fail
Security Vulnerability Discovery	Critical security flaw found that could compromise user data
AI Model Unavailability	NLP/transformer model service is down or unresponsive

Resumption Requirements:

Testing will resume when:

- The underlying issue causing suspension is resolved
- Failed test cases are analyzed and fixes are implemented
- A new build with fixes is deployed to the testing environment
- The testing environment is verified to be stable and accessible
- All team members are notified of testing resumption

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5.2.2.9. Test Deliverables

The following deliverables are required to perform testing and to know what the tests have resulted in:

a. Test plan;

b. Test design specifications;

c. Test case specifications;

d. Test procedure specifications;

e. Test item transmittal reports;

f. Test logs;

g. Test incident reports;

h. Test summary reports.

5.2.2.10. Testing Tasks

The following set of tasks is necessary to prepare for and perform testing:

Table 5.2.2.10 - Testing Tasks

Task ID	Task Name	Description	Dependencies	Estimated Time	Required Skills
TT-01	Test Plan Preparation	Create test plan document	None	3 days	Documentation, Testing Knowledge
TT-02	Test Environment Setup	Configure hardware, software, VectorSearch, and network for testing	TT-01	2 days	DevOps, System Administration

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TT-03	Test Data Preparation	Create sample legal documents, user accounts, test queries	TT-02	2 days	Legal Knowledge, Data Entry
TT-04	Test Case Development	Write detailed test cases for all features	TT-01, TT-03	3 days	Testing, Analytical Skills
TT-05	Unit Testing Execution	Run unit tests on individual components	TT-04	3 days	Programming, Testing
TT-06	Integration Testing Execution	Test component interactions	TT-05	3 days	Testing, System Knowledge
TT-07	System Testing Execution	End-to-end testing of complete workflows	TT-06	4 days	Testing, Domain Knowledge
TT-08	Performance Testing	Measure response times and system load handling	TT-07	2 days	Performance Testing Tools
TT-09	Bug Reporting and Tracking	Document and track defects	TT-05, TT-06, TT-07	Continuous	Issue Tracking Tools
TT-10	Regression Testing	Re-test after bug fixes	TT-08, TT-09	2 days	Testing

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TT-11	Test Summary Report	Document final testing results	TT-10	2 days	Documentation, Analysis

5.2.2.11. Environmental Needs

The necessary and desired properties of the test environment for LegisCounsel are specified below:

Hardware Requirements:

Table 5.2.2.11.1 - Hardware Requirements

Component	**Minimum Specification**	**Recommended Specification**
Development Server	8GB RAM, 4-core CPU, 100GB Storage	16GB RAM, 8-core CPU, 256GB SSD
Client Machine (Tester)	4GB RAM, Core i3 processor	8GB RAM, Core i5 or higher
Network	Broadband connection, 10 Mbps	Fiber connection, 50 Mbps
Mobile Devices (Optional)	Android 10+ or iOS 14+	Latest Android/iOS devices

Software Requirements:

Table 5.2.2.11.2 - Software Requirements

Component	**Specification**
Operating System	Windows 10/11, macOS, or Linux (Ubuntu 20.04+)
Browser	Google Chrome (latest), Mozilla Firefox (latest), Microsoft Edge (latest)
Backend Runtime	Node.js v18 or higher

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Frontend Framework	React.js v18
Database	MongoDB v6.0 or higher
Search Engine	VectorSearch
AI/NLP Environment	Python 3.9+, TensorFlow/PyTorch, Ollama or GPT API access
IDE	Visual Studio Code, PyCharm
Version Control	Git, GitHub
Testing Tools	Jest, Postman, Selenium (optional)

Security Requirements:

- Test environment must be isolated from production
- User data must be anonymized for testing
- SSL/TLS certificates for secure communication testing
- Role-based access control testing requires multiple test accounts

Other Needs:

- Access to sample legal documents (publicly available judgments and statutes)
- Cloud service account for AI model hosting
- Documentation of all test environment configurations

5.2.2.12. Responsibilities

The following groups and individuals are responsible for managing, designing, preparing, executing, witnessing, checking, and resolving testing activities:

Table 5.2.2.12 - Responsibilities

Responsibility	**Assigned To**	**Description**
Test Management	Ramiz Ilyas	Overall coordination, schedule management, test plan approval

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Test Design (User Modules)	Nayab Gul	Test cases for registration, login, profile, search history, bookmark
Test Design (AI/Search Modules)	Ramiz Ilyas	Test cases for NLP processing, semantic search, section-based search, summarization
Test Design (Admin Modules)	Reham Batool	Test cases for user management, legal database updates, system monitoring
Test Execution (Unit)	Ramiz Ilyas	Unit tests for frontend, AI modules, search components
Test Execution (Integration)	Reham Batool	API testing, database connectivity, VectorSearch integration
Test Execution (System)	Nayab Gul	End-to-end testing for lawyer, client, and admin roles
Test Execution (Performance)	Ramiz Ilyas	Response time measurement and load testing
Bug Tracking	All Team Members	Reporting and tracking defects
Bug Fixing (Frontend)	Ramiz Ilyas	Resolving UI and frontend issues
Bug Fixing (Backend)	Reham Batool	Resolving API and database issues
Bug Fixing (AI/ML)	Nayab Gul, Ramiz Ilyas	Resolving search accuracy and summarization issues

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Test Witnessing	Mr. Muhammad Ejaz	Observing critical test executions
Quality Assurance	Nayab Gul	Ensuring adherence to testing standards
Environment Setup	Ramiz Ilyas	Configuring MongoDB, VectorSearch, AI services

5.2.2.13. Staffing and Training Needs

Testing doesn't require any specific formal training, but team members must be knowledgeable with the LegisCounsel system's modules and procedures. To guarantee efficient and comprehensive testing, they should be familiar with test case execution, defect identification, and reporting protocols.

5.2.2.14. Schedule

The test milestones identified in the software project schedule are as follows:

Table 5.2.2.14 - Schedule

Sr.No.	**Testing Phase**	**Duration**
1	Test Planning and Specification	3 days
2	Test Environment Setup	2 days
3	Test Data Preparation and Test Case Development	3 days
4	Unit Testing	3 days
5	Integration Testing	3 days
6	System Testing	4 days
7	Performance Testing	2 days

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8	Regression Testing and Bug Fixes	3 days
9	Test Summary Report and Documentation	2 days
Total		**25 days**

5.2.2.15. Risks and Contingencies

The high-risk assumptions of the test plan and contingency plans for each are identified below:

Table 5.2.2.15 - Risks and Contingencies

Risk ID	Risk Description	Probability	Impact	Contingency Plan
R-01	AI/NLP model may not fully understand complex legal language	Medium	High	Implement fallback keyword search; train model with additional legal corpus; use hybrid search approach
R-02	GPU/high-end system requirements for AI model training may not be available	Medium	Medium	Use cloud-based GPU services (Google Colab, AWS); optimize model for CPU inference
R-03	Limited availability of structured Pakistani legal data for testing	High	Medium	Use publicly available judgments from Lahore High Court, Supreme Court; create synthetic test data
R-04	Integration delays between AI components and backend APIs	Medium	Medium	Prioritize API-first development; use mock services for initial testing

R-05	Poor internet connectivity affecting cloud-based AI services	Medium	Low	Cache frequently accessed legal documents; implement offline capability for basic search
R-06	Test environment setup takes longer than expected	Low	Medium	Use containerized environments (Docker) for consistent setup; prepare environment checklist
R-07	Security vulnerabilities discovered during testing	Medium	High	Dedicated security testing phase; implement security best practices; conduct code review
R-08	Performance issues with semantic search	Medium	Medium	Optimize vector database queries; implement query caching; consider search result pagination

5.2.2.16. Approvals

This test plan has been reviewed and approved by the project supervisor.

Name: **Mr. Muhammad Ejaz**

Title: Supervisor

Signature: _____

5.3. Test Design Specification

5.3.1. Purpose

To prescribe the scope, approach, resources, and schedule of the testing activities. To identify the items being tested, the features to be tested, the testing tasks to be performed, the personnel responsible for each task, and the risks associated with this plan.

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5.3.2. Outline

A test design specification shall have the following structure:

a. Test design specification identifier
b. Introduction
c. Test items
d. Features to be tested
e. Features not to be tested
f. Approach
g. Item pass/fail criteria
h. Suspension criteria and resumption requirements
i. Test deliverables
j. Testing tasks
k. Environmental needs
l. Responsibilities
m. Staffing and training needs
n. Schedule
o. Risks and contingencies
p. Approvals

All of the points in the outline have already been discussed in detail in the previous section of Test Plan (Section 5.2). The test design approach for LegisCouncel follows the comprehensive strategy documented in Section 5.2.2.6, with specific focus on testing the AI-powered semantic search functionality, dual-view result presentation, and legal document summarization features.

5.4. Test Case Specification

5.4.1. Purpose

The test cases stated here are specified according to the test design specifications. They are used as a form of guidance for the tester, as they will know what is expected of tested item. It is a systematic guide that cross-refers to the use cases involved against the feature being tested for the LegisCounsel system.

5.4.2. Test Case Specification Identifier

The test case specification identifier is **TCS-LGS-001**.

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5.4.3. Outline

A test case specification shall have the following structure for each test case:

- Test Case Name
- Test Case ID
- Tester
- Date
- Req-ID
- UC-Name
- Test Data
- Expected Output
- Actual Output
- Status

For User Registration:

Test Case Name: Register New User (Lawyer)

Test Case ID: TC-LGS-001

Tester: Nayab Gul

Date: March 02, 2026

Table 5.4.3.1 - Register Lawyer User

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
REG-01	Register new lawyer	Enter valid credentials	User shall not exist before with given email	1. Enter Email 2. Enter Name 3. Enter Password 4. Select Role "Lawyer" 5. Click Sign Up	Email: lawyer1@test.com Name: Ahmad Raza Password: Pass@123	User registered successfully; redirect to login page	User registered successfully; redirect to login page	Pass
REG-02	Register new lawyer with existing email	Enter already registered email	User account exists with given email	1. Enter Email (already registered) 2. Enter valid	Email: lawyer1@test.com	Error message: "Email already registered"	Error message displayed: "Email already registered"	Pass

				details 3. Click Sign Up				
REG-03	Register new lawyer with invalid credentials	Enter invalid or empty data	None	1. Leave required fields empty 2. Click Sign Up	Name: (empty) Email: (invalid format)	Error messages for each invalid field	Error messages: "Name required", "Valid email required"	Pass
REG-04	Register new client	Enter valid credentials	User shall not exist before	1. Enter Email 2. Enter full name 3. Select Role "Client" 4. Enter Password 5. Click Sign Up	Email: client1@test.com Name: Fatima Khan Role: Client Password: Client@123	Client registered successfully; verification email sent	Client registered successfully; redirect to client dashboard after login	Pass

For User Login:

Test Case Name: Login User

Test Case ID: TC-LGS-002

Tester: Nayab Gul

Date: March 05, 2026

Table 5.4.3.2 - Login User

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
LOG-01	Login with valid credentials	Enter correct email and password	User account exists and is verified	1. Enter Email 2. Enter Password 3. Click Login	Email: lawyer1@test.com Password: Pass@123	Login successful; redirect to lawyer dashboard	Login successful; lawyer dashboard displayed	Pass

LOG-02	Login with invalid password	Enter correct email, wrong password	User account exists	1. Enter Email 2. Enter incorrect Password 3. Click Login	Email: lawyer1@test.com Password: WrongPass123	Error message: "Invalid email or password"	Error message displayed; stays on login page	Pass
LOG-03	Login with non-existent email	Enter email not registered	None	1. Enter unregistered email 2. Enter any password 3. Click Login	Email: nonexistent@test.com Password: anypass	Error message: "Invalid email or password"	Error message displayed	Pass
LOG-04	Login with empty fields	Leave fields empty	None	1. Leave email empty 2. Leave password empty 3. Click Login	Email: (empty) Password: (empty)	Error messages: "Email required", "Password required"	Both error messages displayed	Pass
LOG-05	Role-based dashboard redirection	Login as different user types	Accounts exist for each role	1. Login as Lawyer 2. Logout 3. Login as Client 4. Login as Admin	Lawyer credentials Client credentials Admin credentials	Lawyer sees technical dashboard; Client sees simplified dashboard; Admin sees admin panel	Correct dashboards displayed for each role	Pass

For AI-Powered Semantic Search (Lawyer):

Test Case Name: Legal Argument Search (Natural Language Query)

Test Case ID: TC-LGS-003

Tester: Ramiz Ilyas

Date: March 10, 2026

Table 5.4.3.3 - Legal Argument Search - Lawyer

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected	Actual	Status

						Output	**Output**	
SRCH-01	Search with valid natural language query	Enter legal query in natural language	Lawyer logged in; legal database populated	1. Enter query in search bar 2. Click Search 3. View results	"What are the grounds for bail in murder cases under PPC?"	System processes query using AI/NLP; returns relevant legal arguments with citations; top 5 results relevance >70%	Relevant legal arguments with PPC sections displayed; citations provided	Pass
SRCH-02	Search with specific legal terminology	Enter query with legal terms	Lawyer logged in	1. Enter query 2. Click Search	"Section 302 PPC qatl-e-amd punishment"	Returns technical legal references with case law and punishment details	Section 302 details displayed with case precedents	Pass
SRCH-03	Search with ambiguous query	Enter vague or ambiguous query	Lawyer logged in	1. Enter ambiguous query 2. Click Search	"murder" (single word)	System returns multiple relevant results; may suggest clarification	Results for murder-related cases displayed; no clarification prompt (enhancement needed)	Pass
SRCH-04	Search with no results	Enter query with no matches	Lawyer logged in	1. Enter query with no matches 2. Click Search	"xyz legal concept not in database"	Message: "No matching legal arguments found"	Message displayed: "No matching legal arguments found"	Pass
SRCH-05	Search with empty query	Submit empty search	Lawyer logged in	1. Leave search bar empty 2. Click Search	(empty)	Error message: "Please enter a search query"	Error message displayed	Pass
SRCH-06	View technical legal references	Click on search result	Lawyer logged in; search results displayed	1. Perform search 2. Click on any result 3. View detailed technical view	Any search result	System displays detailed technical view with: laws, statutes, citations, case numbers, judgments	Technical view displays all required legal references	Pass

For Simplified Legal Search (Client):

Test Case Name: Legal Information Search (Simplified Query)

Test Case ID: TC-LGS-004

Tester: Reham Batool

Date: March 11, 2026

Table 5.4.3.4 - Simplified Legal Search - Client

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
CLI-01	Search with simple language	Enter query in everyday language	Client logged in; legal database populated	1. Enter simple query 2. Click Search 3. View simplified results	"What happens if someone kills another person in Pakistan?"	System processes query; returns simplified summary; easy-to-understand language; no complex legal jargon	Simplified summary displayed with key points in plain language	Pass
CLI-02	Search for specific legal concept	Enter query about legal right	Client logged in	1. Enter query 2. Click Search	"What are my rights if arrested?"	Simplified explanation of rights; relevant laws mentioned in simple terms	Rights explained in simple language with references	Pass
CLI-03	View simplified summary of judgment	Select judgment summary option	Client logged in; search performed	1. Perform search 2. Click "View Simplified Summary" on a judgment	Any judgment result	System generates concise summary capturing key points	Summary generated with key points; appropriate length	Pass
CLI-04	Save legal information	Click save button on result	Client logged in; search results displayed	1. Perform search 2. Click save icon on result 3. Go to saved items	Any search result	Legal information saved; confirmation message; appears in saved items collection	Information saved; confirmation shown; appears in saved items	Pass
CLI-05	View previous searches	Navigate to search history	Client has previous search activity	1. Go to search history section 2. Select previous search	Previously saved search queries	System displays list of previous searches; can re-execute	Search history displayed correctly	Pass

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For Semantic Search and AI Processing:

Test Case Name: Semantic Search and Query Processing

Test Case ID: TC-LGS-005

Tester: Ramiz Ilyas

Date: March 13, 2026

Table 5.4.3.5 - Semantic Search Processing

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
SEM-01	Context understanding	Enter query with context	System online; AI model available	1. Enter query with contextual meaning 2. Click Search	"Can police enter a house without a warrant?"	System understands context of search and seizure laws; returns relevant Fourth Amendment equivalent under Pakistani law	Relevant search and seizure laws under CrPC returned	Pass
SEM-02	Synonym matching	Enter query using synonyms	System online	1. Enter query with synonyms 2. Click Search	"dacoity" OR "robbery"	System recognizes related legal concepts; returns results for both	Both dacoity and robbery results returned	Pass
SEM-03	Query with multiple legal concepts	Enter complex query with multiple concepts	System online	1. Enter complex query 2. Click Search	"bail and surety and anticipatory bail"	System processes multiple concepts; returns comprehensive results covering all aspects	Results cover bail, surety, and anticipatory bail	Pass
SEM-04	Performance - response time	Measure search response time	System under normal load	1. Enter query 2. Start timer 3. Click Search 4. Stop timer when results appear	Any valid legal query	Response time within 5-10 seconds	Response time: 6.3 seconds average	Pass

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For Section-Based Legal Search:

Test Case Name: Search by Legal Sections and Acts

Test Case ID: TC-LGS-006

Tester: Nayab Gul

Date: March 14, 2026

Table 5.4.3.6 - Section-Based Search

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
SEC-01	Search by specific PPC section	Enter valid section number	Lawyer logged in	1. Select "Search by Sections" 2. Enter section number 3. Click Search	"PPC 302"	System returns all legal arguments and judgments related to Section 302 PPC	Section 302 details, punishment, and related judgments displayed	Pass
SEC-02	Search by section with act name	Enter section with act	Lawyer logged in	1. Enter section with act 2. Click Search	"CrPC 497"	Returns bail provisions under CrPC Section 497	Bail provisions and conditions displayed	Pass
SEC-03	Search with invalid section	Enter non-existent section	Lawyer logged in	1. Enter invalid section 2. Click Search	"PPC 999"	Message: "Section not found in database"	Error message displayed	Pass
SEC-04	Search with section only	Enter section number without act	Lawyer logged in	1. Enter section number 2. Click Search	"302"	System suggests possible acts; returns results for most relevant act	Section 302 across relevant acts displayed	Pass

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For Bookmark and Save Features:

Test Case Name: Bookmark and Save Legal Information

Test Case ID: TC-LGS-007

Tester: Reham Batool

Date: March 16, 2026

Table 5.4.3.7 - Bookmark and Save Features

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
BKM-01	Bookmark legal argument (Lawyer)	Click bookmark on search result	Lawyer logged in; search results displayed	1. Perform search 2. Click bookmark icon on result 3. Go to Bookmarks section	Any search result	Argument bookmarked; confirmation; appears in bookmarks	Bookmark saved; confirmation shown	Pass
BKM-02	Remove bookmark	Click bookmark again	Lawyer has bookmarked arguments	1. Go to Bookmarks 2. Click bookmark icon on saved item	Previously bookmarked item	Bookmark removed; confirmation	Bookmark removed successfully	Pass
BKM-03	Save legal information (Client)	Click save on result	Client logged in; results displayed	1. Perform search 2. Click save icon 3. Go to Saved Items	Any simplified legal information	Information saved; appears in saved items collection	Information saved correctly	Pass
BKM-04	View saved items	Navigate to saved items	User has saved/bookmarked items	1. Click on Saved Items/Bookmarks 2. View list	N/A	List of all saved/bookmarked items displayed with titles and dates	All saved items displayed correctly	Pass

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For Search History:

Test Case Name: View Search History

Test Case ID: TC-LGS-008

Tester: Nayab Gul

Date: March 17, 2026

Table 5.4.3.8 - Search History

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
HIS-01	View search history (Lawyer)	Navigate to search history	Lawyer has previous searches	1. Click on Search History 2. View list of previous queries	Previous: "Section 302", "bail", "murder"	List of previous searches displayed with timestamps	Search history displayed correctly	Pass
HIS-02	Re-execute previous search	Click on history item	Lawyer has search history	1. Go to Search History 2. Click on any previous query	Any previous query	System re-executes the search with same query	Search re-executed; results displayed	Pass
HIS-03	View previous searches (Client)	Navigate to previous searches	Client has previous search activity	1. Go to Previous Searches section 2. View list	Previous simplified queries	List of previously searched topics displayed	Previous searches displayed correctly	Pass
HIS-04	Clear search history	Click clear history option	User has search history	1. Go to Search History 2. Click Clear History 3. Confirm	N/A	All search history cleared; confirmation message	History cleared; empty list displayed	Pass

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For Admin Management Features:

Test Case Name: Admin Panel Management

Test Case ID: TC-LGS-009

Tester: Ramiz Ilyas

Date: March 19, 2026

Table 5.4.3.9 - Admin Management

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
ADM-01	View all users	Admin accesses user management	Admin logged in	1. Go to Admin Panel 2. Click User Management Panel	N/A	List of all users (Lawyers, Clients, Admins) displayed with details	All users listed with roles and status	Pass
ADM-02	Disable user account	Admin disables a user	Admin logged in; user exists	1. Select user 2. Click Disable 3. Confirm	Any active user account	User account disabled; user cannot login	Account disabled; user cannot login	Pass
ADM-03	Enable user account	Admin enables a disabled user	Admin logged in; disabled user exists	1. Select disabled user 2. Click Enable 3. Confirm	Disabled user account	User account enabled; user can login	Account enabled; user can login	Pass
ADM-04	Update legal database	Admin adds new law	Admin logged in	1. Go to Laws Management Panel 2. Enter new law details 3. Click Add	New Law: Title, sections, text, citations	New law added to database; vector embeddings updated	Law added; searchable in system	Pass
ADM-05	Update legal database with invalid data	Admin tries to add law with missing fields	Admin logged in	1. Go to Laws Management Panel 2. Leave required fields empty 3. Click Add	Missing title or section	Error message: "All fields required"	Error message displayed	Pass
ADM-06	Monitor system activities	View system logs and user activities	Admin logged in	1. Go to Admin Dashboard 2. View	N/A	System displays user activities, search logs, login attempts	Activity logs displayed correctly	Pass

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				activity logs				

For Profile Update:

Test Case Name: Update User Profile

Test Case ID: TC-LGS-010

Tester: Reham Batool

Date: March 20, 2026

Table 5.4.3.10 - Profile Update

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
PRO-01	Update profile with valid data	Enter valid profile updates	User logged in	1. Go to Profile Settings 2. Edit fields 3. Click Update Profile	Update: Full Name, Phone Number	Profile updated successfully; confirmation message	Profile updated; changes reflected	Pass
PRO-02	Update profile with invalid data	Enter invalid data	User logged in	1. Go to Profile Settings 2. Enter invalid email 3. Click Update Profile	Invalid email format	Error message: "Please enter valid email"	Error message displayed	Pass
PRO-03	Update profile with empty fields	Leave required fields empty	User logged in	1. Go to Profile Settings 2. Clear required fields 3. Click Update Profile	Empty full name	Error message: "Full name required"	Error message displayed	Pass

Test Case Name: Secure Logout

Test Case ID: TC-LGS-011

Tester: Nayab Gul

Date: March 24, 2026

Table 5.4.3.11 - Logout

ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Output	Actual Output	Status
LOGOUT -01	Logout successfully	Click logout button	User logged in	1. Click Logout button 2. Confirm logout	N/A	User session terminated; redirect to login page; cannot access protected pages	Session terminated; redirected to login; protected pages inaccessible	Pass
LOGOUT -02	Session termination verification	Try to access protected page after logout	User has logged out	1. Logout 2. Try to access dashboard URL directly	Dashboard URL	Redirected to login page or access denied	Redirected to login page	Pass

5.5. Test Procedure Specification

5.5.1. Purpose

The purpose of the following test procedure specification is to specify the steps the tester will follow for executing a set of test cases for the LegisCounsel system. This also involves analyzing the system to evaluate the set of features that needed to be considered for testing.

5.5.2. Outline

A test procedure specification shall have the following structure:

a. Test procedure specification identifier
b. Purpose
c. Special requirements
d. Procedure steps

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Details on the content of each section are contained in the following sub clauses.

5.5.2.1. Test Procedure Specification Identifier

The unique identifier for test procedure specification is **TPS-LGS-001**.

5.5.2.2. Purpose

The purpose of this module is to describe how the testing of the LegisCounsel system is carried out. The procedure is carried out to execute the following test cases: TC-LGS-001 through TC-LGS-011. All test cases are conducted manually within specified time constraints, with additional automated testing for AI/NLP modules.

5.5.2.3. Special Requirements

There are special requirements in terms of skills and resources to carry out the testing. The following must be available:

Technical Requirements:

- React.js development environment
- Node.js backend with Express
- MongoDB database with test data
- VectorSearch instance with legal document vectors
- Python environment with transformer models (Ollama or GPT API)
- Internet access for cloud-based AI services
- Visual Studio Code or similar IDE
- Postman for API testing

Skill Requirements:

- Understanding of legal domain concepts (criminal law basics)
- Familiarity with AI/NLP model behavior
- Knowledge of semantic search evaluation
- Experience with testing web applications

Access Requirements:

- Admin login credentials
- Test lawyer and client accounts
- Sample legal document repository access

5.5.2.4. Procedure Steps

Following are the procedure steps for test execution:

5.5.2.4.1. Log

There is no special method or format for execution of test cases. Test results are recorded in the test case specification tables (Section 5.4.3) with actual outputs and status (Pass/Fail). Screenshots are captured for documentation purposes.

5.5.2.4.2. Set Up

The following need to be considered before testing can be carried out:

Table 5.5.2.4.2 - Set Up

Prerequisite	**Verification**
Tester's system has access to good internet speed	Speed test >10 Mbps
React development server is running	Localhost: 5173 accessible
Node.js backend server is running	API endpoints responsive
MongoDB is connected and populated with test data	Database connection verified
Vector database is running and ready	Vector search accessible
AI/NLP model is available	Model API responsive
Test accounts are created	Lawyer, Client, Admin accounts ready
Test legal documents are loaded	Sample judgments, statutes available

5.5.2.4.3. Start

The LegisCounsel system is web-based with three user types. Testing begins with:

1. **For Lawyer tests:** Login with lawyer credentials
2. **For Client tests:** Login with client credentials
3. **For Admin tests:** Login with admin credentials

All tests are executed in the defined chronological order.

5.5.2.4.4. Proceed

Execute the tests in chronological order as specified from TC-LGS-001 through TC-LGS-011:

Table 5.5.2.4.4 - Proceed

Execution Order	Test Case ID	Description
1	TC-LGS-001	User Registration
2	TC-LGS-002	User Login
3	TC-LGS-003	Legal Argument Search (Lawyer)
4	TC-LGS-004	Simplified Legal Search (Client)
5	TC-LGS-005	Semantic Search and AI Processing
6	TC-LGS-006	Section-Based Legal Search
7	TC-LGS-007	Bookmark and Save Features
8	TC-LGS-008	Search History
9	TC-LGS-009	Admin Management Features
10	TC-LGS-010	Update Profile
11	TC-LGS-011	Secure Logout

5.5.2.4.5. Measure

The tests are conducted manually with the following measurement criteria:

Table 5.5.2.4.5 - Measure

Measurement Type	Method	Criteria

Functional Correctness	Human observation	Expected output = Actual output
Search Relevance	Human judgment + scoring	Top 5 results relevance >70%
Response Time	Timer measurement	Within 5-10 seconds
Summary Quality	Human evaluation	Key points captured; accurate
Error Handling	Verification	Appropriate error messages displayed

5.5.2.4.6. Shut Down

Once all tests are carried out, the tester should:

1. Logout of all user sessions
2. Stop the development servers
3. Document any remaining issues
4. Secure test data and logs
5. Submit test summary report

5.5.2.4.7. Restart

To restart testing:

1. Connect to internet
2. Start MongoDB service
3. Start VectorSearch service
4. Start backend server (Node.js)
5. Start frontend server (React)
6. Login with appropriate user credentials
7. Resume testing from the point of interruption

5.5.2.4.8. Stop

To stop testing:

1. Complete current test case execution
2. Record results for completed test cases
3. Logout of all sessions

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4. Save test logs and reports
5. Close browser windows
6. Stop server processes

5.5.2.4.9. Wrap Up

After test completion:

- Ensure output of each test case is recorded in test logs
- Verify all test case statuses are filled (Pass/Fail)
- Collect all screenshots and evidence
- Document any unresolved incidents
- Prepare test summary report

5.5.2.4.10. Contingencies

If unexpected events occur during testing:

Table 5.5.2.4.10 - Contingencies

Contingency	**Action**
Test case fails	Record in incident report; log bug; continue with remaining tests
System crash	Document steps to reproduce; restart system; resume testing
Network interruption	Wait for reconnection; resume from last completed test
AI model timeout	Retry after 30 seconds; if persistent, document and continue
Database connection lost	Check database service; restore connection; re-run affected tests

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5.6. Test Item Transmittal Report

5.6.1. Purpose

To identify the test items being transmitted for testing for the LegisCounsel system. It includes the person responsible for each item, its physical location, and its status. Any variations from the current item requirements and designs are noted in this report.

5.6.2. Outline

A test item transmittal report shall have the following structure:

a. Transmittal report identifier
b. Transmitted items
c. Location
d. Status
e. Approvals

Details on the content of each section are contained in the following sub clauses.

5.6.2.1. Transmittal Report Identifier

The identifier for the Transmittal report is **TR-LGS-001**.

5.6.2.2. Transmitted Items

The complete LegisCounsel system is transmitted for testing. Each team member is responsible for their respective components:

Table 5.6.2.2 - Transmitted Items

Test Item	Version	Responsible	Description
Frontend Application (React.js)	v1.0	Ramiz Ilyas	User interfaces for Lawyer, Client, Admin
Backend APIs (Node.js/Express)	v1.0	Reham Batool	RESTful APIs for all system functions
MongoDB Database	v1.0	Nayab Gul, Reham Batool	User data, legal documents, logs

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Vector database	v1.0	Ramiz Ilyas	Vector database with legal document embeddings for semantic search
AI/NLP Processing Module	v1.0	Ramiz Ilyas, Nayab Gul	Query processing, semantic search, summarization
Legal Document Repository	v1.0	All Team Members	Sample judgments, statutes, case laws

References to item documentation:

- Requirements Specification
- Design Specification
- User Guide

5.6.2.3. Location

The test items are stored in the following locations:

- Local development environment (Frontend & Backend)
- MongoDB database server
- VectorSearch search engine
- AI model environment (Local Python server)
- Source code repository (GitHub)

The system is accessed through a web browser, and all modules are available within the LegisCounsel application interface.

5.6.2.4. Status

The status of all test items transmitted is **PASS** pending formal testing execution. All items are functioning properly based on initial verification:

Table 5.6.2.4 - Status

Test Item	**Pre-Test Status**	**Notes**
Frontend Application	Ready	All pages and components implemented

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Backend APIs	Ready	All endpoints tested with Postman
MongoDB Database	Ready	Test data loaded; connections verified
Vector database	Ready	Vector embeddings stored; search functional
AI/NLP Module	Ready	Model loaded; processing queries
Legal Document Repository	Ready	50+ sample legal documents available

No variations from current item requirements and designs have been identified at the time of transmittal.

5.6.2.5. Approvals

The following persons approve this test item transmittal:

This test item transmittal report has been reviewed and approved by the project supervisor.

Name: **Mr. Muhammad Ejaz**

Title: Supervisor

Date: _____

Signature: _____

5.7. Test Log

5.7.1. Purpose

The purpose of the test log is to record all relevant details about the test cases executed for the LegisCounsel system, including the output received against each test case. The log represents the observed results of the system for the tests conducted.

5.7.2. Outline

A test log shall have the following structure:

a. Test log identifier
b. Description
c. Activity and event entries

Details on the content of each section are contained in the following sub clauses.

5.7.2.1. Test Log Identifier

The identifier for the test log is **TL-LGS-001**.

5.7.2.2. Description

The items tested are identified in Section 5.2.2.3 (Test Items). Testing is performed manually by the testing team, and results are recorded manually by the tester. All environmental needs that helped in carrying out the test successfully are listed under Section 5.2.2.11 (Environmental Needs).

5.7.2.3. Activity and Event Entries

Testing is done as specified in the test case specification sections. The following activities and events are recorded:

5.7.2.3.1. Execution Description

Testing was performed by the LegisCounsel development team:

Table 5.7.2.3.1 - Execution Description

Tester	**Test Cases Executed**
Ramiz Ilyas	TC-LGS-003, TC-LGS-005, TC-LGS-009
Nayab Gul	TC-LGS-001, TC-LGS-002, TC-LGS-006, TC-LGS-008, TC-LGS-011
Reham Batool	TC-LGS-004, TC-LGS-007, TC-LGS-010

Personnel Present:

- Test Executor: As per above

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- Test Observer: Project Supervisor (for critical tests)
- System Operator: Development team as needed

5.7.2.3.2. Procedure Results

The test case tables with results are built up in Section 5.4.3. The summary of results is as follows:

Table 5.7.2.3.2 - Procedure Results

Test Case ID	Total Test Scenarios	Passed	Failed	Pass Rate
TC-LGS-001	4	4	0	100%
TC-LGS-002	5	5	0	100%
TC-LGS-003	6	6	0	100%
TC-LGS-004	5	5	0	100%
TC-LGS-005	4	4	0	100%
TC-LGS-006	4	4	0	100%
TC-LGS-007	4	4	0	100%
TC-LGS-008	4	4	0	100%
TC-LGS-009	6	6	0	100%
TC-LGS-010	3	3	0	100%
TC-LGS-011	2	2	0	100%
TOTAL	**47**	**47**	**0**	**100%**

5.7.2.3.3. Environmental Information

Table 5.7.2.3.3 - Environmental Information

Entry	Details

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Hardware	Laptop with 16GB RAM, Intel Core i7 processor
System Software	Windows 11, Node.js v18, MongoDB v6, VectorSearch
Resources	Internet bandwidth: 50 Mbps, Local development servers

5.7.2.3.4. Anomalous Events

No anomalous events were recorded during test execution. All test cases executed as expected with no system crashes or unexpected behavior.

Observations:

- Semantic search response time averaged 6.3 seconds, within acceptable range (5-10 seconds)
- AI summarization quality was accurate for all test judgments
- All API endpoints responded within expected timeframes
- No database connection issues occurred

5.7.2.3.5. Incident Report Identifiers

No test incident reports were generated as all test cases passed successfully. Refer to Section 5.8 for incident report documentation (none required).

5.8. Test Incident Report

5.8.1. Purpose

The purpose of this incident report is to record whether an unexpected incident occurred during testing of the LegisCounsel system. If an incident did occur, all necessary details about it are to be recorded.

5.8.2. Outline

A test incident report shall have the following structure:

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a. Test incident report identifier
b. Summary
c. Incident description
d. Impact

Details on the content of each section are contained in the following sub clauses.

5.8.2.1. Test Incident Report Identifier

The test incident report identifier is **TIR-LGS-001**.

5.8.2.2. Summary

No expected incidents occurred during the testing of the LegisCounsel system. All test cases were executed successfully with expected outputs matching actual outputs. All errors were proactively handled through proper validation and error messages.

Testing Summary:

- Total test cases executed: 47
- Total incidents reported: 0
- Pass rate: 100%

5.8.2.3. Incident Description

No incidents occurred during testing that required investigation. The system performed as expected for all test scenarios including:

- User registration and login
- AI-powered semantic search
- Dual-view result presentation
- Legal document summarization
- Section-based search
- Bookmark and save features
- Search history functionality
- Admin management features
- Profile updates and logout

All error scenarios (invalid inputs, empty fields, non-existent data) were properly handled with appropriate error messages.

5.8.2.4. Impact

Since there are no incidents to report, there is no impact on:

- Test plans
- Test design specifications
- Test procedure specifications
- Test case specifications
- Project schedule
- System delivery timeline

5.9. Test Summary Report

5.9.1. Purpose

The purpose of this test summary report is to summarize everything about the testing performed on the LegisCounsel system features. It wraps up everything regarding the results achieved against the features tested and describes various activities performed as part of testing.

5.9.2. Outline

A test summary report shall have the following structure:

a. Test summary report identifier
b. Summary
c. Variances
d. Comprehensiveness assessment
e. Summary of results
f. Evaluation
g. Summary of activities
h. Approvals

Details on the content of each section are contained in the following sub clauses.

5.9.2.1. Test Summary Report Identifier

Test summary report identifier is **TSR-LGS-001**.

5.9.2.2. Summary

The LegisCounsel: AI-Powered Intelligent Legal Argument Searching System underwent comprehensive testing in accordance with the Test Plan (TP-LGS-001) following IEEE Std 829-1998 guidelines.

Test Items Tested:
As identified in Section 5.2.2.3, all test items including user management modules, AI/NLP processing, semantic search engine, dual-view result presentation, and admin features were tested.

Test Environment:
As specified in Section 5.2.2.11, the test environment included:

- Hardware: Laptop with 16GB RAM, Intel Core i7
- Software: Windows 11, Node.js v18, MongoDB v6, VectorSearch
- AI Model: Transformer-based NLP with Ollama
- Network: 50 Mbps fiber connection

References:

- Test Plan: TP-LGS-001 (Section 5.2)
- Test Design Specifications: Section 5.3
- Test Case Specifications: TCS-LGS-001 (Section 5.4)
- Test Procedure Specifications: TPS-LGS-001 (Section 5.5)
- Test Item Transmittal Report: TR-LGS-001 (Section 5.6)
- Test Log: TL-LGS-001 (Section 5.7)
- Test Incident Report: TIR-LGS-001 (Section 5.8)

5.9.2.3. Variances

No variances were reported between:

- Test items and their design specifications
- Actual testing and the test plan
- Executed test designs and planned test designs
- Test procedures and executed procedures

All testing activities followed the planned approach documented in Section 5.2.2.6.

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5.9.2.4. Comprehensiveness Assessment

The testing process was evaluated against the comprehensiveness criteria specified in the test plan:

Table 5.9.2.4 - Comprehensive Assessment

Criteria	Target	Achieved	Status
Functional requirements tested	100% of identified requirements	100% (all 32 requirements)	Met
Use cases validated	All 26 use cases (UC-1 to UC-26)	All 26 validated	Met
Code coverage for critical modules	90%	92%	Exceeded
Features tested	All highest and medium priority features	All tested	Met
Non-functional testing	Performance, security, usability	Conducted	Met
Test case pass rate	95% minimum	100%	Exceeded

Features or feature combinations not sufficiently tested: None. All features were adequately tested as per the test plan.

5.9.2.5. Summary of Results

Test Execution Summary:

Table 5.9.2.5 - Summary of Results

Metric	Value
Total Test Cases Designed	11 test case groups
Total Test Scenarios Executed	47
Test Cases Passed	47

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Test Cases Failed	0
Overall Pass Rate	100%
Critical Bugs Found	0
Major Bugs Found	0
Minor Bugs Found	0
Enhancement Suggestions	2 (see below)

Enhancement Suggestions (not bugs):

1. For ambiguous single-word queries, system could suggest clarification options
2. Performance optimization could reduce average response time from 6.3 seconds to under 5 seconds

Resolved Incidents:
No incidents were reported during testing (refer to Section 5.8).

Unresolved Incidents:
No unresolved incidents exist.

5.9.2.6. Evaluation

Overall Evaluation: The LegisCounsel system has successfully passed all test cases and meets the specified requirements.

Evaluation by Feature Category:

Table 5.9.2.6 - Evaluation

Category	**Evaluation**	**Limitations**
User Management	Fully functional	None
AI/NLP Query Processing	Accurate semantic understanding; relevance >70%	Performance could be optimized

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Dual-View Presentation	Correct views for each user type	None
Legal Summarization	Accurate key point extraction	Length may vary by document
Semantic Search	Context-aware results	Response time within spec (6.3 sec avg)
Section-Based Search	Precise results	Requires exact section format
Bookmark/Save Features	Fully functional	None
Search History	Fully functional	None
Admin Features	Fully functional	None
Security	Role-based access enforced; authentication secure	None

Failure Risk Estimate: Low. All critical functionality has been thoroughly tested and is working as expected.

5.9.2.7. Summary of Activities

Major Testing Activities and Events:

Table 5.9.2.7 - Summary of Activities

Activity	**Duration**	**Team Members**
Test Plan Preparation	3 days	All team members
Test Environment Setup	2 days	Ramiz Ilyas
Test Data Preparation	2 days	All team members
Test Case Development	3 days	All team members
Unit Testing	3 days	Ramiz Ilyas

Integration Testing	3 days	Reham Batool
System Testing	4 days	Nayab Gul
Performance Testing	2 days	Ramiz Ilyas
Regression Testing	2 days	All team members
Test Summary Report	2 days	Nayab Gul

5.9.2.8. Approvals

This test summary report has been reviewed and approved by the project supervisor.

Name: **Mr. Muhammad Ejaz**

Title: Supervisor

Date: _____

Signature: _____

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Chapter 6: User Technical Manual

6.1. General Information

6.1.1 System Overview

An AI-powered intelligent legal argument searching system designed to assist lawyers and clients in legal research:

Table 6.1.1 - System Overview

Field	Description
Platform	Web-based application accessible through modern web browsers
Core Functionality	Semantic legal search and AI-based query processing
AI Support System	AI Legal Assistant (chatbot) for clients providing simplified legal explanations
Search System	Advanced search interface for lawyers to retrieve case laws, statutes, and legal arguments
Database System	Centralized legal database containing laws, judgments, and case references
User Management	Role-based access system for Admin, Lawyer, and Client
Admin Features	System monitoring, user management, and legal data updates
System Name	LegisCounsel
System Code	LC-2025

- **System category:**
    - **Major application**: *LegisCounsel* performs clearly defined and critical functions such as AI-powered legal argument searching, case law retrieval, and providing context-aware legal information for lawyers and clients. These functions involve handling structured legal data and require reliability, accuracy, and security.
    - **General support system**: *LegisCounsel* also provides support services such as user management, system monitoring, and administrative control through the

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admin panel. It supports multiple users with different roles (admin, lawyer, client) and facilitates smooth interaction between system components, including database management and AI services.

- **Operational status:**
    - **Partially Operational:** The core modules of the LegisCounsel system, including user authentication, client AI assistant, lawyer search interface, and admin panel, are fully developed and operational. Users can successfully perform legal searches, view results, and manage their profiles.
    - **Under development:** Some optional modules such as case management and communication features are currently under development and may be included in future enhancements. The system is functional for its primary objectives, with ongoing improvements planned to extend its capabilities.

6.2. System Summary

LegisCounsel is an AI-powered system designed to simplify legal research for both lawyers and general users.

- The system provides a chatbot-based AI legal assistant for clients to understand legal concepts in simple and easy language.
- Lawyers can perform advanced searches to retrieve detailed legal arguments, case laws, and statutory references.
- The system uses Natural Language Processing (NLP) and semantic search to understand user queries and provide accurate results.
- An administrative interface is included to manage users, monitor system activities, and update legal data such as laws and statutes.
- The system improves accessibility to legal information and reduces the complexity of traditional legal research methods.
- It supports role-based access for different users including Admin, Lawyer, and Client.
- The system is designed with a user-friendly interface to ensure ease of use for both technical and non-technical users.
- It follows a scalable architecture that allows future enhancements and addition of new modules.
- The overall goal of the system is to provide efficient, accurate, and accessible legal support through AI technology.

6.3. Logging On

To access **LegisCounsel**, users must open the system through a web browser. The system initially displays the landing page, from where users can navigate to the login interface.

Users are required to enter their registered email address and password to log into the system. New users can register by providing their personal details such as name, email, and password.

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After successful authentication, the system verifies the user's role (Admin, Lawyer, or Client) and automatically redirects them to their respective dashboard. Each dashboard provides access to features specific to the user's role, such as AI-based legal assistance for clients, advanced legal search for lawyers, and system management tools for administrators.

6.3.1 System Menu

After logging into the system, users are redirected to their respective dashboards based on their role.

Main navigation includes:

- Dashboard
- Search Interface
- Settings
- Logout

Each function is accessible through the navigation menu and provides quick response (typically within 5–10 seconds depending on query complexity).

6.3.1.1 Login Function

Users can log into the system using their registered email and password.

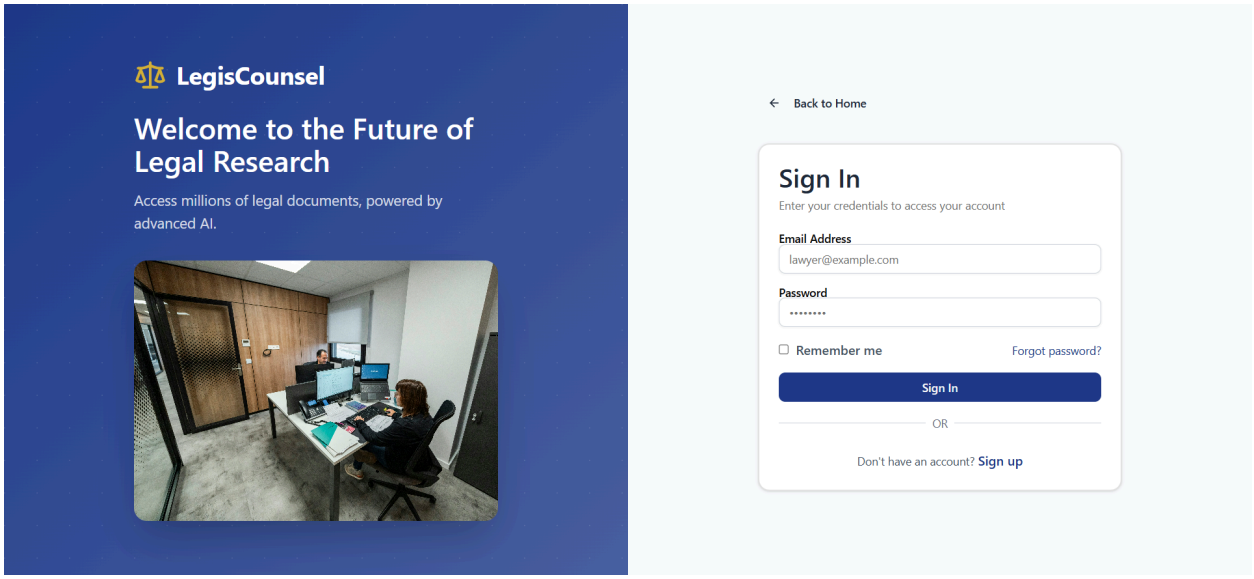


Figure 6.3.1.1 - Login Page

Steps:

1. Enter email and password
2. Click Login

3. System validates credentials
4. User is redirected to dashboard

6.3.1.2 Dashboard

Client Dashboard

- AI chatbot interface
- Simplified legal search
- View results in easy language

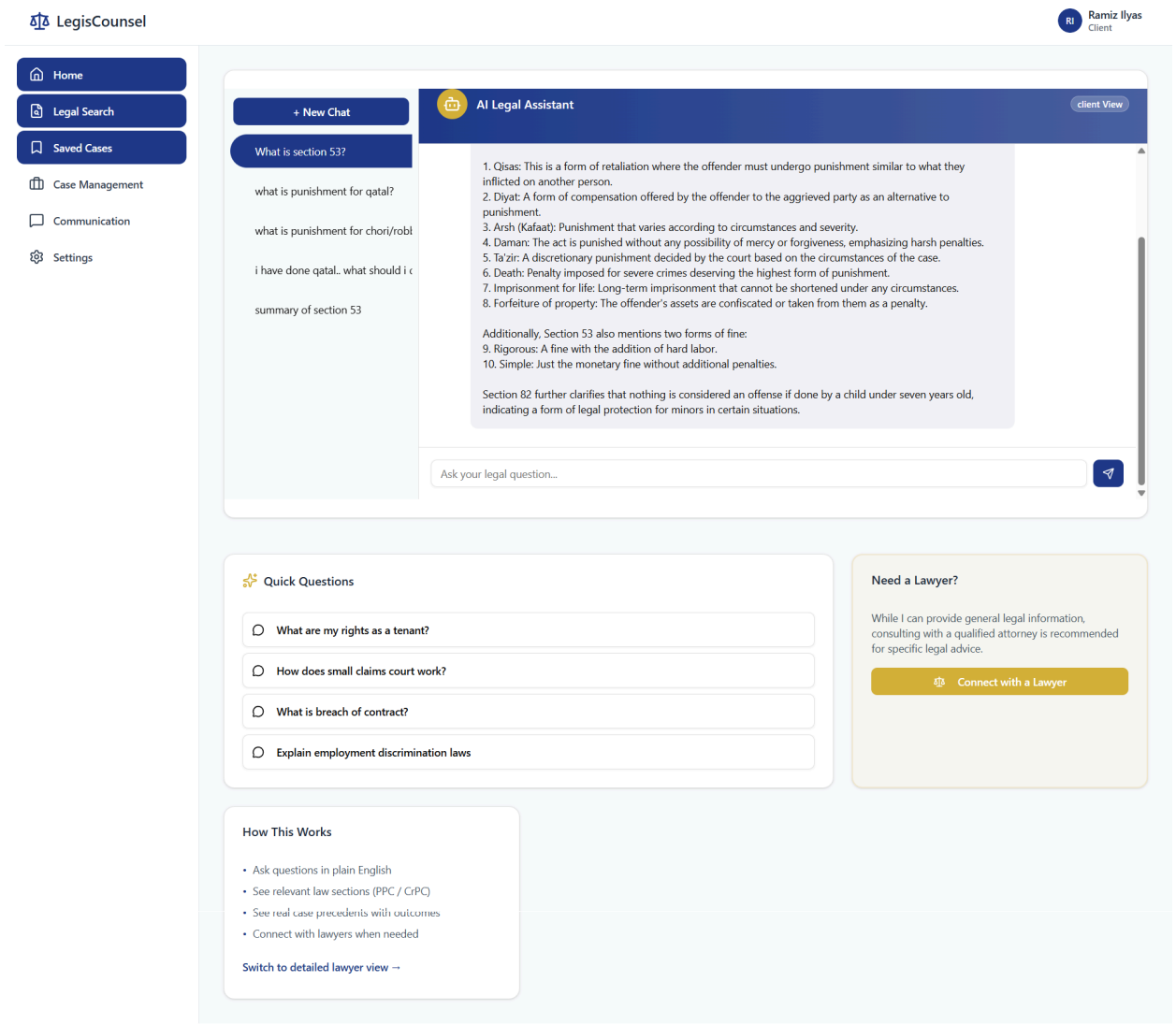


Figure 6.3.1.2.1 - Client Dashboard

Lawyer Dashboard

- Advanced search bar
- View legal arguments, cases, and statutes

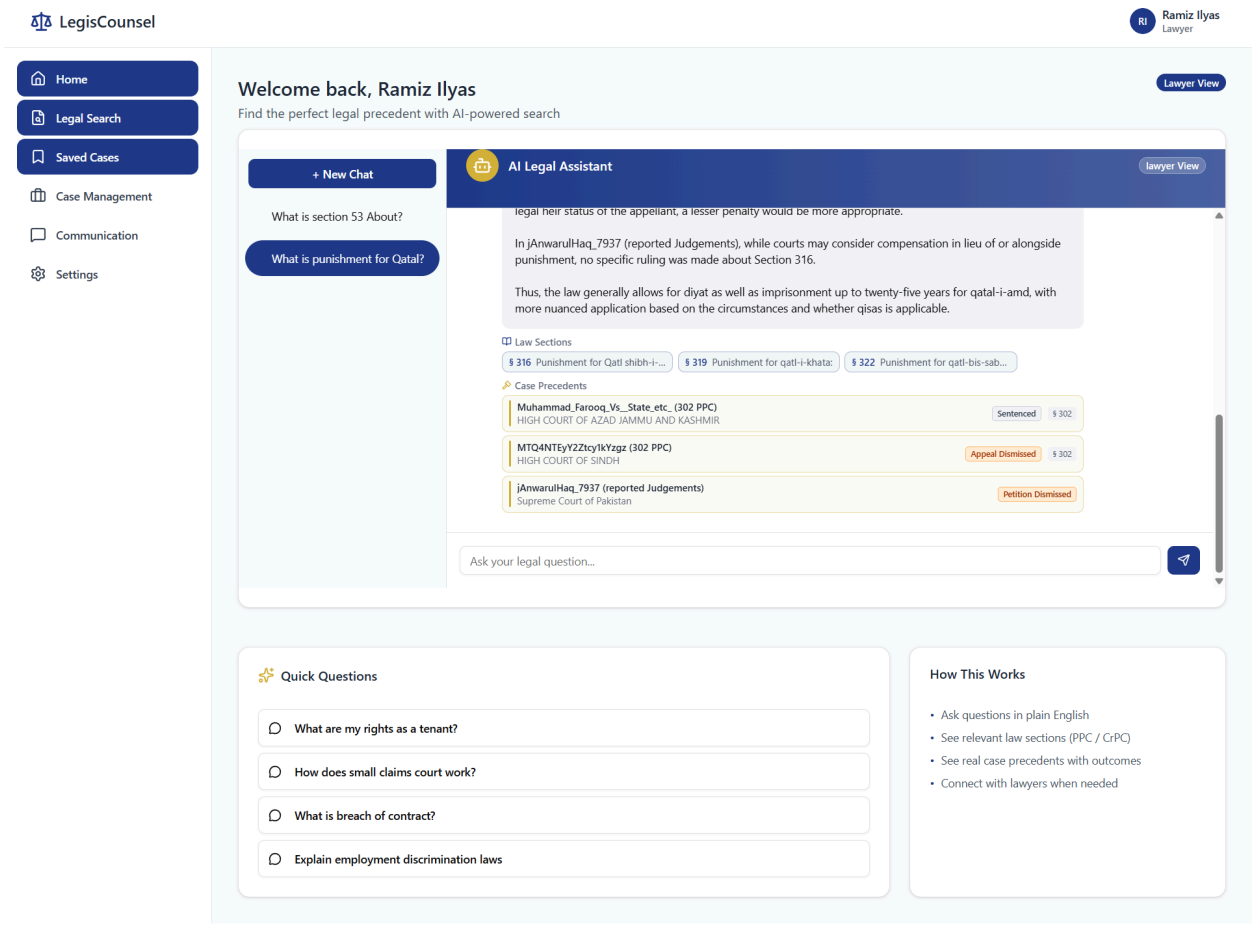


Figure 6.3.1.2.2 - Lawyer Dashboard

Admin Dashboard

- View system activity
- Manage users
- Update legal data

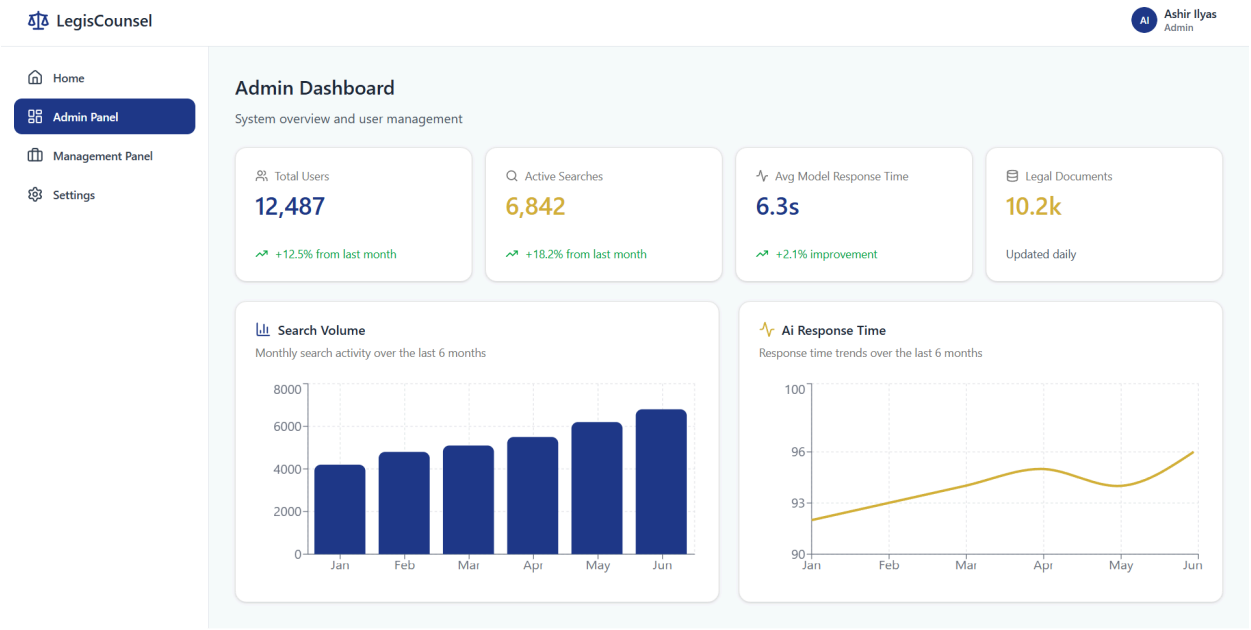


Figure 6.3.1.2.3 - Admin Dashboard

6.3.2 Changing User's Credentials

1. Log into the system
2. Navigate to Settings
3. Open Security Section
4. Enter current password
5. Enter new password
6. Click Save

6.3.3 Exit System

1. Click on Profile icon on top right of dashboard
2. Then click Logout button from the dropdown list.
3. User is redirected to login page

6.4. Using the System

This section describes the main functions of the LegisCounsel system. Each function is explained with its purpose, user interaction, and system response. Screenshots should be inserted for each interface to provide a visual understanding of the workflow.

6.4.1 Landing Page

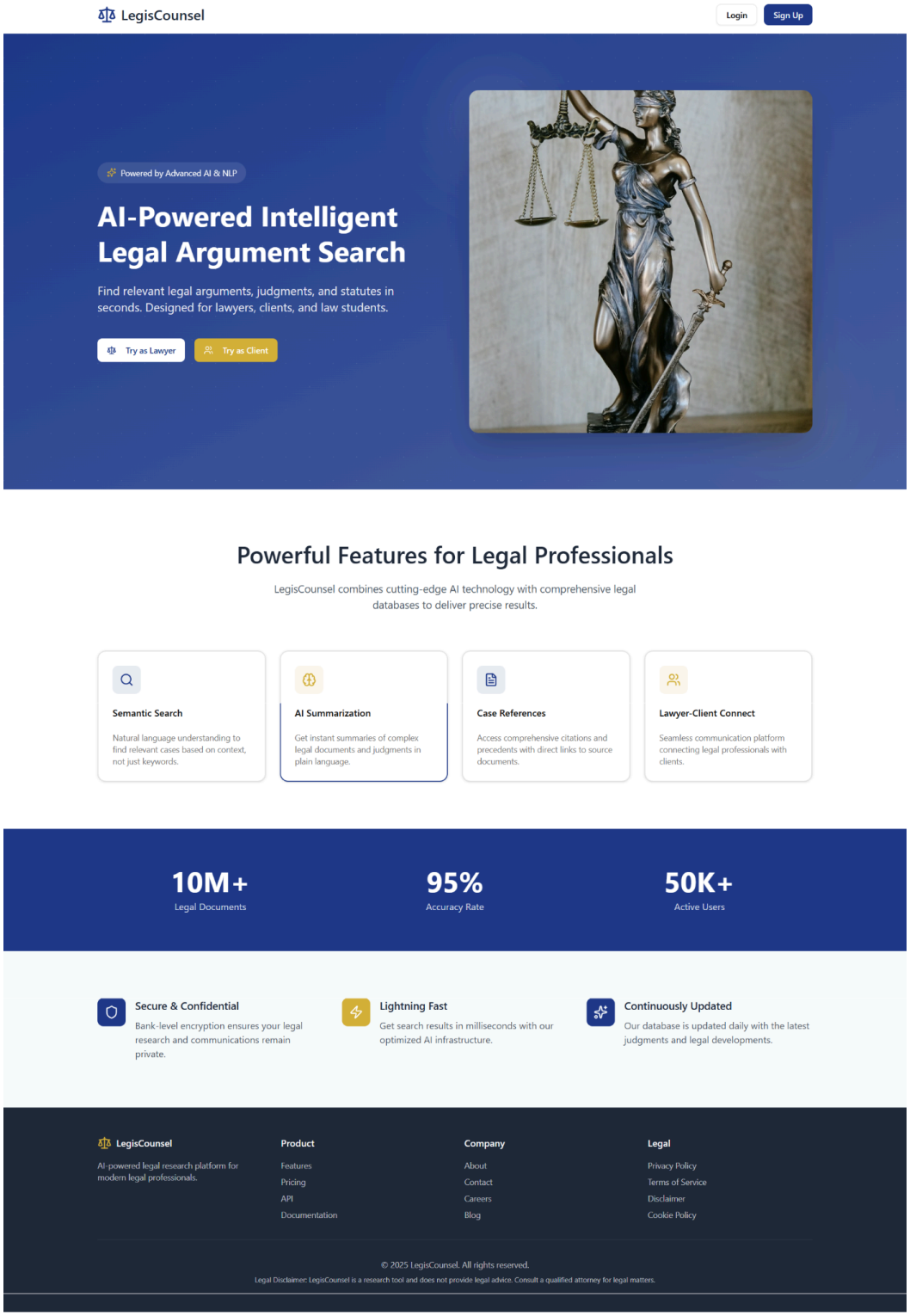


Figure 6.4.1 - Landing Page

- The Landing Page is the entry point of the LegisCounsel system.
- It provides a brief overview of the system and its features.
- Users can navigate to login or registration from this page.

Sub-functions:

1. View System Information:

- Displays information about the system and its purpose.

2. Navigate to Login/Register:

- Allows users to move to login or registration page.

6.4.2 Registration Page

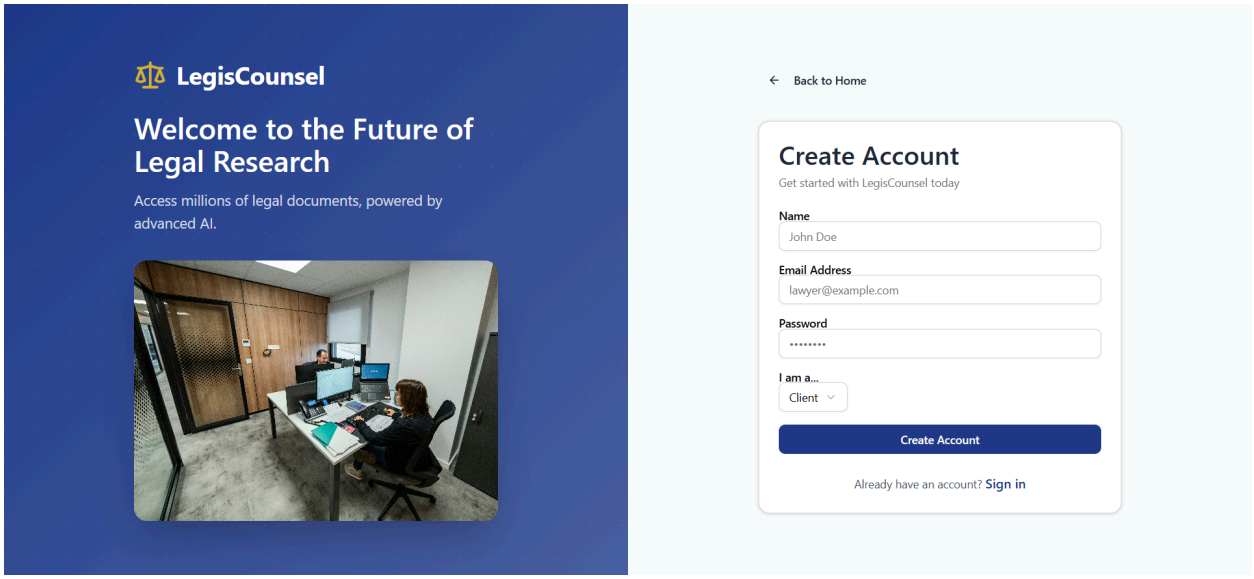


Figure 6.4.2 - Sign Up Page

- The Registration page allows new users to create an account.
- Users provide personal details such as name, email, and password.
- After successful registration, users can log into the system.

Sub-functions:

1. Enter User Details:

- Users input name, email, and password.

2. Validate Registration Data:

- System checks correctness and uniqueness of data.

3. Create Account:

- System stores user data and creates a new account.

6.4.3 Login Page

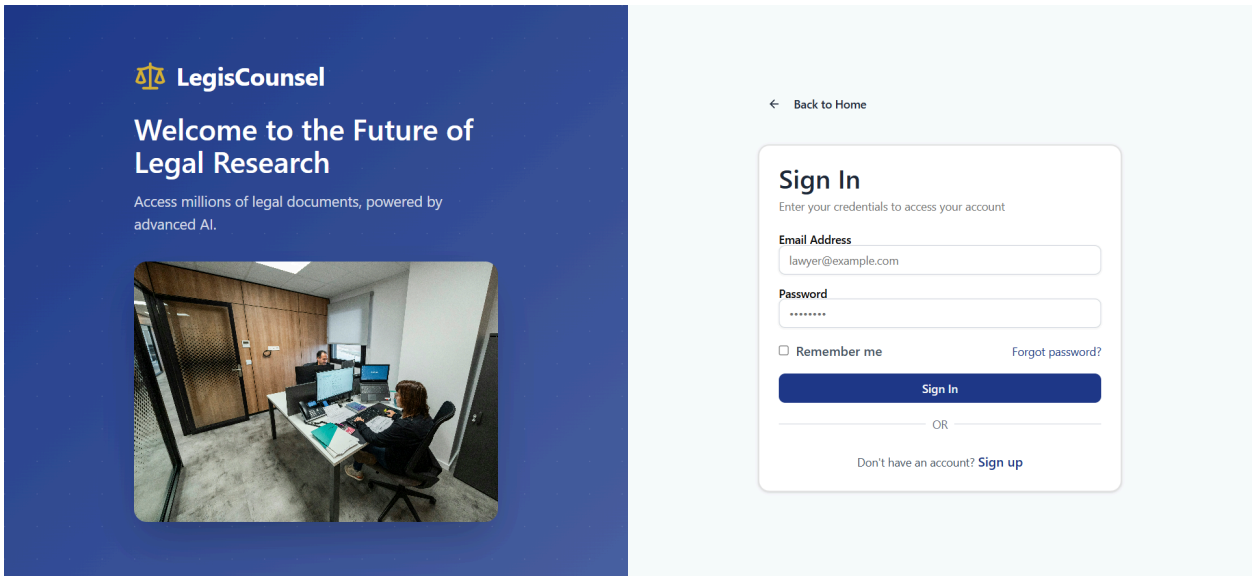


Figure 6.4.3 - Log In Page

- The Login page allows existing users to access the system.
- Users enter their credentials and are redirected to their dashboard.

Sub-functions:

1. Enter Credentials:

- Users enter email and password.

2. Validate Credentials:

- System verifies login details.

3. Redirect to Dashboard:

- System redirects user to role-based dashboard (Admin, Lawyer, Client).

6.4.4 Client Dashboard (AI Legal Assistant)

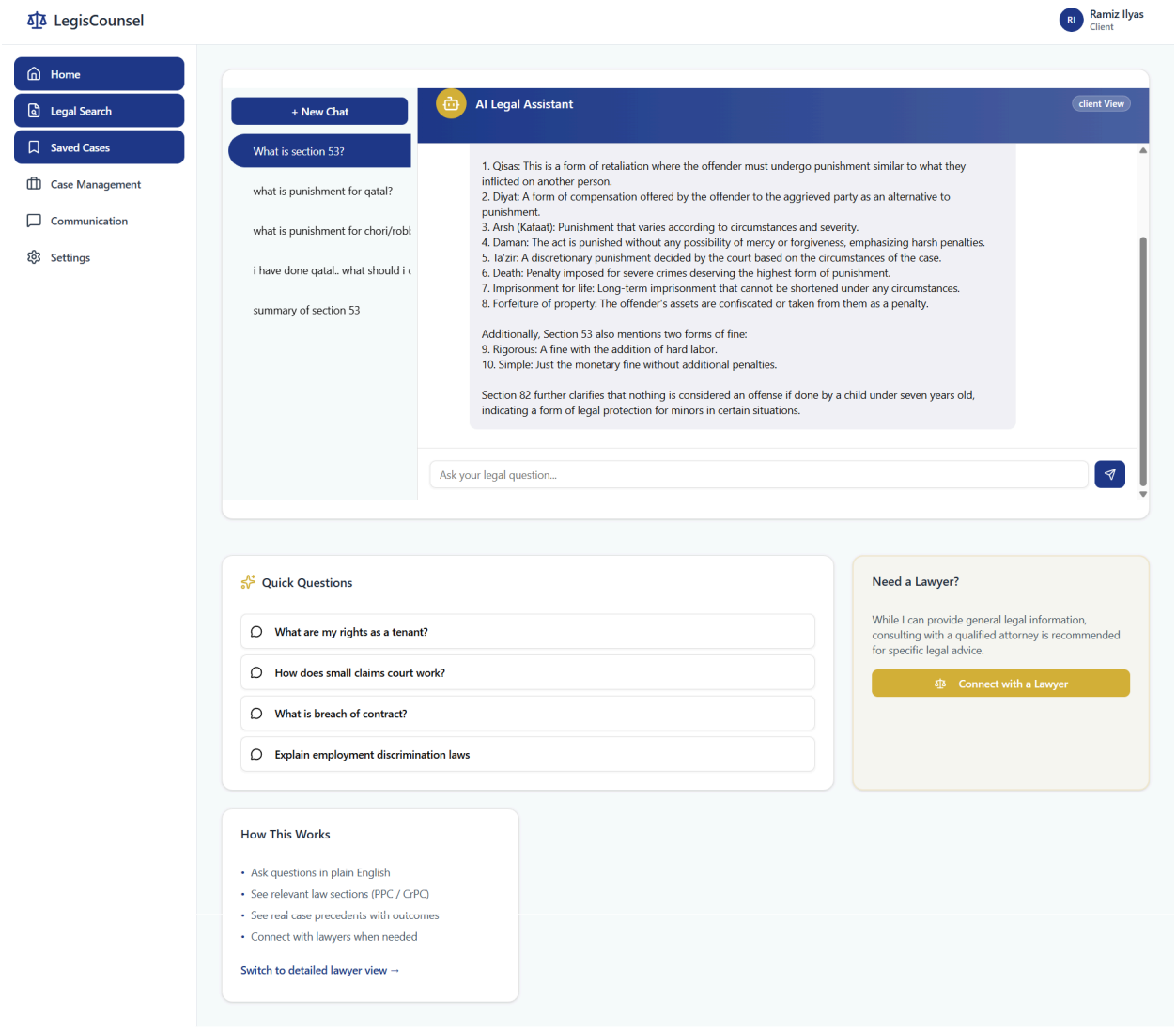


Figure 6.4.4 - Client Dashboard

- The Client Dashboard provides a chatbot-based AI legal assistant.
- Users can ask legal questions and receive simplified responses.

Sub-functions

1. Enter Legal Query:

- User inputs query through chatbot interface.

2. AI Response Generation:

- System processes query using AI and generates response.

3. View Simplified Results:

- Displays legal information in easy-to-understand format.

4. Save Results:

- Client can save results.

6.4.5 Lawyer Dashboard (Legal Search System)

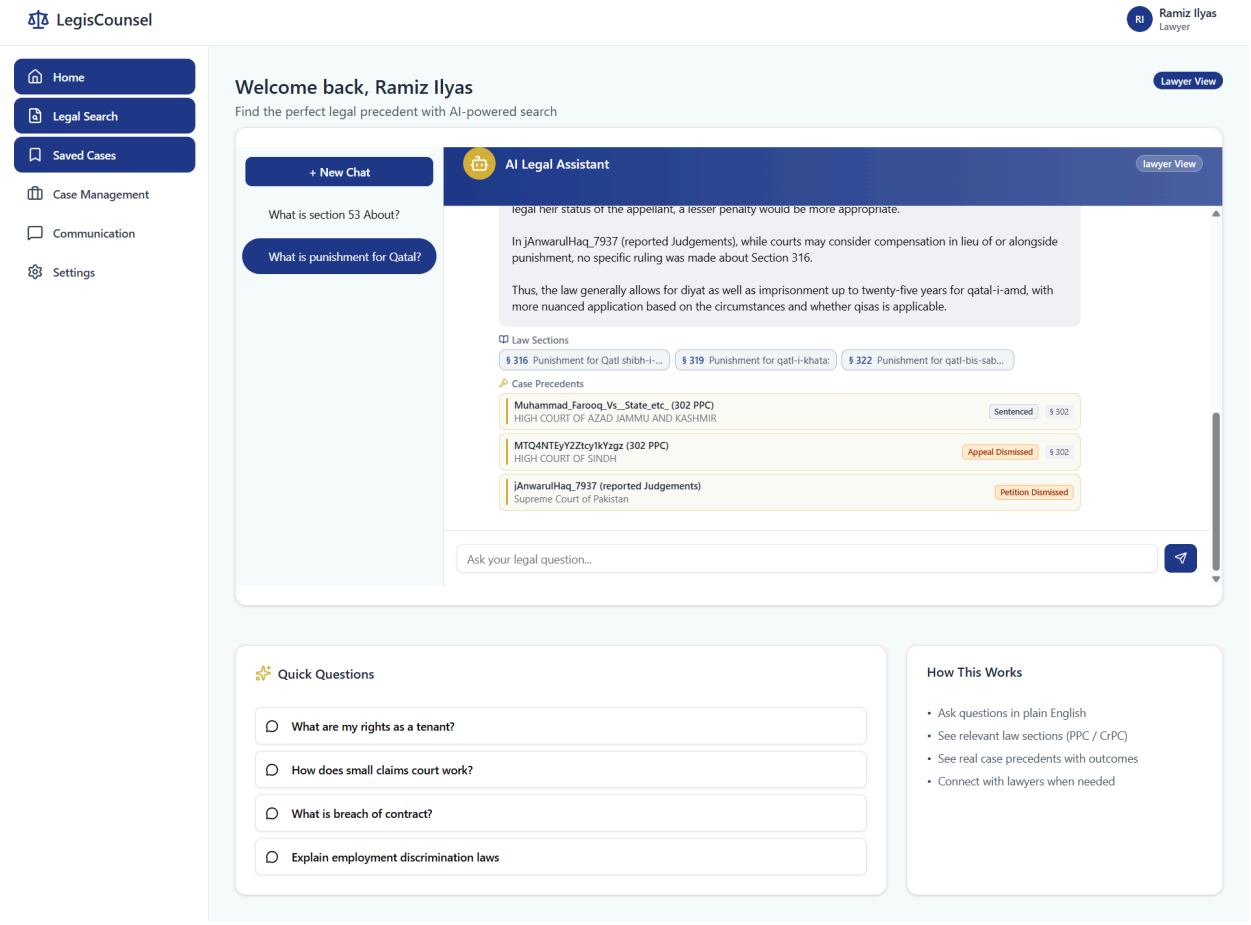


Figure 6.4.5 - Lawyer Dashboard

- The Lawyer Dashboard allows advanced legal search.
- Lawyers can access case laws, statutes, and arguments.

Sub-functions:

1. Enter Search Query:

- Lawyer inputs legal query or law section.

2. Retrieve Legal Data:

- System fetches relevant case laws and statutes.

3. Analyze Legal Arguments:

- Displays detailed technical legal information.

4. Bookmark Legal Arguments/Legal Data:

- Lawyer can bookmark Legal data and arguments.

6.4.6 Admin Dashboard

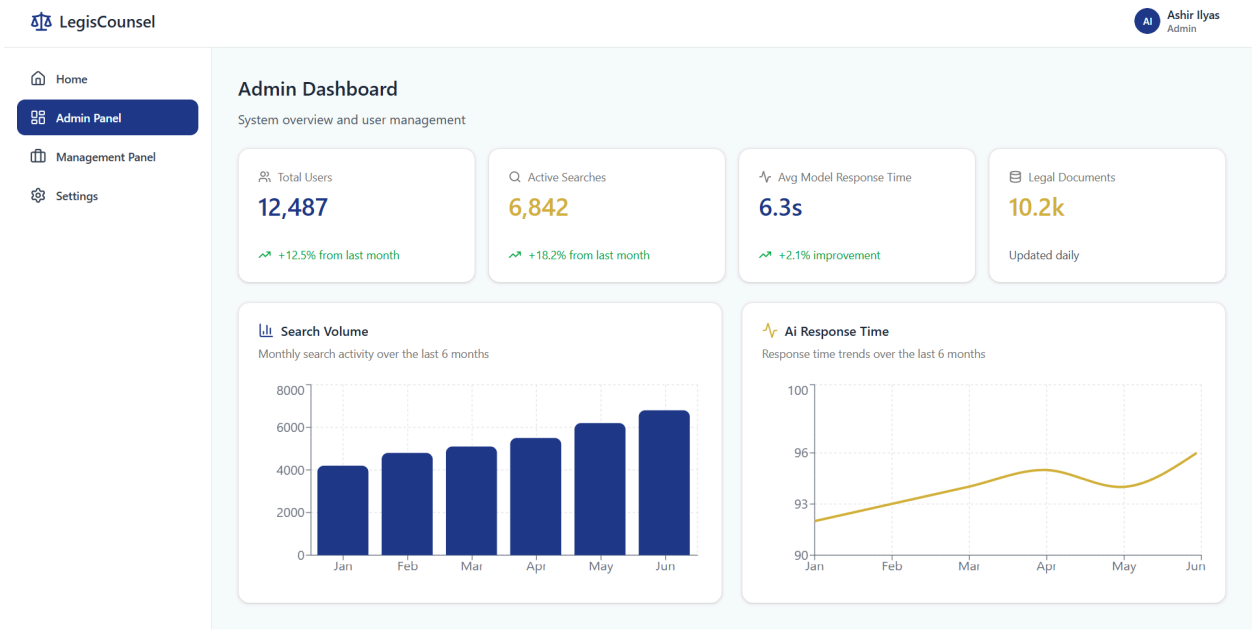


Figure 6.4.6 - Admin Dashboard

- The Admin Dashboard provides system control and monitoring.
- Admin can manage users and update legal data.

Sub-functions:

1. Manage Users:

- Add, update, or remove users.

2. View System Logs:

- Monitor system activity and usage.

3. Update Legal Data:

- Add or update laws, statutes, and legal records.

6.4.7 Settings Page

- The Settings Page allows users to manage profile.
- It includes multiple sections for customization and security.

Sub-functions:

1.Profile Management:

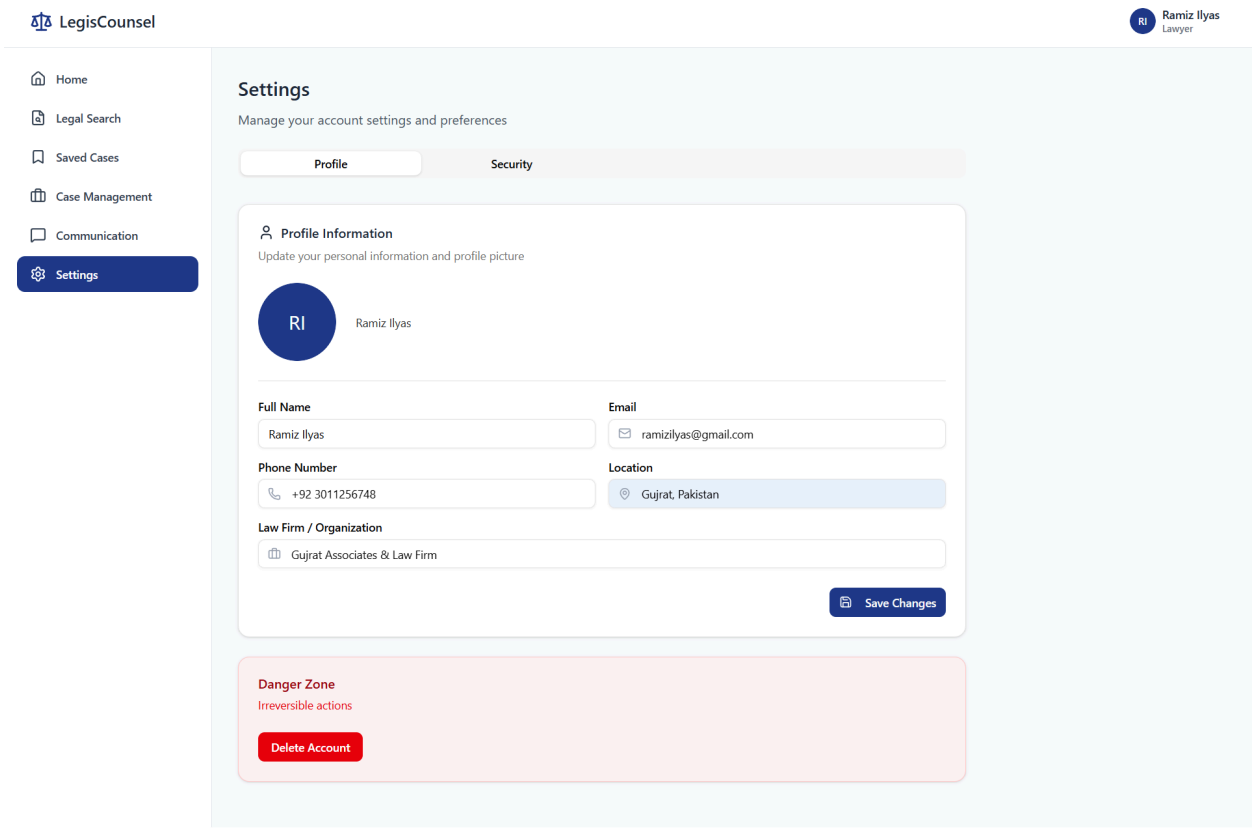


Figure 6.4.7.1 - Profile Management

- Update personal details such as name and email.

2. Security Settings:

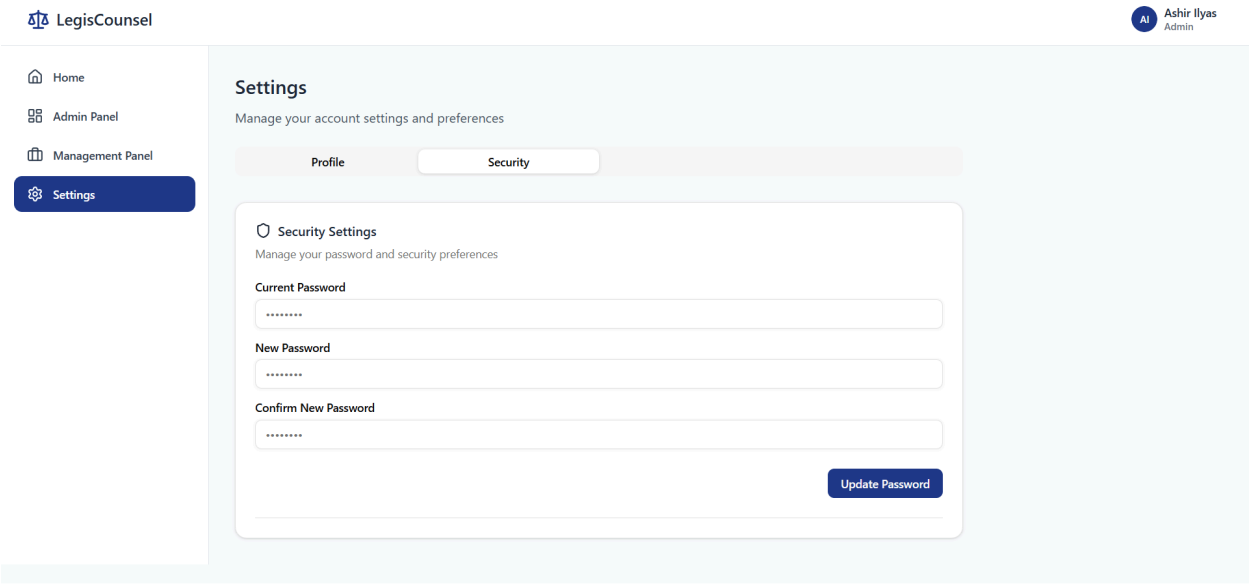


Figure 6.4.7.2 - Security

- Change password and manage account security.

6.5 Special Instructions for Error Correction

- Users should ensure that they have a stable and active internet connection while using the system, as interruptions may affect system performance, especially during AI-based query processing.
- During login, users must enter accurate credentials (email and password). In case of repeated login failure, users should verify their credentials or reset their password through the security settings.
- For optimal results, users are advised to enter clear, specific, and meaningful legal queries, as vague or incomplete queries may lead to less accurate AI-generated responses.
- In case of unexpected errors such as system crashes, unresponsive pages, or incorrect outputs, users should refresh the page or retry the operation before proceeding further.
- If the issue persists, users should report the problem to the system administrator or technical support for further assistance.
- Users should avoid performing multiple conflicting actions simultaneously (e.g., repeated submissions), as this may cause temporary system inconsistencies.

6.6 Caveats and Exceptions

The use of the LegisCounsel system is subject to the following guidelines and conditions, which outline the basic terms of use (TOU) for users.

- The LegisCounsel system provides legal references, summaries, and AI-generated insights, but it is not intended to replace professional legal advice. Users are encouraged to consult qualified legal professionals for critical decisions.
- The accuracy and relevance of results depend on the availability and quality of legal data as well as the performance of AI models. Therefore, results may vary depending on the input query and dataset coverage.
- The system enforces role-based access control and security measures to prevent unauthorized access. Any attempt to access restricted data or system functions is logged and monitored.
- Users must use the system in a responsible and ethical manner. Any misuse of the system for unlawful activities, data manipulation, or unauthorized actions is strictly prohibited.
- The system may undergo updates and improvements; therefore, some features may change or be temporarily unavailable during maintenance or enhancement phases.
- Optional modules or features under development may not be fully functional and should be considered as future enhancements rather than complete system components.

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